Hochschild Homology of the Cusp Calculus

Notes on Melrose–Nistor, *Homology of pseudodifferential operators I. Manifolds with boundary*

Expository notes on Melrose–Nistor

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Abstract

These notes are a reader's guide to the paper of R. Melrose and V. Nistor [14], which computes the continuous Hochschild homology of the algebra of *cusp* pseudodifferential operators of integral order with Laurent coefficients at the boundary of a compact manifold X, together with the ideals and quotients cut out by the two natural filtrations. The paper then interprets the index of a fully elliptic operator as a Hochschild 1-cocycle and writes it as a sum of a symbolic (Atiyah–Singer type) term and an eta invariant of the indicial family. We work the model cases first: closed manifolds, the half-line $[0,1)$ with its suspended boundary algebra, the interval and the cylinder. After that we follow the sections of [14] in order and supply the arguments that the paper compresses: H-unitality, the weighted de Rham lemmas, the spectral sequences and their convergence, the analytic continuation behind the four trace functionals, the boundary maps, and the index formula. We also record a number of corrections. The most important are these. The homology statements in [14, (16)] tacitly assume that ∂X is connected and that $\dim X \geq 2$; for N boundary components the boundary map in degree two has rank $N-1$, not 0. The signs of the symbolic terms in the index theorem [14, (93)] are inconsistent with the paper's own formula [14, (87)] and with the closed case [14, (3), (5)]. We verify the corrected formula on explicit examples.

1 Overview and roadmap

1.1 The algebras

Let X be a compact manifold with boundary, $\dim X = n$, and let $x \in C^\infty(X)$ be a boundary defining function. The function x fixes a *cusp structure* (Section 4.1), and with it the algebra $\Psi_c^{\mathbb{Z}}(X)$ of classical cusp pseudodifferential operators of integral order. Conjugation by powers of x preserves each $\Psi_c^m(X)$, so

$$x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X) = \bigcup_{k\in\mathbb{Z}} \bigcup_{m\in\mathbb{Z}} x^k \Psi_c^m(X)$$

is an algebra. MN write $x^{-\mathbb{Z}}$ to stress that this second filtration, by powers of x, runs in the opposite direction to the order filtration. Each filtration has a residual part, and the doubly residual operators form the ideal

$$\mathcal{I} = x^\infty \Psi_c^{-\infty}(X) = \bigcap_{k,m} x^k \Psi_c^m(X).$$

Its elements are the operators with kernels in $\dot{C}^\infty(X^2)$, the smooth functions on X^2 vanishing to infinite order at $\partial(X^2)$. Following [14, (8)–(10)] we set

$$\begin{aligned} \mathcal{A} &= x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X)/\mathcal{I}, & \mathcal{I}_\partial &= x^{-\mathbb{Z}}\Psi_c^{-\infty}(X)/\mathcal{I}, & \mathcal{I}_\sigma &= x^\infty \Psi_c^{\mathbb{Z}}(X)/\mathcal{I}, \\ \mathcal{A}_\sigma &= \mathcal{A}/\mathcal{I}_\partial, & \mathcal{A}_\partial &= \mathcal{A}/\mathcal{I}_\sigma, & \mathcal{A}_{\partial,\sigma} &= \mathcal{A}/(\mathcal{I}_\sigma + \mathcal{I}_\partial). \end{aligned}$$

In words:

- $\mathcal{I}_\sigma$ consists of *complete symbols vanishing to infinite order at the boundary*;
- $\mathcal{A}_\sigma$ consists of *complete symbols with Laurent singularities at the boundary*;
- $\mathcal{I}_\partial$ consists of *Laurent series of smoothing indicial operators*;
- $\mathcal{A}_\partial$ consists of *Laurent series of indicial operators of finite order*;
- $\mathcal{A}_{\partial,\sigma}$ consists of *Laurent series in x of complete symbols at the boundary*.

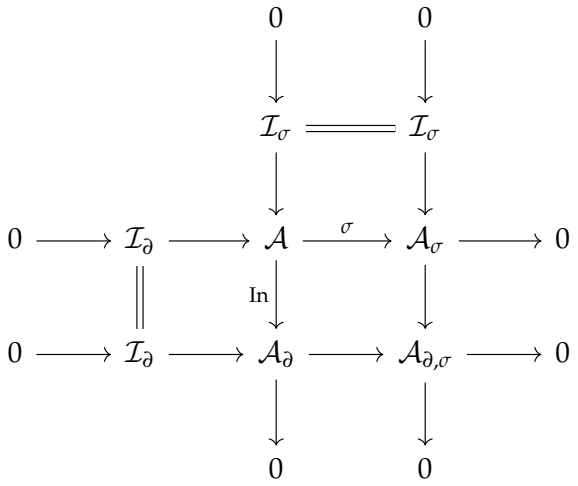


Figure 1: The extensions relating the algebras of [14]. Rows are the indicial (boundary) extensions and columns the symbol (interior) extensions. In addition there are the extensions $0 \to \mathcal{I} \to x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X) \to \mathcal{A} \to 0$, $0 \to \mathcal{I} \to x^{-\mathbb{Z}}\Psi_c^{-\infty}(X) \to \mathcal{I}_\partial \to 0$ and $0 \to \mathcal{I}_\sigma + \mathcal{I}_\partial \to \mathcal{A} \to \mathcal{A}_{\partial,\sigma} \to 0$. Since $\mathcal{I}_\sigma \cap \mathcal{I}_\partial = 0$ in $\mathcal{A}$, the central square is a pull-back.

The maps between them form the diagram of Figure 1, which reproduces the diagram on [14, p. 4].

The two quotient maps out of $\mathcal{A}$ are the *full symbol* σ, which records the complete symbol expansion as a Laurent series in x, and the *indicial map* In, which records the Taylor series at ∂X of the kernel at the cusp front face. After choices (a quantization, a boundary defining function and a normal fibration) there are isomorphisms [14, (13)–(15)]

$$\begin{gathered} \mathcal{I}_\sigma \cong \dot{C}^\infty({}^cS^*X)[q]], \qquad \mathcal{A}_\sigma \cong x^{-\mathbb{Z}}C^\infty({}^cS^*X)[q]], \\ \mathcal{I}_\partial \cong \Psi_{\mathrm{sus}}^{-\infty}(\partial X)[[x], \qquad \mathcal{A}_\partial \cong \Psi_{\mathrm{sus}}^{\mathbb{Z}}(\partial X)[[x], \qquad \mathcal{A}_{\partial,\sigma} \cong C^\infty({}^cS^*_{\partial X}X)[[x, q^{-1}]. \end{gathered} \tag{1}$$

Here $q = \sigma_1(Q)$ is the symbol of a positive elliptic $Q \in \Psi_c^1(X)$, and $[q]]$ denotes Laurent series in q^{-1}, with powers of q bounded above. Likewise $[[x]$ denotes Laurent series in x, with powers of x bounded below. The products are star products. In the indicial algebras the Laurent variable does not commute with the suspended operators:

$$[x, \partial_t] = x^2, \qquad t = 1/x, \tag{2}$$

where t is the linear variable of the suspension $\mathbb{R}_t \times \partial X$. The relation (2) is the *cusp twist*. We explain it in Section 4.5; it is responsible for most of the differences from the b-calculus (Section 18).

1.2 The main homological results

Write $Y = {}^cS^*X$ for the cosphere bundle of the cusp cotangent bundle; it is a compact manifold with boundary of dimension $2n-1$, and its boundary is $Z = {}^cS^*_{\partial X}X$, the part over ∂X. Let $\mathbb{S}_\sigma$ and $\mathbb{S}_\partial$ be circles. They record the homogeneity in the fibres and the Laurent variable x, and carry the classes $d\theta_\sigma = [d\log r]$ and $d\theta_\partial = [d\log x]$. All cohomology has complex coefficients, and Hochschild homology is the continuous homology of Section 5.

MN state their results for Hochschild *cohomology* [14, (16)]. In the form proved in these notes they read as follows.

Theorem 1.1 (MN (16), with the number of boundary components restored)**.** *Let X be connected, $n = \dim X \geq 2$, and let ∂X have $N \geq 1$ components $H_1, \dots, H_N$. Then Y is connected, and $Z = \bigsqcup_j Z_j$ with each $Z_j = {}^c S^*_{H_j} X$ connected. The continuous Hochschild cohomology groups are:*

$$
\begin{aligned}
\mathrm{HH}^k(\mathcal{I}) &\cong \begin{cases} \mathbb{C}, & k = 0, \\ 0, & k > 0, \end{cases} \\
\mathrm{HH}^k(\mathcal{I}_\sigma) &\cong H^k(Y \times \mathbb{S}_\sigma), \\
\mathrm{HH}^k(\mathcal{A}_\sigma) &\cong H^k_{\mathrm{rel}}(Y \times \mathbb{S}_\sigma) \oplus H^k(Z \times \mathbb{S}_\sigma), \\
\mathrm{HH}^k(\mathcal{A}_{\partial,\sigma}) &\cong H^k(Z \times \mathbb{S}_\sigma \times \mathbb{S}_\partial), \\
\mathrm{HH}^k(\mathcal{I}_\partial) &\cong \begin{cases} \mathbb{C}^N, & k = 0, 1, \\ 0, & k \geq 2, \end{cases} \\
\mathrm{HH}^k(\mathcal{A}_\partial) &\cong \begin{cases} H^k(Z \times \mathbb{S}_\sigma \times \mathbb{S}_\partial)/\mathbb{C}^N, & k = 1, 2, \\ H^k(Z \times \mathbb{S}_\sigma \times \mathbb{S}_\partial), & \textit{otherwise,} \end{cases} \\
\mathrm{HH}^k(\mathcal{A}) &\cong \begin{cases} \mathbb{C} \oplus H^1_{\mathrm{rel}}(Y \times \mathbb{S}_\sigma) \oplus H^1(Z \times \mathbb{S}_\sigma)/\mathbb{C}^N, & k = 1, \\ \big(H^2_{\mathrm{rel}}(Y \times \mathbb{S}_\sigma) \oplus H^2(Z \times \mathbb{S}_\sigma)\big)/\mathbb{C}^{N-1}, & k = 2, \\ H^k_{\mathrm{rel}}(Y \times \mathbb{S}_\sigma) \oplus H^k(Z \times \mathbb{S}_\sigma), & \textit{otherwise.} \end{cases}
\end{aligned}
$$

The subspaces $\mathbb{C}^N$ are spanned by the classes $d\theta_\sigma \cdot 1_{Z_j}$ (for $k = 1$) and $d\theta_\partial \wedge d\theta_\sigma \cdot 1_{Z_j}$ (for $k = 2$), and $\mathbb{C}^{N-1}$ is the image of $\{\sum_j c_j\, d\theta_\sigma 1_{Z_j}\}$ under the de Rham connecting map $H^1(Z \times \mathbb{S}_\sigma) \to H^2_{\mathrm{rel}}(Y \times \mathbb{S}_\sigma)$; it is $(N-1)$-dimensional. Finally $\mathrm{HH}^k(x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X)) \cong \mathrm{HH}^k(\mathcal{A})$ for $k \neq 1$, while $\mathrm{HH}^1(x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X))$ is the quotient of $\mathrm{HH}^1(\mathcal{A})$ by the line spanned by the index cocycle.

For $N = 1$ these are exactly the formulas of [14, (16)]. The corresponding homology groups are obtained by Lefschetz duality on the $2n$-manifold $Y \times \mathbb{S}_\sigma$ and Poincaré duality on the closed manifolds $Z \times \mathbb{S}_\sigma$ and $Z \times \mathbb{S}_\sigma \times \mathbb{S}_\partial$. For example [14, Props. 2, 3]

$$
\mathrm{HH}_k(\mathcal{I}_\sigma) \cong H^{2n-k}_{\mathrm{rel}}(Y \times \mathbb{S}_\sigma), \qquad \mathrm{HH}_k(\mathcal{A}_\sigma) \cong H^{2n-k}(Y \times \mathbb{S}_\sigma) \oplus H^{2n-k-1}(Z \times \mathbb{S}_\sigma).
$$

The precise sources of the hypotheses are discussed in Corrections 9.4 and 10.9. The proof of Theorem 1.1 occupies Sections 6–12.

1.3 The index formula

A matrix A of elements of $x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X)$ is *fully elliptic* if its image in (a matrix algebra over) $\mathcal{A}$ is invertible. For an operator of order m this means two things: A is elliptic as a cusp operator, and its indicial family $\mathrm{In}(A, \tau)$ is invertible for every real τ. Such an operator is Fredholm on $\dot{C}^\infty(X)$, and Fedosov's formula reads [14, (7)]

$$
\mathrm{Ind}(A) = \dim\ker A - \dim\ker A^* = \mathrm{Tr}([A, B]) = \mathrm{IF}([A], [A]^{-1}), \tag{3}
$$

where B is any inverse of A modulo $\mathcal{I}$. The *index functional* IF is the image of the trace on $\mathcal{I}$ under the connecting map of $0 \to \mathcal{I} \to x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X) \to \mathcal{A} \to 0$. MN express it through the following data:

- the extensions $\widehat{\mathrm{Tr}}_\partial$ and $\widehat{\mathrm{Tr}}_\sigma$ of the traces on the two ideals $\mathcal{I}_\partial$ and $\mathcal{I}_\sigma$;
- the two exterior derivations $D_{\log x} = [\log x, \cdot\,]$ and $D_{\log Q} = [\log Q, \cdot\,]$.

In the form proved here (Proposition 15.3),

$$\mathrm{IF}(A,B) = \tfrac{1}{2}\big(\widehat{\mathrm{Tr}}_\partial + \widehat{\mathrm{Tr}}_\sigma\big)\big(B\widetilde{D}(A) + \widetilde{D}(A)B\big), \qquad \widetilde{D} = [\log x - \log Q, \cdot\,]. \tag{4}$$

This is [14, (87)]. In [14, (21)] the derivation is printed with the opposite sign (Correction 15.4). Inserting $B = A^{-1}$ splits the index into four terms. If A is translation invariant near the boundary with normal indicial family, two of them vanish, and one obtains the pseudodifferential APS formula (Theorem 15.12):

$$\mathrm{Ind}(A) = \mathrm{AS}(A) - \tfrac{1}{2}\overline{\eta}(A), \qquad \mathrm{AS}(A) = \mathrm{Tr}_{\mathrm{R}}\big(A[\log Q, A^{-1}]\big), \qquad \overline{\eta}(A) = 2\,\overline{\mathrm{Tr}}\big(A_0^{-1}[t, A_0]\big). \tag{5}$$

Here $A_0 = \mathrm{In}(A)$, $t = 1/x$ is the suspension variable, $\overline{\mathrm{Tr}}$ is Melrose's regularized trace on the suspended algebra, and in terms of the indicial variable

$$\overline{\eta}(A) = \frac{1}{\pi i}\,\overline{\int_{\mathbb{R}}}\,\mathrm{Tr}\Big(A_0(\tau)^{-1}\frac{dA_0(\tau)}{d\tau}\Big)\,d\tau.$$

For a Dirac operator in APS normal form this is the Atiyah–Patodi–Singer eta invariant of the boundary operator (Corollary 15.9). In MN, [14, (22)–(24)] and [14, (100)], the Atiyah–Singer term is written $\mathrm{Tr}_{\mathrm{R}}(A^{-1}[\log Q, A])$, which is $-\mathrm{AS}(A)$. The sign is wrong; see Correction 15.6 and the checks in Section 16.

1.4 Organization and how to read these notes

Section 2 fixes conventions and gives the dictionary to MN's notation. It also lists the background facts that we quote, and it collects the corrections. Section 3 treats the closed case (Brylinski–Getzler, Wodzicki), which is the template for everything else. Section 4 introduces the cusp calculus, starting from the half-line. Section 5 assembles the homological toolkit. Sections 6–12 follow [14, §§1–8] and prove Theorem 1.1. Sections 13–15 follow [14, §§9–12] and prove the index formula. Section 16 contains worked examples. Section 17 summarizes [14, §§13–15]. Section 18 compares the cusp and b-calculi and cites the companion notes [6].

Figure 2 shows the logical dependencies.

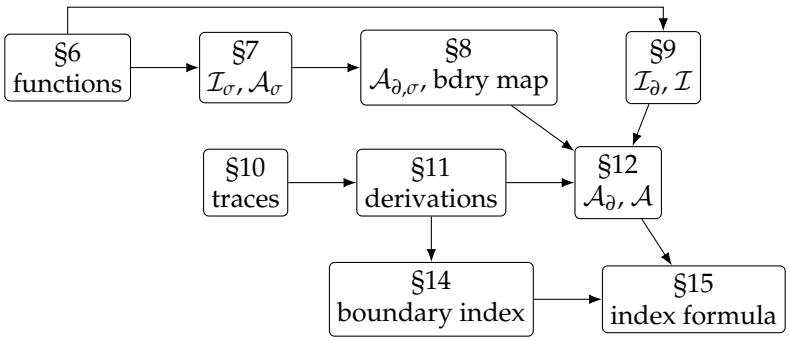


Figure 2: Logical structure of the main part of the notes.

2 Conventions, dictionary, background and corrections

2.1 Coordinates and Fourier conventions

Near a boundary point we use coordinates (x, y), $y \in \mathbb{R}^{n-1}$, where x is the fixed boundary defining function. The *cusp cotangent* coordinates (x, y, τ, η) are defined by writing a cusp covector as

$$\alpha = \tau \, \frac{dx}{x^2} + \eta \cdot dy, \tag{6}$$

as in [14, (139)]. Thus τ is the symbol of $x^2 D_x$, where $D = -i\partial$. The canonical symplectic form is $\omega = d\alpha = d\tau \wedge \frac{dx}{x^2} + \sum_j d\eta_j \wedge dy_j$. It agrees with the standard form $\sum d\xi \wedge dx$ in the interior, where $\tau = x^2\xi$. The Poisson bracket is

$$\{a, b\} = \partial_\tau a \, x^2 \partial_x b - x^2 \partial_x a \, \partial_\tau b + \partial_\eta a \cdot \partial_y b - \partial_y a \cdot \partial_\eta b, \tag{7}$$

and the principal symbol of a commutator is $\sigma([A, B]) = -i\{\sigma(A), \sigma(B)\}$. (In [14, (139)] the last term is misprinted as $\partial_y a \cdot \partial_\tau b$.) Formula (7) follows from the usual bracket $\partial_\xi a \, \partial_x b - \partial_x a \, \partial_\xi b$ by the change of variables $\tau = x^2\xi$. Indeed $\partial_\xi|_x = x^2 \partial_\tau$ and $\partial_x|_\xi = \partial_x|_\tau + 2x^{-1}\tau \, \partial_\tau$, and the two terms involving $2x^{-1}\tau$ cancel.

Near the boundary we use the *suspension variable*

$$t = \frac{1}{x} \in (1, \infty), \qquad x^2 \partial_x = -\partial_t, \qquad \frac{dx}{x^2} = -dt. \tag{8}$$

Fourier transformation is $\hat{u}(\xi) = \int e^{-it\xi} u(t)\, dt$, with inverse $u(t) = (2\pi)^{-1} \int e^{it\xi} \hat{u}(\xi)\, d\xi$, as in the companion notes [17, 6].

2.2 The indicial variable

MN define the indicial family by conjugation [14, Def. 4]:

$$\mathrm{In}(A, \tau) = \left(e^{i\tau/x} A e^{-i\tau/x} \right) \Big|_{\partial X}, \qquad \tau \in \mathbb{R}. \tag{9}$$

Then $\mathrm{In}(x^2 D_x, \tau) = \tau$, so the indicial variable is the boundary value of the cusp covariable of (6). For a translation invariant operator on $\mathbb{R}_t \times \partial X$ with convolution kernel $\kappa(t - t', y, y')$, one finds

$$\mathrm{In}(A, \tau) = \int_{\mathbb{R}} e^{i\tau s} \kappa(s, \cdot, \cdot)\, ds = \hat{\kappa}(-\tau), \qquad \mathrm{In}(D_t, \tau) = -\tau, \qquad \mathrm{In}([t, A], \tau) = -i\, \partial_\tau \, \mathrm{In}(A, \tau). \tag{10}$$

The last identity holds because $[t, A]$ has kernel $(t - t')\kappa(t - t')$. Under the transcendental blow-up $x_b = e^{-1/x}$ of Section 4.6, the b-indicial family of the companion notes, $I(A, \lambda) = \int e^{-iz\lambda} \kappa_A(e^z)\, dz$ with $z = \log(x_b / x_b') = -(t - t')$, becomes (10) with $\tau = \lambda$. So *the cusp indicial variable τ of MN is the b-indicial variable λ of [17, 6].* In the b-calculus λ may be complex. In the cusp calculus only real τ occurs.

2.3 Densities and traces

Kernels are taken with respect to a cusp density, equal to $x^{-2} dx \, \nu_\partial = dt \, \nu_\partial$ near the boundary, where ν_∂ is a smooth positive density on ∂X. We use the following functionals; complex parameters are written s (for powers of Q) and z (for powers of x).

functional	definition	MN
Tr on $\mathcal{I}$	integral of the kernel on the diagonal	Tr
Tr_σ on $\mathcal{I}_\sigma$	$\operatorname{Res}_{s=0}\operatorname{Tr}(AQ^{-s})$ (Wodzicki residue)	(49)
$\operatorname{Tr}_\partial$ on $\mathcal{I}_\partial$	$\operatorname{Res}_{z=0}\operatorname{Tr}(Ax^z)$	Lemma 10
$\operatorname{Tr}_{\partial,\sigma}$ on $\mathcal{A}$	coefficient of $(sz)^{-1}$ in $Z(A;s,z)$	(44)
$\widehat{\operatorname{Tr}}_\sigma$, $\widehat{\operatorname{Tr}}_\partial$	the mixed coefficients of $Z(A;s,z)$	(44)
$\overline{\operatorname{Tr}}$ on $\Psi^{\mathbb{Z}}_{\mathrm{sus}}(\partial X)$	$(2\pi)^{-1}\overline{\int}\operatorname{Tr}\operatorname{In}(B,\tau)\,d\tau$	(57)–(59)

Here $Z(A;s,z)=\frac{1}{2}\big(\operatorname{Tr}(Ax^zQ^{-s})+\operatorname{Tr}(AQ^{-s}x^z)\big)$. **MN write τ for our s** in [14, (43)–(62)], and also use τ for the cusp covariable in [14, (139)]. We keep τ for the covariable throughout.

2.4 Hochschild conventions

The Hochschild boundary is $b=\sum_{i=0}^{k}(-1)^i d_i$, where d_i multiplies a_i and a_{i+1} for $i<k$ and $d_k(a_0\otimes\cdots\otimes a_k)=a_ka_0\otimes a_1\otimes\cdots\otimes a_{k-1}$. The bar differential is $b'=\sum_{i<k}(-1)^i d_i$. For a derivation D,

$$i_D(a_0\otimes a_1\otimes\cdots\otimes a_k)=a_0D(a_1)\otimes a_2\otimes\cdots\otimes a_k \qquad [14, §6],$$

and on cochains $i_D\varphi=\varphi\circ i_D$. For a 0-cochain φ this reads

$$(i_D\varphi)(A,B)=\varphi\big(AD(B)\big).$$

2.5 Background facts quoted

We do not reprove the analytic foundations of the cusp calculus or Wodzicki's excision theorem. Every step that uses one of the following facts says so.

(B1) *Cusp calculus.* The cusp operators form a filtered algebra. It has the symbol sequence [14, (11)], the indicial sequence [14, (12)], a global quantization map [14, Prop. 24], and the Taylor series isomorphism at the front face [14, Prop. 25]. Conjugation by x^z preserves orders. References: [14, App. A], [9, 12].

(B2) *Fredholm theory.* A fully elliptic element of $x^{-\mathbb{Z}}\Psi^{\mathbb{Z}}_c(X;E)$ is Fredholm on $\dot{C}^\infty(X;E)$ and on the cusp Sobolev spaces. It has a generalized inverse in the calculus, and its kernel and cokernel lie in $\dot{C}^\infty$ [14, p. 4, §12], [9].

(B3) *Complex powers and meromorphy.* If $Q\in\Psi^1_c(X)$ is positive, elliptic and invertible, then Q^{-s} is an entire family of order $-s$. For holomorphic families $A(s,z)$ of fixed order, the function $\operatorname{Tr}(A(s,z)x^zQ^{-s})$ is meromorphic, with at most simple poles on the lines $s=m+n-k$ and $z=-p+1-j$ [14, Lemma 4, App. B].

(B4) *Wodzicki excision.* A short exact sequence of topological algebras whose kernel is topologically H-unital gives a long exact sequence in continuous Hochschild homology [21].

(B5) *Connes' theorem.* The Hochschild–Kostant–Rosenberg map gives $\mathrm{HH}_k(C^\infty(M))\cong C^\infty(M;\Lambda^k)$ for compact manifolds, possibly with boundary or corners [4]. A self-contained proof for closed M is in [13, Ch. 10]; for manifolds with boundary the proof is the same.

(B6) *Künneth and Morita.* The topological Künneth formula holds for nuclear algebras [7]. Hochschild homology is Morita invariant for full corners $eM_N(A)e$ [4, 8].

(B7) *The suspended algebra* [11]. The regularized trace $\overline{\mathrm{Tr}}$ is a trace on $\Psi^{\mathbb{Z}}_{\mathrm{sus}}$. The eta invariant is $\eta(A) = (\pi i)^{-1}\overline{\int}\,\mathrm{Tr}(A^{-1}\partial_\tau A)\,d\tau$, and for Dirac operators it is the APS eta invariant.

(B8) *Wodzicki residue* [19, 5]: it is a trace, given by the local formula $(2\pi)^{-n}\int_{S^*M} a_{-n}\,\nu$, and independent of Q.

2.6 Corrections and unstated hypotheses

For convenience we list here the points where we disagree with the text of [14]. Each is proved in the place indicated.

1. *Connected boundary.* [14, (16)], Proposition 6, Lemma 12 and Proposition 12 treat $\mathcal{I}_\partial$ as if ∂X were connected. For N components, $\mathrm{HH}^0(\mathcal{I}_\partial) \cong \mathrm{HH}^1(\mathcal{I}_\partial) \cong \mathbb{C}^N$, the double residue splits into N traces, and the boundary map $\mathrm{HH}^1(\mathcal{I}_\partial) \to \mathrm{HH}^2(\mathcal{A}_\sigma)$ has rank $N-1$ (Correction 9.4).

2. *Dimension one.* For $n = 1$ the boundary cosphere bundle Z_j consists of two points. Then the residue traces Tr_σ and $\mathrm{Tr}_{\partial,\sigma}$ split into two (Correction 10.9); the boundary trace Tr_∂ on $\mathcal{I}_\partial$ does not, and the claims "HH^0 is one-dimensional" fail.

3. *The cusp de Rham complex.* The remark after [14, (33)] says that the cusp de Rham complex has b-cohomology. In fact it has an extra summand $H^{k-1}(\partial M)$ (Correction 6.8). This is not used elsewhere.

4. *The cut-off in Lemma 1.* The function ϕ in [14, (26)] must be supported in $(-1, 1)$ (Remark 6.3).

5. *The derivation $\widetilde{D}$.* [14, (21)] and [14, (83)] have opposite signs; (4) is correct (Correction 15.4).

6. *Signs of $\overline{\mathrm{AS}}$ and β in [14, (93)].* With the definitions [14, (90), (91)] the index theorem reads $\mathrm{Ind} = -\overline{\mathrm{AS}} - \frac{1}{2}\overline{\eta} - \beta_Q + \gamma$. The symbolic term in [14, (23), (100)] has the wrong sign (Correction 15.6).

7. *The eta term.* The last line of the proof of [14, Prop. 17] should read $\overline{\eta}(A) = 2\overline{\mathrm{Tr}}(A_0^{-1}[t, A_0])$. The proposition needs Q to be asymptotically translation invariant (Correction 15.8).

8. *The residue on the suspended algebra.* The identification of the residue with $\overline{\mathrm{Tr}}([t, B])$ in [14, Prop. 8] fails, for instance when ∂X is a point; the same parity argument applies to Q_0^{-n} on any boundary (Remark 10.13). We use the residue itself.

9. *The compactification in Appendix A.* The map [14, (130)] gives $[-1, 1]$ the wrong smooth structure at the endpoints for the cusp calculus. One must compactify by $1/z$ (Remark 4.2).

10. *The cusp metric in Appendix B.* The metric $dx^2/x^4 + h/x^2$ of [14, App. B] is a scattering metric, and its Laplacian is not cusp-elliptic. An exact cusp metric is $dx^2/x^4 + h$ (Remark 4.6).

11. *Minor misprints.* We note these as they occur. Examples: $i_{\log r}$ for $i_{\log x}$ in Lemma 13, and A^z for a^z in the proof of Lemma 9. Also the factor 2 in [14, (80)]: in fact $C(0) = [\log x, \log Q]$.

To avoid a clash with the eta invariant, we write β_Q for the functional MN call β in [14, (91)].

3 The closed case: Brylinski–Getzler and Wodzicki

Everything in [14] is modelled on the case $\partial X = \varnothing$, which MN recall in their introduction [14, (1)–(5)]. We treat it first, with proofs of the formulas that will be generalized later.

3.1 The symbol algebra and its homology

Let M be a closed manifold of dimension n, let $\Psi^{\mathbb{Z}}(M)$ be the algebra of classical pseudodifferential operators of integral order, and let $\mathcal{A}_M = \Psi^{\mathbb{Z}}(M)/\Psi^{-\infty}(M)$. A quantization identifies $\mathcal{A}_M$ with formal sums $\sum_{j\le m} a_j$ of smooth functions a_j on $T^*M \setminus 0$, homogeneous of degree j, with a star product

$$a \star b = \sum_{l\ge 0} P_l(a,b), \qquad P_0(a,b) = ab, \qquad P_1(a,b) - P_1(b,a) = -i\{a,b\},$$

where P_l is bidifferential and lowers the total degree by l.

Theorem 3.1 (Brylinski–Getzler, Wodzicki [3, 20])**.** $\mathrm{HH}_k(\mathcal{A}_M) \cong H^{2n-k}(S^*M \times \mathbb{S}_\sigma)$, *and dually* $\mathrm{HH}^k(\mathcal{A}_M) \cong H^k(S^*M \times \mathbb{S}_\sigma)$.

The proof is the model for Sections 7–8. We give it in general form in Section 5; see also [13, Ch. 10], whose argument we follow. In outline:

1. filter Hochschild chains by total order;
2. the associated graded algebra is commutative, so E^1 is given by homogeneous differential forms (HKR and Connes, (B5));
3. the first differential is the Poisson differential $-i\delta$, and the symplectic star operator turns it into the de Rham differential;
4. homogeneous forms of non-zero degree are exact, and forms of degree 0 on $T^*M \setminus 0 \cong S^*M \times (0,\infty)$ give $H^*(S^*M) \oplus H^{*-1}(S^*M)\, dr/r$, which is $H^*(S^*M \times \mathbb{S}_\sigma)$;
5. the E^2 term lies on one line, so the spectral sequence degenerates and converges.

Question 3.2. *Where does the circle* $\mathbb{S}_\sigma$ *come from, and why must one take operators of* all *integral orders?*

Answer. The fibres of $T^*M \setminus 0$ are cones, $S^{n-1} \times (0,\infty)_r$. Homogeneous forms see the radial factor only through dr/r, which is invariant under dilation. So the homogeneous de Rham complex is that of $S^*M \times \mathbb{S}_\sigma$, where $\mathbb{S}_\sigma = (0,\infty)/e^{\mathbb{Z}}$ is the quotient of the radial half-line by integral dilations [14, §1]. The class $d\theta_\sigma = [dr/r]$ corresponds to $i_{\log Q}$ (Proposition 11.3). If only operators of order ≤ 0 were allowed, the chains would not be closed under the shifts of order that produce dr/r. The homology is then infinite dimensional; see [14, §15] and Section 17.3. MN make the same point on [14, p. 3]. □

3.2 The residue trace, the exterior derivation and the index

Let $Q \in \Psi^1(M)$ be positive, elliptic and invertible. For $A \in \Psi^m(M)$ the function $\mathrm{Tr}(AQ^{-s})$ is holomorphic for $\mathrm{Re}\, s > m + n$ and extends meromorphically with simple poles. Wodzicki's residue is $\mathrm{Tr}_{\mathrm{R}}(A) = \mathrm{Res}_{s=0} \mathrm{Tr}(AQ^{-s})$ [14, (2)]. It is a trace, independent of Q, vanishes on $\Psi^{-\infty}$, and is given locally by (B8):

$$\mathrm{Tr}_{\mathrm{R}}(A) = (2\pi)^{-n} \int_{S^*M} a_{-n}\, \nu, \qquad \nu = \iota_R \frac{\omega^n}{n!} \text{ up to the orientation convention,} \tag{11}$$

where R is the radial vector field. The *regularized trace* $\widehat{\mathrm{Tr}}(A)$ is the constant term of $\mathrm{Tr}(AQ^{-s})$ at $s = 0$. The formula

$$D_{\log Q}(A) = [\log Q, A] = \frac{d}{ds}\Big(Q^s A Q^{-s}\Big)\Big|_{s=0}$$

defines a derivation of $\Psi^{\mathbb{Z}}(M)$, lowering order by one, which is not inner [14, §6].

Lemma 3.3 (Trace defect). *For $A, B \in \Psi^{\mathbb{Z}}(M)$,*

$$\widehat{\mathrm{Tr}}([A,B]) = \mathrm{Tr_R}(A[\log Q, B]) = -\,\mathrm{Tr_R}([\log Q, A]B), \qquad \mathrm{Tr_R}([\log Q, C]) = 0.$$

Proof. For $\operatorname{Re} s$ large, the trace property gives $\mathrm{Tr}([A,B]Q^{-s}) = \mathrm{Tr}(A[B,Q^{-s}])$. Write

$$[B,Q^{-s}] = BQ^{-s} - Q^{-s}B = s\,M(s)Q^{-s}, \qquad M(s) = \frac{B - Q^{-s}BQ^{s}}{s}.$$

Then M is an entire family of fixed order with $M(0) = \frac{d}{ds}(-Q^{-s}BQ^{s})|_{s=0} = [\log Q, B]$. Hence $\mathrm{Tr}([A,B]Q^{-s}) = s\,\mathrm{Tr}(AM(s)Q^{-s})$. The function $\mathrm{Tr}(AM(s)Q^{-s})$ has at most a simple pole at $s=0$, with residue $\mathrm{Tr_R}(AM(0))$. So the left side is regular at $s=0$, with value $\mathrm{Tr_R}(A[\log Q, B])$. For the last identity take $A = \mathrm{Id}$: the function $\mathrm{Tr}([\mathrm{Id}, B]Q^{-s})$ vanishes identically, so $\mathrm{Tr_R}([\log Q, B]) = 0$. Applying this to $C = AB$ gives $\mathrm{Tr_R}([\log Q, A]B) + \mathrm{Tr_R}(A[\log Q, B]) = 0$. □

Proposition 3.4 (The index functional, MN (3)–(5)). *The connecting map of $0 \to \Psi^{-\infty} \to \Psi^{\mathbb{Z}} \to \mathcal{A}_M \to 0$ sends the trace to the Hochschild 1-cocycle $\mathrm{IF}(A,B) = \mathrm{Tr_R}(A[\log Q, B])$ on $\mathcal{A}_M$. If $A \in \Psi^m(M)$ is elliptic with parametrix B, then*

$$\mathrm{Ind}(A) = \mathrm{Tr}([A,B]) = \mathrm{Tr_R}\big(A[\log Q, A^{-1}]\big).$$

Proof. The trace on $\Psi^{-\infty}$ extends to the functional $\widehat{\mathrm{Tr}}$ on $\Psi^{\mathbb{Z}}$. Its Hochschild coboundary $(b\widehat{\mathrm{Tr}})(A,B) = \widehat{\mathrm{Tr}}([A,B])$ vanishes if A or B is smoothing, because $\mathrm{Tr_R}$ vanishes on $\Psi^{-\infty}$. By Lemma 3.3 it equals $\mathrm{Tr_R}(A[\log Q, B])$. Lemma 5.2 below then identifies it with the image of $[\mathrm{Tr}]$ under the connecting map. For the index: $1 - BA$ and $1 - AB$ are smoothing, $\mathrm{Ind}\,A = \mathrm{Tr}(1-BA) - \mathrm{Tr}(1-AB) = \mathrm{Tr}([A,B])$ (Fedosov), and $\mathrm{Tr}([A,B]) = \widehat{\mathrm{Tr}}([A,B])$ because $[A,B]$ is smoothing. □

Since $A[\log Q, A^{-1}] = -[\log Q, A]A^{-1}$, we also have

$$\mathrm{Ind}(A) = -\,\mathrm{Tr_R}\big([\log Q, A]A^{-1}\big) = -\,\mathrm{Tr_R}\big(A^{-1}[\log Q, A]\big).$$

Keep this sign in mind; it is the source of Correction 15.6.

3.3 Example: the circle

Let $M = S^1$, with modes $e_k = e^{ik\theta}$, and let Q act by $Qe_k = \langle k\rangle e_k$, where $\langle k\rangle = (1+k^2)^{1/2}$.

Homology. $S^*S^1 = S^1 \times \{\pm 1\}$, so $S^*S^1 \times \mathbb{S}_\sigma$ is two tori, and Theorem 3.1 gives

$$\dim \mathrm{HH}_k(\mathcal{A}_{S^1}) = 2,\ 4,\ 2 \qquad (k = 0,1,2).$$

In degree 0 there are *two* independent residue traces,

$$\mathrm{Tr}^{\pm}_{\mathrm{R}}(A) = \frac{1}{2\pi}\int_0^{2\pi} a_{-1}(\theta, \pm 1)\, d\theta,$$

one for each direction of ξ. Their sum is $\mathrm{Tr_R}$. This is the $n = 1$ phenomenon that will reappear at the boundary (Correction 10.9).

Index. Let Π be the orthogonal projection onto $\operatorname{span}\{e_k : k \geq 0\}$; it is a classical operator of order 0. Put

$$A = e^{i\theta}\Pi + (1-\Pi), \qquad B = e^{-i\theta}\Pi + (1-\Pi).$$

Then $Ae_k = e_{k+1}$ for $k \geq 0$ and $Ae_k = e_k$ for $k < 0$. So A is injective and its range misses exactly e_0, whence $\operatorname{Ind} A = -1$. A direct check gives $BA = 1$ and $1 - AB = R$, where R is the rank one operator $e_0 \mapsto e_0 - e_{-1}$. Therefore $\operatorname{Tr}([A, B]) = -\operatorname{Tr} R = -1$, as Fedosov's formula requires. Now compute the residue formula. For $k \geq 1$,

$$A[\log Q, B]e_k = A\big(\log\langle k-1\rangle - \log\langle k\rangle\big)e_{k-1} = \big(\log\langle k-1\rangle - \log\langle k\rangle\big)e_k = \Big(-\frac{1}{k} + O(k^{-2})\Big)e_k.$$

For $k \leq -1$ the operator $A[\log Q, B]$ vanishes on e_k, and e_0 contributes a finite rank term. Hence

$$\operatorname{Tr}_{\mathrm{R}}\big(A[\log Q, B]\big) = \operatorname{Res}_{s=0} \sum_{k\geq 1} \Big(-\frac{1}{k} + O(k^{-2})\Big)\langle k\rangle^{-s} = -1 = \operatorname{Ind} A.$$

In symbol terms, $A[\log Q, B]$ has $a_{-1}(\theta, \xi) = -\xi^{-1}$ for $\xi > 0$ and 0 for $\xi < 0$. Only $\operatorname{Tr}_{\mathrm{R}}^{+}$ contributes.

Discussion. The example fixes the sign convention for everything that follows:

$$\operatorname{Ind} A = \operatorname{Tr}_{\mathrm{R}}\big(A[\log Q, A^{-1}]\big), \qquad \textit{not} \qquad \operatorname{Tr}_{\mathrm{R}}\big(A^{-1}[\log Q, A]\big).$$

The second expression equals $+1$ here.

4 The cusp calculus

This section collects the facts about cusp operators that [14] uses. It follows [14, App. A–C], [9] and [12]. We begin with the half-line, where everything is explicit.

4.1 Cusp structures and cusp vector fields

A *cusp structure* on X is a choice of boundary defining function x, modulo two kinds of change: multiplication by a positive constant on each boundary component, and addition of a term in $x^2C^\infty(X)$ [14, App. A]. Equivalently, it is a trivialization of the normal bundle of ∂X, up to constant multiples on each component. The *cusp vector fields* are [14, (6), (114)]

$$\mathcal{V}_c(X) = \{V \in C^\infty(X; TX) : \ Vx \in x^2C^\infty(X)\}.$$

Lemma 4.1 (MN Lemma 20 and Corollary 4)**.** *In coordinates (x, y) near a boundary point, $V \in \mathcal{V}_c(X)$ if and only if $V = a\,x^2\partial_x + \sum_j b_j\partial_{y_j}$ with a, b_j smooth. If x' is another boundary defining function, then $\mathcal{V}_c$ is the same for x and x' exactly when x' defines the same cusp structure.*

Proof. Write $V = \alpha\partial_x + \sum \beta_j\partial_{y_j}$. Then $Vx = \alpha$, so $V \in \mathcal{V}_c$ if and only if $\alpha = ax^2$. For the second statement write $x' = e^f x$, with f smooth. Then

$$(x^2\partial_x)x' = x^2e^f(1 + x\partial_x f), \qquad \partial_{y_j}x' = x\,e^f\partial_{y_j}f.$$

The first expression lies in $x'^2C^\infty = x^2C^\infty$. The second lies in x^2C^∞ if and only if $\partial_y f = O(x)$ at ∂X, that is, if and only if $f|_{\partial X}$ is locally constant. In that case $x' = cx + O(x^2)$ on each component, which is the stated equivalence. □

The cusp vector fields form a Lie algebra and a $C^\infty(X)$-module. Near the boundary they are locally free on $x^2\partial_x, \partial_{y_j}$. Hence there is a vector bundle ${}^cTX$ with $\mathcal{V}_c = C^\infty(X; {}^cTX)$. Its dual is ${}^cT^*X$, with local frame $dx/x^2, dy_j$. Over the interior ${}^cTX = TX$, and globally ${}^cTX \cong TX$ as

abstract bundles. So the cosphere bundle $Y = {}^cS^*X = ({}^cT^*X \setminus 0)/\mathbb{R}^+$ is diffeomorphic to S^*X. Its boundary is $Z = {}^cS^*_{\partial X}X$, the sphere bundle of ${}^cT^*_{\partial X}X = T^*\partial X \times \mathbb{R}_\tau$. If $n \geq 2$, each component Z_j is an S^{n-1}-bundle over H_j, hence connected. If $n = 1$, each Z_j is the pair of points $\tau = \pm 1$.

MN also show that any two cusp structures are related by a diffeomorphism isotopic to the identity [14, Lemma 19]. So the algebras for different structures are isomorphic. The proof integrates a vector field transverse to the boundary that interpolates between the two defining functions.

4.2 The model: the half-line

Let $X = [0,1)_x$. In the variable $t = 1/x \in (1,\infty)$, the cusp vector field is $x^2\partial_x = -\partial_t$ by (8). So the cusp structure makes the end $x \to 0$ into a *cylindrical end* $t \to \infty$ of a complete manifold. The dictionary:

on $[0,1)_x$	on $(1,\infty)_t$
$C^\infty([0,1))$	smooth functions with expansions in powers of $1/t$
$\dot{C}^\infty([0,1))$	functions rapidly decreasing as $t \to \infty$
$x^{-\mathbb{Z}}C^\infty([0,1))$	functions with expansions $\sum_{j \geq -p} c_j t^{-j}$
cusp density dx/x^2	dt
exact cusp metric dx^2/x^4	dt^2
$\sum_j a_j(x)(x^2D_x)^j$	$\sum_j a_j(1/t)(-D_t)^j$
$\mathrm{In}(P,\tau)$	$\sum_j a_j(0)\tau^j$

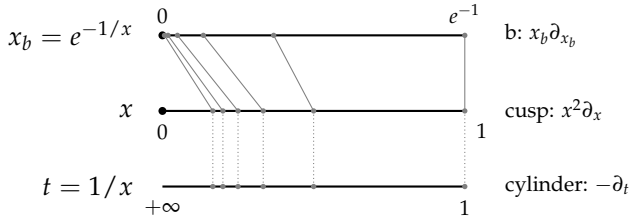


Figure 3: The half-line in three coordinates. The cusp vector field $x^2\partial_x$ is $-\partial_t$ in $t = 1/x$, the generator of translations of a cylindrical end, and it is the b-vector field $x_b\partial_{x_b}$ in the transcendental coordinate $x_b = e^{-1/x}$ (Section 4.6). The gray marks are the points $t = 1, 2, \ldots, 6$: equally spaced in t, at $x = 1/t$, and at $x_b = e^{-t}$, where they crowd exponentially towards 0. All powers of x_b vanish to infinite order in x.

A cusp differential operator is thus a differential operator on the half-line $(1,\infty)_t$ whose coefficients have expansions in $1/t$. Its indicial family is its "value at $t = \infty$", Fourier transformed in t. The pseudodifferential operators near the boundary are the operators

$$Au(t) = \frac{1}{2\pi}\iint e^{i(t-t')\xi}\, a\big(1/t,\xi\big)u(t')\,dt'\,d\xi,$$

where $a(x,\xi)$ is a classical symbol in ξ, smooth in x up to $x = 0$. The kernel is smooth off the diagonal and rapidly decreasing in $t - t'$ [14, Props. 22, 23]. The translation invariant operators (with $a = a(\xi)$) form the suspended algebra $\Psi^{\mathbb{Z}}_{\mathrm{sus}}(\mathrm{pt})$, the convolution operators by kernels that are classical conormal at 0 and Schwartz at infinity. It is *commutative*.

Remark 4.2 (The compactification in [14], (130)). MN identify $\mathbb{R}$ with $(-1,1)$ via $z \mapsto z/(1+z^2)^{1/2}$. Near $z = +\infty$ this gives $1 - z/\langle z\rangle = \frac{1}{2}z^{-2} + O(z^{-4})$, so the smooth structure it induces on $[-1,1]$ is that of ρ^2 with $\rho = 1/z$. In that structure ∂_z is not a cusp vector field, and translations do not act smoothly. The identification [14, Prop. 22] and the description of Ψ_{sus} on [14, p. 44] need the compactification by $\rho = 1/|z|$. For example one can use $z \mapsto \frac{2}{\pi}\arctan z$, for which $1 - \frac{2}{\pi}\arctan z = \frac{2}{\pi}\arctan(1/z)$ is a smooth defining function in ρ. Then $\partial_z = -\rho^2\partial_\rho$ is a cusp vector field, and the translation $z \mapsto z + c$ acts by $\rho \mapsto \rho/(1 + c\rho)$, which is smooth. With this change the statements of [14, App. A] hold as written.

Example 4.3. The operator $P = x^2 D_x + c$ on $[0,1)$ is $-D_t + c$, and $\mathrm{In}(P,\tau) = \tau + c$. Its inverse on $\mathbb{R}_t$, when $c \notin \mathbb{R}$, is convolution with an exponentially decreasing kernel. The indicial family is invertible for all real τ precisely when $c \notin \mathbb{R}$. This is the cusp version of "no indicial roots on the real line". There is no condition off the real axis; compare Section 18.

4.3 Kernels and the cusp double space

Globally, kernels are described on a blown-up double space [14, App. A]. First blow up the corners $H \times H$ of X^2 to obtain the b-double space X_b^2, whose front face has the coordinate $s = x/x'$. The cusp structure picks out the point $s = 1$ on each fibre of the b-front face, more precisely the fibre diagonal $\Phi_2(H) = \{s = 1\} \times H^2$. Blowing these up gives

$$X_c^2 = [X_b^2;\, \{\Phi_2(H)\}_H], \qquad \mu = \frac{1}{x} - \frac{1}{x'} = t - t' \quad \text{on the new front face } \mathrm{ff}_c.$$

The interior of ff_c is $\mathbb{R}_\mu \times H^2$; see Figure 4. The lifted diagonal meets the boundary of X_c^2 only in ff_c, at $\mu = 0$ [14, Lemma 22]. Then $\Psi_c^m(X)$ consists of the kernels on X_c^2 that are classical conormal of order m at the lifted diagonal, smooth up to ff_c, and rapidly vanishing at all other boundary faces [14, Def. 2]. In particular their restriction to ff_c is rapidly decreasing as $|\mu| \to \infty$. The kernels in $\Psi_c^{-\infty}$ that vanish to infinite order also at ff_c are those of $\mathcal{I}$ [14, Def. 1].

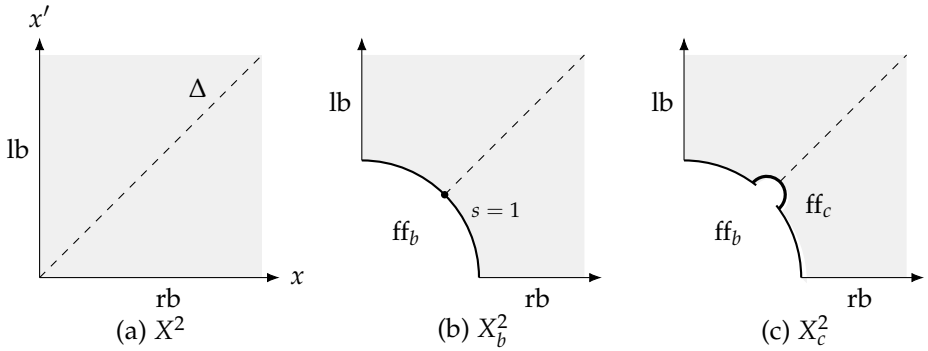


Figure 4: Near a corner $H \times H$ of X^2 (suppressing H^2). (b) The b-blow-up replaces the corner by the front face ff_b with coordinate $s = x/x'$; the diagonal meets it at $s = 1$. (c) The cusp double space blows up the fibre diagonal $\{s = 1\}$ once more. The new face ff_c carries the coordinate $\mu = 1/x - 1/x' = t - t' \in [-\infty, \infty]$, the difference of the suspension variables, and cusp kernels restrict to it as convolution kernels in μ, rapidly decreasing at $\mu = \pm\infty$. b-kernels are instead smooth on ff_b and convolution kernels in $\log s$.

4.4 Symbols, quantization and the star product

The principal symbol gives exact sequences [14, (11)]

$$0 \to \Psi_c^{m-1}(X) \to \Psi_c^m(X) \xrightarrow{\sigma_m} \mathcal{G}^m({}^cT^*X) \to 0,$$

where $\mathcal{G}^m({}^cT^*X)$ is the space of smooth functions on ${}^cT^*X \setminus 0$ that are homogeneous of degree m. The symbol of a product is the product of the symbols, and $\sigma_{m+m'-1}([A,B]) = -i\{\sigma_m(A), \sigma_{m'}(B)\}$ with the bracket (7). A global quantization $x^{-\mathbb{Z}}S^{\mathbb{Z}}({}^cT^*X) \to x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X)$ exists [14, Prop. 24]. It induces the isomorphisms (1) for $\mathcal{I}_\sigma$ and $\mathcal{A}_\sigma$, and a star product

$$a \star b = \sum_{j\geq 0} P_j(a,b), \qquad P_0(a,b) = ab, \qquad P_1(a,b) - P_1(b,a) = -i\{a,b\} \qquad \text{[14, (137)–(138)]},$$

with P_j bidifferential of total homogeneity $-j$. In local coordinates with the left quantization, $P_j(a,b) = \sum_{|\alpha|=j}(\alpha!)^{-1}\partial_\zeta^\alpha a\, {}^cD^\alpha b$, where $\zeta = (\tau,\eta)$ and ${}^cD = (x^2D_x, D_y)$ up to lower order terms in the x-derivatives. Only P_0 and the antisymmetric part of P_1 will matter.

4.5 The indicial family, the suspended algebra and the twist

For a compact manifold H without boundary, $\Psi_{\mathrm{sus}}^m(H)$ is the algebra of pseudodifferential operators on $\mathbb{R}_t \times H$ that are invariant under translation in t and whose convolution kernels decrease rapidly as $|t| \to \infty$ [11]. Through (10) it is the same as the algebra of families $\mathbb{R} \ni \tau \mapsto B(\tau) \in \Psi^m(H)$ that are classical symbols jointly in τ and in the cotangent variables of H. The smoothing part is $\Psi_{\mathrm{sus}}^{-\infty}(H) \cong \mathcal{S}(\mathbb{R})\widehat{\otimes}\Psi^{-\infty}(H)$. The indicial map gives [14, (12)]

$$0 \to x\Psi_c^m(X) \to \Psi_c^m(X) \xrightarrow{\mathrm{In}} \Psi_{\mathrm{sus}}^m(\partial X) \to 0.$$

More precisely, a choice of normal fibration identifies a collar with $[0,\epsilon)\times\partial X$. The Taylor series of the kernel at ff_c then gives [14, Prop. 25]

$$x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X)/x^\infty\Psi_c^{\mathbb{Z}}(X) \;\cong\; \Psi_{\mathrm{sus}}^{\mathbb{Z}}(\partial X)[[x]], \qquad A \sim \sum_{j\geq -p} x^j A_{(j)}. \tag{12}$$

Here the powers of x are written *on the left*, and MN write A_{-j} for our $A_{(j)}$ [14, (53)]. The product on the right is not the naive one, because x does not commute with Ψ_{sus}.

Lemma 4.4 (The cusp twist)**.** *Let $g \in \Psi_{\mathrm{sus}}^m(\partial X)$, acting on functions of (t,y) with $t = 1/x$ near the boundary. Write g' for the element with $\mathrm{In}(g',\tau) = \partial_\tau \mathrm{In}(g,\tau)$. Then:*

1. *$[t,g] = -i\,g'$;*
2. *$[x,g] = i\,xg'x = i\,x^2g' + O(x^3)$, and in particular $[x,\partial_t] = x^2$;*
3. *$[\log x, g] = -\sum_{k\geq 1}\frac{1}{k}\,x^k\,\mathrm{ad}_t^k(g)$, where $\mathrm{ad}_t^k(g)$ has indicial family $(-i\partial_\tau)^k\,\mathrm{In}(g)$.*

The associated graded algebra of $\Psi_{\mathrm{sus}}(\partial X)[[x]]$ for the x-filtration is commutative modulo $\Psi(\partial X)$. Its Poisson bracket is $\{x,\tau\} = -x^2$, that is, $\{t,\tau\} = 1$.

Proof. (1) follows from (10), since $[t,g]$ has kernel $(t-t')\kappa(t-t')$. For (2),

$$[t^{-1},g] = t^{-1}g - gt^{-1} = -t^{-1}(tg-gt)t^{-1} = -t^{-1}[t,g]t^{-1} = i\,t^{-1}g't^{-1}.$$

Moving the right-hand factor x past g' costs a further power of x, by the same identity. For $g = \partial_t$ we have $\mathrm{In}(\partial_t, \tau) = -i\tau$, so $g' = -i$ and $[x, \partial_t] = x^2$. For (3), the kernel of $[\log x, g] = -[\log t, g]$ is $-(\log t - \log t')\kappa(t - t')$. With $s = t - t'$,

$$\log t - \log t' = -\log\Big(1 - \frac{s}{t}\Big) = \sum_{k\geq 1} \frac{1}{k}\,\frac{s^k}{t^k},$$

and $t^{-k} = x^k$ acts on the left. The kernel $s^k\kappa(s)$ is that of $\mathrm{ad}_t^k(g)$, so its indicial family is $(-i\partial_\tau)^k\,\mathrm{In}(g)$ by (10). The bracket of the leading symbols follows from (2): $\sigma([x,\tau]) = -i\{x,\tau\}$ and $[x,\tau]$ has leading term ix^2, so $\{x,\tau\} = -x^2$. Then $\{t,\tau\} = -x^{-2}\{x,\tau\} = 1$. □

Discussion. **What the twist means.** In $\mathcal{I}_\partial$ and $\mathcal{A}_\partial$ the "Laurent variable" x is the function $1/t$ on the suspension line, and the coefficients are functions of D_t. So these algebras are quantizations of functions of (t,τ) near $t = +\infty$, and t and τ are canonically conjugate. In a Laurent series in x, a derivative in τ costs one power of x. This is why the associated graded algebra is commutative and why the first differential of the x-adic spectral sequence is a Poisson differential, exactly as for the order filtration (Section 9). In the b-calculus the corresponding relation is $x^{-1}Ax = \alpha(A)$, where α is the complex shift $\lambda \mapsto \lambda - i$. This automorphism does *not* become trivial on the associated graded algebra; see Section 18.

4.6 The transcendental blow-up and the b-calculus

The b-vector fields are spanned by $x_b\partial_{x_b}$ and ∂_y. Put [14, (143)]

$$x_b = e^{-1/x}, \qquad \text{so that} \qquad \frac{dx_b}{dx} = \frac{x_b}{x^2}, \qquad x_b\partial_{x_b} = x^2\partial_x.$$

Thus b-differential operators in (x_b, y) pull back to cusp differential operators in (x, y). A change $x_b' = a(y)x_b + O(x_b^2)$ of the b-defining function changes x to $x' = x/(1 - x\log a) + O(x^\infty) = x + x^2\log a + O(x^3)$. If a is constant on each component, this is the same cusp structure. In general the two structures are related by Lemma 4.1 up to the isotopy of [14, Lemma 19]. MN quote that b-pseudodifferential operators lift to cusp operators and that the lifts are dense [14, Prop. 26].

Two features of the lift explain why it is useful.

1. The b-indicial family $I(A,\lambda)$ of [17] becomes the cusp indicial family with $\tau = \lambda \in \mathbb{R}$ (Section 2.2).

2. A b-operator is $A_0 + x_bA_1 + \ldots$, with A_0 dilation invariant. After the lift, A_0 is translation invariant in t, and $x_b = e^{-1/x}$ vanishes to infinite order in x. So the Laurent series (12) of a lifted b-operator is just $\mathrm{In}(A)$, with no higher terms.

A b-operator whose indicial family is invertible on $\mathbb{R}$ (that is, Fredholm on L^2_b) therefore lifts to a fully elliptic cusp operator. The cusp index formula applies to it, and its eta term is the b-eta invariant of the indicial family. The b-Laurent algebra $x_b^{-\mathbb{Z}}\Psi_b^{\mathbb{Z}}$ would require invertibility for *all* $\lambda \in \mathbb{C}$ (Section 18). This is the "problem with completeness" mentioned in [14, p. 3]. Note also that Laurent series in x_b and Laurent series in $x = 1/\log(1/x_b)$ are different algebras, so the lift does not identify the two Hochschild computations [6].

Question 4.5. *Why does the cusp calculus avoid the completeness problem, if the two calculi have isomorphic norm closures [14, p. 3]?*

Answer. The norm closures see only the boundary behaviour on the real line. The C^∞ algebras used for Hochschild homology see more. In the small b-calculus, indicial families are *entire* functions of $\lambda \in \mathbb{C}$. Inverting an operator modulo $x_b^\infty \Psi_b^{-\infty}$ means inverting its Taylor series at ff_b, so the inverse of the indicial family must again be entire. Hence $I(A,\lambda)$ must be invertible for *every* $\lambda \in \mathbb{C}$ [6, Lemma "invertible"]. Dirac operators never satisfy this. In the cusp calculus conjugation by x acts through the twist of Lemma 4.4, which only adds terms of higher order in x. So only invertibility on $\mathbb{R}$ is needed. □

4.7 Ellipticity, cusp metrics and complex powers

An element of $x^{-p}\Psi_c^m(X)$ is *elliptic* if its principal symbol (after removing x^{-p}) is invertible on ${}^cT^*X \setminus 0$. It is *fully elliptic* if, in addition, its leading indicial family is invertible for all real τ; in the Laurent setting this is invertibility in $\mathcal{A}$ (B2).

Remark 4.6 (Exact cusp metrics). [14, App. B] writes an exact cusp metric as $dx^2/x^4 + h/x^2$. The orthonormal frames of that metric are spanned by $x^2\partial_x$ and $x\partial_y$, which are the *scattering* vector fields. Its Laplacian is a cusp differential operator, but its cusp symbol $\tau^2 + x^2|\eta|^2$ degenerates at $x = 0$. So it is not cusp-elliptic, and $(\Delta+1)^{-s/2}$ is not a cusp family of order $-s$. The metric compatible with $\mathcal{V}_c$ is

$$g = \frac{dx^2}{x^4} + h = dt^2 + h \quad \text{near } \partial X,$$

a metric with cylindrical ends. Its Laplacian has cusp symbol $\tau^2 + |\eta|_h^2$. This is the metric to use in [14, §5].

For a positive invertible elliptic $Q \in \Psi_c^1(X)$, such as $(\Delta_g + 1)^{1/2}$ for the metric of Remark 4.6, the complex powers Q^{-s} form an entire family of order $-s$ (B3). The formal logarithm

$$\log Q = \frac{d}{ds} Q^s \Big|_{s=0}$$

is not in the calculus, but commutators with it are. If Q is asymptotically translation invariant, $\mathrm{In}(Q) = Q_0 = (D_t^2 + \Delta_{\partial X} + 1)^{1/2}$, and this Q_0 is the regularizer used in [14, (60)].

5 The Hochschild homology toolkit

This section collects the homological algebra used in [14], in the form in which we use it. Several statements are adapted from the companion notes [6] and from [13, Ch. 10]; we say so in each case.

5.1 Chains of topological algebras

For a Fréchet algebra $\mathfrak{A}$, the space of k-chains is the completed projective tensor product $C_k(\mathfrak{A}) = \mathfrak{A}^{\widehat{\otimes}(k+1)}$ [14, (36)]. Most of our algebras are increasing unions $\mathfrak{A} = \bigcup_N \mathfrak{A}_N$ of nuclear Fréchet spaces with $\mathfrak{A}_N\mathfrak{A}_M \subset \mathfrak{A}_{N+M}$; for example $\mathfrak{A}_N = x^{-N}\Psi_c^N(X)$. For these we take

$$C_k(\mathfrak{A}) = \varinjlim_N \mathfrak{A}_N^{\widehat{\otimes}(k+1)}.$$

Ideals and quotients carry the induced filtrations [6, §"The chain model"]. The basic examples are

$$C^\infty(M)^{\widehat{\otimes}(k+1)} = C^\infty(M^{k+1}), \qquad \dot{C}^\infty(M)^{\widehat{\otimes}(k+1)} = \dot{C}^\infty(M^{k+1})$$

for compact manifolds with boundary [14, proof of Lemma 1]. Here $\dot{C}^\infty(M^{k+1})$ denotes functions vanishing to infinite order at all boundary faces of the manifold with corners M^{k+1}. Since nuclear completed tensor products preserve short exact sequences of Fréchet spaces [18], for a closed ideal $J \subset \mathfrak{A}$ the sequence

$$0 \to \mathcal{K}_\bullet \to C_\bullet(\mathfrak{A}) \to C_\bullet(\mathfrak{A}/J) \to 0 \tag{13}$$

is exact. Here $\mathcal{K}_k$ is the closed span of the chains with at least one entry in J [6, Lemma "Exactness of chain spaces"].

5.2 H-unitality and excision

An algebra J is *H-unital* if its bar complex $(C_\bullet(J), b')$ is acyclic [21]. It is *topologically* H-unital if there is a continuous contraction E with $b'E + Eb' = 1$. A unital algebra is H-unital, with $E(a_0 \otimes \cdots \otimes a_k) = 1 \otimes a_0 \otimes \cdots \otimes a_k$.

Background fact 5.1 (Wodzicki's excision theorem, (B4))**.** If $J \subset \mathfrak{A}$ is a closed ideal that is topologically H-unital, then $C_\bullet(J) \to \mathcal{K}_\bullet$ is a quasi-isomorphism. Hence (13) gives a long exact sequence

$$\cdots \to \mathrm{HH}_k(J) \to \mathrm{HH}_k(\mathfrak{A}) \to \mathrm{HH}_k(\mathfrak{A}/J) \xrightarrow{\partial} \mathrm{HH}_{k-1}(J) \to \cdots .$$

To compute $\partial[c]$, lift the cycle c to a chain $\tilde{c}$ of $\mathfrak{A}$. Then $b\tilde{c} \in \mathcal{K}$, and $\partial[c]$ is its class.

Idea of the proof. The obstruction to $C(J) \to \mathcal{K}$ being a quasi-isomorphism lies in chains whose *interior* entries come from $\mathfrak{A}$ while some entry lies in J. Wodzicki filters $\mathcal{K}$ by the position of the first entry from J. The associated graded pieces are of the form (bar complex of J) $\otimes$ (chains of $\mathfrak{A}$). They are acyclic precisely when J is H-unital. The topological version needs the contraction to be continuous, and this is what Lemma 5.4 provides.

Lemma 5.2 (Connecting maps on cocycles)**.** *Under the hypotheses of 5.1, let φ be a $(k-1)$-cocycle on J. Let $\hat{\varphi}$ be a continuous $(k-1)$-cochain on $\mathfrak{A}$ that extends φ and such that $b\hat{\varphi}$ vanishes whenever one of its arguments lies in J. Then $b\hat{\varphi}$ descends to a k-cocycle on $\mathfrak{A}/J$ that represents $\partial^*[\varphi]$; that is, $\langle \varphi, \partial[c] \rangle = \langle b\hat{\varphi}, c \rangle$.*

Proof. This is [6, Lemma "Connecting maps on cocycles"]. The cochain $\hat{\varphi}$ restricted to $\mathcal{K}$ is a cocycle there, because $b\hat{\varphi}$ vanishes on $\mathcal{K}$. It restricts to φ on $C(J)$, so it represents $[\varphi]$ under the quasi-isomorphism. Hence $\langle \varphi, \partial[c] \rangle = \langle \hat{\varphi}, b\tilde{c} \rangle = \langle b\hat{\varphi}, \tilde{c} \rangle$, and the last pairing depends only on the image c of $\tilde{c}$. □

This lemma is how every boundary map in [14] is computed: extend a trace from an ideal by analytic continuation, and compute the coboundary of the extension. MN's formula (17) is the case of the traces on $\mathcal{I}_\sigma$ and $\mathcal{I}_\partial$.

5.3 Derivations

Lemma 5.3 (MN §6)**.** *Let D be a derivation of $\mathfrak{A}$ and $i_D(a_0 \otimes a_1 \otimes \cdots \otimes a_k) = a_0 D(a_1) \otimes a_2 \otimes \cdots \otimes a_k$. Then $bi_D + i_D b = 0$. If $D = [h, \cdot\,]$ with $h \in \mathfrak{A}$ ($\mathfrak{A}$ unital), then*

$$i_D = bR_h - R_h b, \qquad R_h(a_0 \otimes \cdots \otimes a_k) = a_0 h \otimes a_1 \otimes \cdots \otimes a_k.$$

So i_D induces a map $\mathrm{HH}_k \to \mathrm{HH}_{k-1}$ that vanishes for inner derivations. Dually, $b(i_D\varphi) = -i_D(b\varphi)$ for cochains.

Proof. Both identities are checked on elementary tensors. We do $k = 2$; the general case is the same bookkeeping. On the one hand,

$$i_D b(a_0 \otimes a_1 \otimes a_2) = a_0 a_1 D(a_2) - a_0 D(a_1 a_2) + a_2 a_0 D(a_1) = -a_0 D(a_1) a_2 + a_2 a_0 D(a_1),$$

while $b i_D(a_0 \otimes a_1 \otimes a_2) = b(a_0 D(a_1) \otimes a_2) = a_0 D(a_1) a_2 - a_2 a_0 D(a_1)$. The sum vanishes. For the inner case,

$$b R_h(a_0 \otimes a_1 \otimes a_2) - R_h b(a_0 \otimes a_1 \otimes a_2) = a_0 h a_1 \otimes a_2 - a_0 a_1 h \otimes a_2 = a_0 [h, a_1] \otimes a_2,$$

since the terms $a_0 h \otimes a_1 a_2$ and $a_2 a_0 h \otimes a_1$ occur in both $b R_h$ and $R_h b$ and cancel. □

In [14] the derivations $D_{\log Q}$ and $D_{\log x}$ are commutators with elements *not* in the algebra [14, (63)–(64)]. Changing Q or x changes them by commutators with $\log(Q'/Q) \in \Psi_c^0(X)$ or $\log(x'/x) \in C^\infty(X)$, which are inner. So the maps $i_{\log Q}$ and $i_{\log x}$ on homology do not depend on the choices. The ideals $\mathcal{I}_\sigma$, $\mathcal{I}_\partial$, $\mathcal{I}$ are preserved by both derivations, and R_h maps chains of an ideal to chains of the same ideal. So the same independence holds for the ideals.

5.4 Explicit H-unitality contractions

Lemma 5.4. *Let V, W be nuclear Fréchet spaces with a continuous pairing $\langle \cdot, \cdot \rangle : W \times V \to \mathbb{C}$. Let $\mathcal{K} = V \widehat{\otimes} W$ be the algebra with product $(v \otimes w)(v' \otimes w') = \langle w, v' \rangle v \otimes w'$, and fix $\psi \in W$, $\tilde{\psi} \in V$ with $\langle \psi, \tilde{\psi} \rangle = 1$. Then*

$$E\big((v \otimes w) \otimes c\big) = (v \otimes \psi) \otimes (\tilde{\psi} \otimes w) \otimes c$$

is a continuous contraction of the bar complex. Hence the algebras

$$\Psi^{-\infty}(H) = C^\infty(H) \widehat{\otimes} C^\infty(H), \qquad \mathcal{I} = \dot{C}^\infty(X) \widehat{\otimes} \dot{C}^\infty(X)$$

are topologically H-unital. Moreover $\mathrm{HH}_0(\mathcal{K}) \cong \mathbb{C}$, spanned by the trace $v \otimes w \mapsto \langle w, v \rangle$, and $\mathrm{HH}_k(\mathcal{K}) = 0$ for $k > 0$.

Proof. The contraction is [6, Lemma "Explicit contractions"]: $d_0 E = 1$ because $(v \otimes \psi)(\tilde{\psi} \otimes w) = v \otimes w$, and $d_{i+1} E = E d_i$ for $i \geq 0$, so $b' E + E b' = 1$. The homology statement is Morita invariance for the smoothing kernels [6, Lemma "Morita invariance for smoothing kernels"], applied with $\mathfrak{A} = \mathbb{C}$. □

5.5 The Hochschild–Kostant–Rosenberg map and Connes' theorem

For a commutative algebra the map [14, after (26)]

$$\chi(f_0 \otimes f_1 \otimes \cdots \otimes f_k) = \frac{1}{k!} f_0 \, df_1 \wedge \cdots \wedge df_k$$

satisfies $\chi \circ b = 0$. Conversely the antisymmetrization $\varphi(f_0; f_1, \ldots, f_k) = \sum_{\sigma \in \mathfrak{S}_k} \mathrm{sgn}(\sigma) f_0 \otimes f_{\sigma(1)} \otimes \cdots \otimes f_{\sigma(k)}$ is a cycle with $\chi\varphi(f_0; f_1, \ldots, f_k) = f_0 \, df_1 \wedge \cdots \wedge df_k$.

Background fact 5.5 (Connes' theorem, (B5))**.** For a compact manifold M, possibly with boundary or corners, χ induces isomorphisms $\mathrm{HH}_k(C^\infty(M)) \cong C^\infty(M; \Lambda^k)$ and $\mathrm{HH}_k(\dot{C}^\infty(M)) \cong \dot{C}^\infty(M; \Lambda^k)$. The same holds, fibrewise, for smooth functions on a manifold with values in a commutative nuclear algebra of Laurent or formal series.

A self-contained proof for closed M is [13, Thm. "Connes"]. It has three steps: reduction to normalized chains, a local argument near the diagonal using Taylor expansion and the polynomial HKR theorem (proved by a Koszul resolution), and patching. Boundaries of normalized chains form a $C^\infty(M^{k+1})$-module, and this makes the patching possible. Nothing changes for manifolds with boundary. For $\dot{C}^\infty(M)$ all the constructions preserve infinite-order vanishing at the boundary faces.

5.6 Filtered complexes and the convergence of spectral sequences

Let $C_k = \{\sum_{p \le P} c_p : \ c_p \in C_{p,k}\}$ be a complex of formal sums, bounded above in p, with differential $b = \sum_{l \ge 0} b_l$, where $b_l : C_{p,k} \to C_{p-l,k-1}$. The filtration $F_p = \{c : \ c_q = 0 \text{ for } q > p\}$ is exhaustive, separated and complete. Put $E^1_{p,k} = H_k(C_{p,\bullet}, b_0)$. Then b_1 induces $d^1 : E^1_{p,k} \to E^1_{p-1,k-1}$, and we write $E^2 = H(E^1, d^1)$.

Proposition 5.6 (Degeneration and convergence [13, Prop. "convergence"])**.** *Suppose there is an integer n such that $E^2_{p,k} = 0$ whenever $p \neq k - n$. Then every class in $H_k(C, b)$ has a representative in $F_{k-n}C_k$, and*

$$\Theta : H_k(C, b) \to E^2_{k-n,k}, \qquad [c] \mapsto [c_{k-n}],$$

is an isomorphism. All differentials d^r, $r \ge 2$, vanish.

Proof. We recall the key step. Suppose $c \in F_pC_k$ satisfies $b_0c_p = 0$ and $b_1c_p + b_0c_{p-1} = 0$, and the class of c_p in $E^2_{p,k}$ vanishes. Then $[c_p] = d^1[x]$ for a b_0-cycle $x \in C_{p+1,k+1}$, so $c_p - b_1x = b_0y$ with $y \in C_{p,k+1}$. The components of $b(x + y)$ of order $p + 1$ and p are $b_0x = 0$ and $b_1x + b_0y = c_p$, so $c - b(x + y) \in F_{p-1}$. Since E^2 is concentrated on the line $p = k - n$, repeating this step lowers the order of any cycle to $k - n$. It also shows that a cycle with vanishing E^2-class is a boundary: the corrections form a series in decreasing orders, which converges by completeness. Surjectivity is the same argument applied to the obstruction $bc^{(r)}$, whose order decreases. Finally d^r maps (p, k) to $(p - r, k - 1)$, off the line $p = k - n$ for $r \ge 2$. The details are in [13, Ch. 10]. □

Remark 5.7 (Pole orders). For the symbol algebras with Laurent singularities at the boundary ($\mathcal{A}_\sigma$ and $\mathcal{A}_{\partial,\sigma}$) the chain spaces are unions over pole bounds N of complete spaces. Proposition 5.6 applies in each piece provided the corrections x, y in its proof can be chosen *without increasing the pole bound*. MN do not discuss this; they say convergence "follows directly, as in the boundaryless case" [14, p. 12]. The corrections are built from three things: the HKR lift φ, which preserves pole orders; the de Rham homotopies of Section 6, which pair Taylor coefficients of the same x-degree and so preserve pole orders; and the symplectic star, whose factors $x^{\pm 2}$ cancel between $*$ and $*^{-1}$. The one point that requires care is the inverse of b_0 on boundaries of the commutative complex, where Connes' patching argument multiplies by non-smooth Laurent factors. A complete treatment of the parallel point for the b-calculus, by mixed complexes, is given in [6, §"Convergence and degeneration"]. It applies here with the cusp Poisson structure in place of the b-structure. We do not repeat it, and in what follows *convergence for $\mathcal{A}_\sigma$ and $\mathcal{A}_{\partial,\sigma}$ is to be read as resting on that argument*. For $\mathcal{I}_\sigma$ (rapid vanishing at the boundary) and for the closed case no issue arises.

5.7 The Poisson differential and the symplectic star

Let $V \to M$ be a real vector bundle with a symplectic structure on $V \setminus 0$ that is homogeneous of degree 1 under dilations. The examples are T^*M and ${}^cT^*X$ with the forms of Section 2.1. Let Π be

the Poisson bivector and ι_Π contraction with it. The Koszul–Brylinski differential is $\delta = \iota_\Pi d - d\iota_\Pi$, which maps k-forms of homogeneity p to $(k-1)$-forms of homogeneity $p-1$. On functions,

$$\delta(a_0\, da_1 \wedge \cdots \wedge da_k) = \sum_j (-1)^{j+1}\{a_0, a_j\}\, da_1 \wedge \ldots \widehat{da_j} \cdots \wedge da_k + \sum_{j<l} (-1)^{j+l} a_0\, d\{a_j, a_l\} \wedge \cdots .$$

Proposition 5.8 (First differential [13, Prop. "d1"]). *For a star product with $P_0(a,b) = ab$ and $P_1(a,b) - P_1(b,a) = -i\{a,b\}$, the differential d^1 of the order spectral sequence is $-i\delta$ under the HKR identification of E^1 with homogeneous forms.*

The point of the proof is that the symmetric part of P_1 cancels in pairs on antisymmetrized chains, and the antisymmetric part produces δ term by term; see [13, Ch. 10].

Proposition 5.9 (Brylinski [2]). *Let $*$ be the symplectic star operator, defined by $\alpha \wedge *\beta = \Lambda(\alpha,\beta)\,\omega^n/n!$. Then $\delta = (-1)^{k-1} *^{-1} d *$ on k-forms, and $*$ maps forms of degree k and homogeneity p to forms of degree $2n-k$ and homogeneity $p+n-k$. Moreover $\iota_{H_f}(*\eta) = \pm *(df \wedge \eta)$.*

For ${}^cT^*X$ the volume form is $\omega^n/n! = x^{-2}\times$(a smooth volume). So on smooth forms $*$ introduces a factor $x^{\pm 2}$ [14, p. 12]. In the Laurent algebras this is harmless. In $\mathcal{I}_\sigma$, where everything vanishes to infinite order at the boundary, it is harmless as well.

Proposition 5.10 (Homogeneous de Rham cohomology [14, Lemma 3]). *Let $\Omega^\bullet_{(c)}$ be the forms on $V \setminus 0$ of homogeneity c. If $c \neq 0$ the complex $(\Omega^\bullet_{(c)}, d)$ is acyclic. For $c = 0$, writing forms as $\alpha' + \alpha'' \wedge dr/r$ with r a positive homogeneous function of degree one,*

$$H^j(\Omega^\bullet_{(0)}, d) \cong H^j(SV) \oplus H^{j-1}(SV)\,\frac{dr}{r} \cong H^j(SV \times \mathbb{S}_\sigma).$$

The same holds for Laurent forms, flat forms, and formal Laurent forms over manifolds with boundary, with the corresponding de Rham cohomology of SV (Section 6).

Proof. Let R be the radial vector field. It commutes with dilations, so ι_R preserves homogeneity, and $\mathcal{L}_R = c$ on $\Omega_{(c)}$. If $c \neq 0$ and $d\alpha = 0$, Cartan's formula gives $\alpha = d(\iota_R\alpha/c)$. If $c = 0$, the forms α' and α'' are pulled back from SV, and $d(\alpha' + \alpha'' \wedge dr/r) = d\alpha' + d\alpha'' \wedge dr/r$. The operator ι_R preserves Laurent, flat or formal behaviour in x, so the same proof applies to those classes of forms. □

5.8 Morita invariance

If $e \in M_N(\mathfrak{A})$ is an idempotent not contained in any proper two-sided ideal, then $eM_N(\mathfrak{A})e$ is Morita equivalent to $\mathfrak{A}$ and has the same Hochschild homology (B6). The trace map $\mathrm{Tr}_* : C(M_N(\mathfrak{A})) \to C(\mathfrak{A})$ gives the isomorphism [14, Prop. 13], [8]. For an algebra of the form $\mathcal{K}\widehat{\otimes}\mathfrak{A}$ as in Lemma 5.4, the generalized trace gives the isomorphism $\mathrm{HH}(\mathcal{K}\widehat{\otimes}\mathfrak{A}) \cong \mathrm{HH}(\mathfrak{A})$ even when $\mathfrak{A}$ is only H-unital [6, Lemma "Morita invariance for smoothing kernels"]. We use this with $\mathcal{K} = \Psi^{-\infty}(\partial X)$ to remove the boundary smoothing operators in Section 9.

6 Functions and weighted de Rham cohomology (MN §1)

Question 6.1. *Why do MN allow arbitrary negative powers of x, that is, Laurent coefficients, instead of working with $\Psi^{\mathbb{Z}}_c(X)$ itself?*

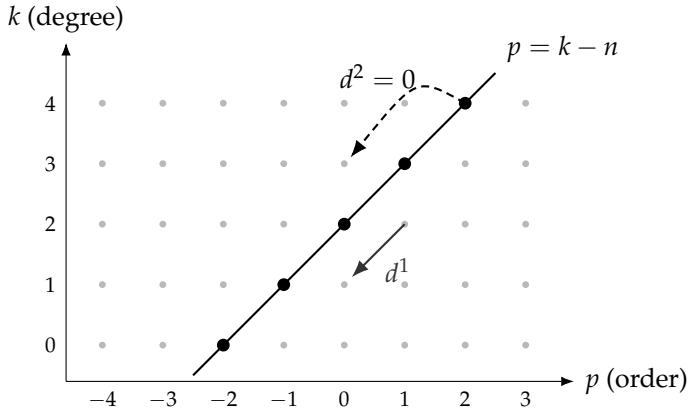


Figure 5: The spectral sequence of the order filtration for a symbol algebra, drawn for $n = 2$. Gray dots mark the positions (p, k) where $E^1_{p,k}$ (homogeneous k-forms of degree p) can be non-zero. The differential d^1 moves along lines parallel to $p = k - n$. After the symplectic star and Proposition 5.10, E^2 lives only on the line $p = k - n$ (black dots), and the higher differentials leave that line, so they vanish. Adapted from [13, Ch. 10].

Answer. For the same reason that one allows operators of all integral orders. The order filtration and the x-filtration are treated symmetrically: in $x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}$ both Q and x are invertible modulo the residual ideals. The homology is then finite dimensional, and each filtration contributes a circle, $\mathbb{S}_\sigma$ for the order and $\mathbb{S}_\partial$ for the powers of x, with classes $d\log r$ and $d\log x$. Without negative powers, closed forms such as $\beta \wedge dx/x^2$ would no longer be exact (Correction 6.8), and the homology would acquire infinite-dimensional pieces, as it does for order zero (Section 17.3). On the analytic side, $x^{-p}A$ has the same index as A when both are fully elliptic [14, p. 31], so nothing is lost. □

Throughout this section M is a compact manifold with boundary, and x is a boundary defining function with a fixed collar $[0, \epsilon)_x \times \partial M$. We consider the three algebras of the Taylor sequence [14, (25)],

$$0 \to \dot{C}^\infty(M) \to x^{-\mathbb{Z}} C^\infty(M) \to C^\infty(\partial M)[[x]] \to 0, \tag{14}$$

where the last map takes the Taylor series at ∂M in the collar and $C^\infty(\partial M)[[x]]$ is the algebra of formal Laurent series $\sum_{k \geq -p} f_k x^k$.

6.1 H-unitality of the flat functions

Lemma 6.2 (MN Lemma 1)**.** *$\dot{C}^\infty(M)$ is topologically H-unital.*

Proof. The k-chains are $\dot{C}^\infty(M^{k+1})$. Choose $\phi \in C_c^\infty((-1,1))$ with $\phi(0) = 1$, and for $f \in \dot{C}^\infty(M^{k+1})$ define

$$(Ef)(z^{(0)}, z^{(1)}, \ldots, z^{(k+1)}) = \phi\Big(\frac{x^{(0)} - x^{(1)}}{x^{(0)} + x^{(1)}}\Big) f(z^{(1)}, \ldots, z^{(k+1)}), \tag{15}$$

where $x^{(j)}$ is x on the j-th factor [14, (26)].

Ef is flat. Put $w = (x^{(0)} - x^{(1)})/(x^{(0)} + x^{(1)}) \in [-1, 1]$. On the support of $\phi(w)$ we have $|w| \leq 1 - \delta$ for some $\delta > 0$. Hence $x^{(0)}$ and $x^{(1)}$ are comparable there: $c\, x^{(1)} \leq x^{(0)} \leq C\, x^{(1)}$. The function $\phi(w)$ is smooth away from $x^{(0)} = x^{(1)} = 0$. Each derivative of $\phi(w)$ is bounded by a power of $(x^{(0)} + x^{(1)})^{-1}$, which is bounded by a power of $1/x^{(1)}$ on the support. Since f vanishes

to infinite order as $x^{(1)} \to 0$, every derivative of Ef is $O((x^{(1)})^\infty) = O((x^{(0)})^\infty)$ there. At the faces $x^{(j)} = 0$, $j \geq 1$, flatness is inherited from f. So $E : \dot{C}^\infty(M^{k+1}) \to \dot{C}^\infty(M^{k+2})$ continuously.

E is a contraction. The face d_0 restricts to $z^{(0)} = z^{(1)}$, where $w = 0$, so $d_0 Ef = f$. For $i \geq 1$, the face d_i identifies $z^{(i)}$ and $z^{(i+1)}$, and does not touch the factor $\phi(w)$ unless $i = 1$. When $i = 1$ it replaces $z^{(1)}$ by itself, so in all cases $d_i E = E d_{i-1}$. Therefore

$$b'Ef = d_0 Ef + \sum_{i\geq 1} (-1)^i d_i Ef = f - E\Big(\sum_{i\geq 0} (-1)^i d_i f\Big) = f - Eb'f. \qquad \square$$

Remark 6.3 (The support of ϕ). MN only ask that $\phi \in C_c^\infty(\mathbb{R})$ with $\phi(0) = 1$. This is not enough. If $\phi(-1) \neq 0$, then at a point with $x^{(0)} = 0 < x^{(1)}$ we have $w = -1$, and $Ef = \phi(-1) f(z^{(1)}, \dots)$ need not vanish, so $Ef \notin \dot{C}^\infty(M^{k+2})$. The support condition $\operatorname{supp}\phi \subset (-1, 1)$ is what makes $x^{(0)}$ and $x^{(1)}$ comparable.

The algebra $x^{-\mathbb{Z}} C^\infty(M)$ is unital, and so is $C^\infty(\partial M)[[x]]$. By excision (B4), (14) gives a long exact sequence in Hochschild homology.

6.2 Hochschild homology of the three function algebras

The HKR map χ is defined on the chains of all three algebras [14, (27)]:

$$\begin{aligned}
&\chi : \dot{C}^\infty(M^{k+1}) \to \dot{C}^\infty(M; \Lambda^k),\\
&\chi : x_0^{-\mathbb{Z}} \cdots x_k^{-\mathbb{Z}} C^\infty(M^{k+1}) \to x^{-\mathbb{Z}} C^\infty(M; \Lambda^k),\\
&\chi : C^\infty((\partial M)^{k+1})[[x_0, \dots, x_k] \to C^\infty(\partial M; \Lambda^k)[[x] \oplus C^\infty(\partial M; \Lambda^{k-1})[[x] \, \frac{dx}{x^2}.
\end{aligned}$$

(MN print M for ∂M in the last target.) In the last line a k-form on the formal neighbourhood of ∂M is split into its tangential part and its dx-part. Any of dx, dx/x, dx/x^2 may be used as the normal basis element, since the coefficients are Laurent series. MN use dx/x^2 to match the cusp geometry. We use dx/x when computing cohomology.

Lemma 6.4 (MN Lemma 2)**.** *The three HKR maps are isomorphisms on Hochschild homology. The long exact sequence of* (14) *splits into the short exact sequences [14, (28)]*

$$0 \to \dot{C}^\infty(M; \Lambda^k) \to x^{-\mathbb{Z}} C^\infty(M; \Lambda^k) \to C^\infty(\partial M; \Lambda^k)[[x] \oplus C^\infty(\partial M; \Lambda^{k-1})[[x] \, \frac{dx}{x} \to 0.$$

Proof. For $\dot{C}^\infty(M)$ the HKR isomorphism is Connes' theorem in its flat form (B5). For the formal Laurent series, the Künneth formula (B6) reduces the problem to $C^\infty(\partial M)$ (Connes) and to $\mathbb{C}[[x]]$. The algebra $\mathbb{C}[[x]]$ has the Koszul resolution

$$0 \to \mathbb{C}[[x, x']] \xrightarrow{x - x'} \mathbb{C}[[x, x']] \to \mathbb{C}[[x]] \to 0$$

by free modules over $\mathbb{C}[[x]] \widehat{\otimes} \mathbb{C}[[x]] = \mathbb{C}[[x, x']]$. It gives $\mathrm{HH}_0 = \mathbb{C}[[x]]$, $\mathrm{HH}_1 = \mathbb{C}[[x]]\, dx$, and $\mathrm{HH}_k = 0$ for $k \geq 2$. Inverting the central element x is an exact localization, so the same holds for $\mathbb{C}[[x]$ with Laurent coefficients. Now the HKR maps commute with the maps of (14), since restriction to Taylor series commutes with d. By the five lemma, the HKR map for the middle algebra $x^{-\mathbb{Z}} C^\infty(M)$ is also an isomorphism. Finally, the right-hand map at the level of forms is surjective by Borel's lemma (every formal Laurent form is the Taylor series of a Laurent form). So the connecting maps vanish, and the long exact sequence breaks into the displayed short exact sequences. $\square$

6.3 Weighted de Rham cohomology

The de Rham differential acts on all three spaces of forms in Lemma 6.4.

Lemma 6.5 (Formal Laurent forms)**.** *The cohomology of the de Rham complex of formal Laurent forms $\sum_{j\geq -p} x^j(\alpha_j + \beta_j \wedge dx/x)$, with α_j, β_j forms on ∂M, is $H^k(\partial M) \oplus H^{k-1}(\partial M)\, dx/x$. It is spanned by the classes with $j = 0$.*

Proof. For a k-form α on ∂M, $d(x^j\alpha) = x^j\big(d\alpha + (-1)^k j\,\alpha \wedge dx/x\big)$, and $d(x^j\beta \wedge dx/x) = x^j\, d\beta \wedge dx/x$. So the complex splits into the subcomplexes of fixed x-degree j. On degree j the differential is $(\alpha, \beta) \mapsto (d\alpha,\ d\beta + (-1)^k j\alpha)$. This is the mapping cone of multiplication by j on the de Rham complex of ∂M. For $j \neq 0$ that multiplication is invertible, so the cone is acyclic; explicitly $h(\alpha, \beta) = ((-1)^{k-1}j^{-1}\beta, 0)$ is a contracting homotopy. For $j = 0$ the complex is the direct sum of the de Rham complex of ∂M and its shift. □

Proposition 6.6 (MN Proposition 1)**.** *The de Rham cohomology of the three complexes in Lemma 6.4 is*

$$H^k_{\mathrm{rel}}(M), \qquad H^k(M) \oplus H^{k-1}(\partial M), \qquad H^k(\partial M) \oplus H^{k-1}(\partial M),$$

and the long exact sequence is [14, (29)]

$$\cdots \to H^k_{\mathrm{rel}}(M) \to H^k(M) \oplus H^{k-1}(\partial M) \xrightarrow{\iota^* \oplus \mathrm{Id}} H^k(\partial M) \oplus H^{k-1}(\partial M) \to H^{k+1}_{\mathrm{rel}}(M) \to \cdots,$$

the direct sum of the long exact sequence of the pair $(M, \partial M)$ and the identity on $H^{k-1}(\partial M)$.

Proof. Flat forms. The complex $\dot{C}^\infty(M; \Lambda^\bullet)$ contains the compactly supported forms on the interior, and the inclusion is a quasi-isomorphism [14, p. 9]. In the collar a flat form is $\alpha + \beta \wedge dx$ with α, β flat at $x = 0$. The homotopy $h(\alpha + \beta \wedge dx) = \pm \int_0^x \beta$, integrated from the boundary and cut off away from the collar, preserves flatness. It shows that a closed flat form is cohomologous to one that vanishes near ∂M. The cohomology is therefore $H^k_c(M^\circ) = H^k_{\mathrm{rel}}(M)$.

Formal Laurent forms. This is Lemma 6.5.

Laurent forms. Let $\chi_0 \in C^\infty(M)$ be a cut-off equal to 1 near ∂M and supported in the collar. Consider

$$\Psi : C^\infty(M; \Lambda^k) \oplus C^\infty(\partial M; \Lambda^{k-1}) \to x^{-\mathbb{Z}} C^\infty(M; \Lambda^k), \qquad (\gamma, \beta) \mapsto \gamma + \chi_0\, \beta \wedge \frac{dx}{x}.$$

It is a chain map for the differential $(\gamma, \beta) \mapsto (d\gamma, d\beta)$, because $d\chi_0 \wedge \beta \wedge dx = 0$ ($d\chi_0$ is a multiple of dx in the collar). It fits into a map of short exact sequences: from [14, (30)],

$$0 \to \dot{C}^\infty(M; \Lambda^\bullet) \to C^\infty(M; \Lambda^\bullet) \oplus C^\infty(\partial M; \Lambda^{\bullet-1}) \to C^\infty(\partial M; \Lambda^\bullet \oplus \Lambda^{\bullet-1}) \to 0,$$

where the second map is restriction to ∂M plus the identity, to the sequence of Lemma 6.4. On the outer terms the induced maps in cohomology are the identity and the inclusion of the $j = 0$ part of Lemma 6.5. Both are isomorphisms, so Ψ is a quasi-isomorphism by the five lemma. The cohomology of the source is $H^k(M) \oplus H^{k-1}(\partial M)$, by the de Rham theorem for manifolds with boundary. The long exact sequence of the source sequence is the sequence of the pair together with the identity on $H^{k-1}(\partial M)$. □

Example 6.7. For $M = [0, 1)$ (non-compact, but the computation is local), the Laurent de Rham cohomology is $H^0 = \mathbb{C}$ and $H^1 = \mathbb{C}\, dx/x$. For the compact interval $M = [0, 1]$ with $x = s(1 - s)$, it is $H^0 = \mathbb{C}$ and $H^1 = \mathbb{C}^2$, spanned by dx/x localized near either end. The relative cohomology is $H^1_{\mathrm{rel}}([0, 1]) = \mathbb{C}$, and the map $H^1_{\mathrm{rel}} \to H^1 \oplus H^0(\partial M)$ is zero, as it must be.

6.4 b- and cusp de Rham complexes

The Laurent forms $x^{-\mathbb{Z}}C^\infty(M;\Lambda^k)$ coincide with the Laurent b-forms and the Laurent cusp forms. The smooth b-forms $C^\infty(M;{}^b\Lambda^k)$, locally generated by dx/x and dy, and the smooth cusp forms $C^\infty(M;{}^c\Lambda^k)$, generated by dx/x^2 and dy, are subcomplexes [14, (32)–(33)]. By the proof of Proposition 6.6 the b-complex has the cohomology of the Laurent complex,

$$ {}^bH^k(M) \cong H^k(M) \oplus H^{k-1}(\partial M) $$

(Nest–Tsygan [15], quoted in [14]).

Correction to MN 6.8 (The cusp de Rham complex). After [14, (33)] it is stated that the cusp de Rham complex also has cohomology ${}^bH^k(M)$. This is not so: its cohomology is

$$ {}^cH^k(M) \cong H^k(M) \oplus H^{k-1}(\partial M) \oplus H^{k-1}(\partial M). $$

The extra classes are $\beta \wedge dx/x^2$, with β closed on ∂M. They are exact in the Laurent complex ($\beta \wedge dx/x^2 = \pm d(\beta/x)$) but not in the smooth cusp complex.

Proof. Expand a smooth cusp form near ∂M as $\sum_{j\geq 0} x^j(\alpha_j + \beta_j \wedge dx/x^2)$. From $d(x^j\alpha) = x^j\,d\alpha + (-1)^k j\,x^{j+1}\alpha \wedge dx/x^2$ we see that the differential pairs α_j with β_{j+1}, with coefficient j. For $j \geq 1$ this pairing is invertible, and those pairs cancel as in Lemma 6.5. What survives is α_0, β_0 and β_1. Since α_{-1} is not available, β_0 is not hit. This gives $H^k(\partial M) \oplus H^{k-1}(\partial M) \oplus H^{k-1}(\partial M)$ for the formal boundary complex. A form $\chi_0\beta \wedge dx/x^2$ with $d\beta = 0$ is closed on M, so the connecting map vanishes on the new classes, and the five-lemma argument of Proposition 6.6 gives the result. On $[0,1)$: dx/x^2 and dx/x are closed smooth cusp 1-forms, and neither is df for a smooth f. □

This plays no role in [14], which only uses the Laurent complex.

6.5 Homogeneous Laurent forms

Let $V \to M$ be a real vector bundle and r a positive function on $V \setminus 0$, homogeneous of degree one. Combining Proposition 5.10 with Proposition 6.6 gives [14, (35)]: the de Rham cohomology of formal sums of homogeneous Laurent forms on $V \setminus 0$, with degrees bounded above, is

$$ {}^bH^k(SV \times \mathbb{S}_\sigma) = {}^bH^k(SV) \oplus {}^bH^{k-1}(SV)\,\frac{dr}{r} \cong H^k(SV \times \mathbb{S}_\sigma) \oplus H^{k-1}(\partial SV \times \mathbb{S}_\sigma)\,\frac{dx}{x}. \qquad (16) $$

The same holds with flat forms (relative cohomology) and with formal Laurent forms at the boundary (cohomology of $\partial SV \times \mathbb{S}_\sigma \times \mathbb{S}_\partial$).

7 The symbol algebras $\mathcal{A}_\sigma$ and $\mathcal{I}_\sigma$ (MN §2)

Proposition 7.1 (MN Proposition 2). $\mathrm{HH}_k(\mathcal{A}_\sigma) \cong {}^bH^{2n-k}(Y \times \mathbb{S}_\sigma)$, *that is,*

$$ \mathrm{HH}_k(\mathcal{A}_\sigma) \cong H^{2n-k}(Y) \oplus H^{2n-k-1}(Y)\,d\theta_\sigma \oplus H^{2n-k-1}(Z)\,\frac{dx}{x} \oplus H^{2n-k-2}(Z)\,d\theta_\sigma \wedge \frac{dx}{x}. $$

Equivalently, $\mathrm{HH}_k(\mathcal{A}_\sigma) \cong H^{2n-k}(Y \times \mathbb{S}_\sigma) \oplus H^{2n-k-1}(Z \times \mathbb{S}_\sigma)$, *and by duality* $\mathrm{HH}^k(\mathcal{A}_\sigma) \cong H^k_{\mathrm{rel}}(Y \times \mathbb{S}_\sigma) \oplus H^k(Z \times \mathbb{S}_\sigma)$.

Proof. Chains and filtration. By the quantization (B1),

$$\mathcal{A}_\sigma \cong \varinjlim_{m,N} x^{-N} \prod_{j\le m} \mathcal{G}^j({}^cT^*X),$$

with the star product [14, (37)]. Completing tensor products gives the chains [14, (38)]: formal sums $\sum_{p\le P} c_p$, where c_p is a function on $({}^cT^*X\setminus 0)^{k+1}$ with Laurent singularities in each x_j, homogeneous in each factor, of total homogeneity p. The order filtration is independent of the quantization, and the Hochschild differential decomposes as $b=\sum_l b_l$ with b_l coming from P_l. So we are in the setting of Section 5.6.

E^1. The differential b_0 is the Hochschild differential of the commutative algebra $x^{-\mathbb{Z}}\mathcal{G}({}^cT^*X)$. By Lemma 6.4, applied fibrewise and combined with Connes' theorem on the sphere bundle (B5), $E^1_{p,k}\cong x^{-\mathbb{Z}}\mathcal{G}^p({}^cT^*X;\Lambda^k)$, the homogeneous Laurent k-forms of degree p.

d^1. By Proposition 5.8, $d^1=-i\delta$, where δ is the Koszul–Brylinski differential of the cusp Poisson structure (7). This is the extension by continuity of the Poisson differential of the interior [14, p. 12].

E^2. The cusp symplectic form is $\omega = d\tau\wedge dx/x^2 + d\eta\wedge dy$, homogeneous of degree one, with only a finite order singularity at $x=0$. The symplectic star operator therefore maps $x^{-\mathbb{Z}}\mathcal{G}^p({}^cT^*X;\Lambda^k)$ isomorphically onto $x^{-\mathbb{Z}}\mathcal{G}^{p+n-k}({}^cT^*X;\Lambda^{2n-k})$ and intertwines δ with $\pm d$ (Proposition 5.9). In local coordinates, $*(f\,dx^I\,d\zeta^J)=\pm x^{\pm 2} f\,dx^{J^\complement}d\zeta^{I^\complement}$, which explains the factor x^2 in [14, p. 12]. Hence

$$E^2_{p,k}\cong H^{2n-k}\big(x^{-\mathbb{Z}}\mathcal{G}^{p+n-k}({}^cT^*X;\Lambda^\bullet),d\big),$$

which vanishes for $p\neq k-n$ by Proposition 5.10. For $p=k-n$ it equals ${}^bH^{2n-k}(Y\times \mathbb{S}_\sigma)$ by (16).

Degeneration and convergence. E^2 is concentrated on the line $p=k-n$, so Proposition 5.6 applies (with Remark 5.7).

Duality. $Y\times\mathbb{S}_\sigma$ is a compact oriented $2n$-manifold with boundary $Z\times\mathbb{S}_\sigma$. It is oriented by the symplectic structure and the orientation of the circle. Lefschetz duality gives $H^{2n-k}(Y\times\mathbb{S}_\sigma)^*\cong H^k_{\mathrm{rel}}(Y\times\mathbb{S}_\sigma)$, and Poincaré duality on the closed $(2n-1)$-manifold $Z\times\mathbb{S}_\sigma$ gives $H^{2n-k-1}(Z\times\mathbb{S}_\sigma)^*\cong H^k(Z\times\mathbb{S}_\sigma)$. □

Discussion. **The singular symplectic form.** Over the interior, ${}^cT^*X=T^*X$ and ω is the usual symplectic form. At the boundary, $\omega=d\tau\wedge dx/x^2+\cdots$ blows up to second order. Its inverse, the Poisson bivector $x^2\partial_\tau\wedge\partial_x+\cdots$, *vanishes* to second order. This is why δ extends smoothly to the boundary and why the commutator of two cusp operators is again a cusp operator. On Laurent forms the factor $x^{\pm2}$ is invisible, and the computation is word for word the closed one. What changes is the de Rham cohomology of the Laurent forms, which picks up the boundary term $H^{*-1}(Z)\,dx/x$ (Proposition 6.6).

Proposition 7.2 (MN Proposition 3)**.** *$\mathcal{I}_\sigma$ is H-unital,* $\mathrm{HH}_k(\mathcal{I}_\sigma)\cong H^{2n-k}_{\mathrm{rel}}(Y\times\mathbb{S}_\sigma)$, *and* $\mathrm{HH}^k(\mathcal{I}_\sigma)\cong H^k(Y\times\mathbb{S}_\sigma)$.

Proof. The computation is that of Proposition 7.1 with flat forms in place of Laurent forms. By Proposition 6.6 these give relative cohomology. Here no pole orders occur, so the convergence argument of the closed case applies verbatim. For H-unitality, filter the bar complex by order. Its E^1 term is the bar homology of the commutative algebra $\dot{C}^\infty$-valued in homogeneous functions. This vanishes by Lemma 6.2, whose contraction (15) only involves the base variables and preserves homogeneity. By completeness the bar complex of $\mathcal{I}_\sigma$ is acyclic. Alternatively, use the contraction (15) directly with a quantization in which Id has full symbol 1 and which is a left quantization in the collar. Then the cut-off factor in (15) is independent of the fibre variable in slot 0, and $P_j(\phi(w),\cdot)=0$ for $j\ge 1$ [14, p. 13]. □

Remark 7.3. One can also obtain Proposition 7.2 from the closed case: the operators with kernels supported in the interior are dense in $\mathcal{I}_\sigma$, and $\mathcal{I}_\sigma$ is an ideal in $\Psi^{\mathbb{Z}}(2X)/\Psi^{-\infty}(2X)$ for the double $2X$ [14, (47)]. In Section 10 this identification shows that Tr_σ is Wodzicki's residue.

8 The boundary symbol algebra and the boundary sequence (MN §3)

Proposition 8.1 (MN Proposition 4). $\mathrm{HH}_k(\mathcal{A}_{\partial,\sigma}) \cong H^{2n-k}(Z \times \mathbb{S}_\sigma \times \mathbb{S}_\partial)$ *and* $\mathrm{HH}^k(\mathcal{A}_{\partial,\sigma}) \cong H^k(Z \times \mathbb{S}_\sigma \times \mathbb{S}_\partial)$.

Proof. $\mathcal{A}_{\partial,\sigma} \cong C^\infty(Z)[[x, q^{-1}]$ consists of formal Laurent series in x of complete symbols over ∂X. The proof of Proposition 7.1 applies with formal Laurent forms on the formal neighbourhood of ${}^cT^*_{\partial X}X \setminus 0$. The homogeneous de Rham cohomology is generated over $H^*(Z)$ by 1, $d\log r$, $d\log x$ and $d\log x \wedge d\log r$ [14, p. 13] (Lemma 6.5 and Proposition 5.10). The total dimension of $Z \times \mathbb{S}_\sigma \times \mathbb{S}_\partial$ is $2n$. Since $Z \times \mathbb{S}_\sigma \times \mathbb{S}_\partial$ is closed, Poincaré duality gives the cohomology statement. □

Proposition 8.2 (MN Proposition 5). *The ideal $\mathcal{I}_\sigma$ is H-unital. Under the isomorphisms of Propositions 7.1, 7.2 and 8.1, the long exact sequence of $0 \to \mathcal{I}_\sigma \to \mathcal{A}_\sigma \to \mathcal{A}_{\partial,\sigma} \to 0$ is the direct sum of the long exact sequence of the pair $(Y \times \mathbb{S}_\sigma, Z \times \mathbb{S}_\sigma)$ (in degrees $2n-k$) and the identity on $H^{2n-k-1}(Z \times \mathbb{S}_\sigma)\, dx/x$:*

1. $\mathrm{HH}_k(\mathcal{I}_\sigma) \to \mathrm{HH}_k(\mathcal{A}_\sigma)$ *is $\iota^* \oplus 0$, relative to absolute cohomology;*
2. $\mathrm{HH}_k(\mathcal{A}_\sigma) \to \mathrm{HH}_k(\mathcal{A}_{\partial,\sigma})$ *is restriction to $Z \times \mathbb{S}_\sigma$ on the first summand and the identity on the summand with dx/x;*
3. *the connecting map $\partial : \mathrm{HH}_k(\mathcal{A}_{\partial,\sigma}) \to \mathrm{HH}_{k-1}(\mathcal{I}_\sigma)$ is a non-zero universal multiple of the de Rham connecting map $H^{2n-k}(Z \times \mathbb{S}_\sigma) \to H^{2n-k+1}_{\mathrm{rel}}(Y \times \mathbb{S}_\sigma)$ on the summand without $d\theta_\partial$, and vanishes on the summand $H^{2n-k-1}(Z \times \mathbb{S}_\sigma)\, d\theta_\partial$.*

Proof. Excision applies because $\mathcal{I}_\sigma$ is H-unital (Proposition 7.2). The maps in (1) and (2) come from algebra morphisms that preserve the order filtration. They therefore induce maps of spectral sequences, and on E^2 they are the maps induced on homogeneous forms by inclusion and restriction. This gives (1) and (2) [14, p. 14].

For (3) we follow MN. By Wodzicki's theorem the connecting map is that of $0 \to \mathcal{K} \to C(\mathcal{A}_\sigma) \to C(\mathcal{A}_{\partial,\sigma}) \to 0$. First let $X = H \times [0,1]$ be a product, with cusp structure given by $x(1-x)$. Then $\mathcal{A}_{\partial,\sigma}$ splits as a sum of the two boundary algebras, $\partial = \partial_0 \oplus \partial_1$, and by symmetry $\partial_1 = -\partial_0$. MN use the subalgebra of operators that commute with $x^2(1-x)^2\partial_x$ to construct a splitting $s_0 : \mathcal{A}^{(0)}_{\partial,\sigma} \to \mathcal{A}_\sigma$ that is an algebra morphism; this is special to the product situation [14, p. 14]. Let $\varphi \in C^\infty([0,1])$ be 1 near 0 and 0 near 1. Define

$$s(f_0 \otimes \cdots \otimes f_k) = s_0(f_0)\varphi \otimes s_0(f_1) \otimes \cdots \otimes s_0(f_k),$$
$$\sigma(f_0 \otimes \cdots \otimes f_k) = s_0(f_0)[\varphi, s_0(f_1)] \otimes s_0(f_2) \otimes \cdots \otimes s_0(f_k).$$

Since s_0 is multiplicative, $bs - sb$ can be computed term by term, and the only defect comes from moving φ past $s_0(f_1)$. Up to the convention for the cyclic face this gives $\sigma = [b, s]$. The chain $\sigma(z)$ lies in $\mathcal{K}$, because $[\varphi, \cdot\,]$ is supported where $d\varphi \neq 0$, in the interior. So σ_* is the connecting map. It lowers order by one, and on leading symbols

$$\chi\big(\sigma(z)\big) = -i\, \iota_{H_\varphi}\chi(z) \qquad \text{for antisymmetric } z,$$

because $\sigma([\varphi, f]) = -i\{\varphi, f\} = -iH_\varphi f$. By Proposition 5.9, $\iota_{H_\varphi}(*\eta) = \pm * (d\varphi \wedge \eta)$. So if $\chi(z) = *\eta$ with $d\eta = 0$, the image class is $\pm i[d\varphi \wedge \eta] = \pm i[d(\varphi\eta)]$, where η is extended to the collar as a

product. This is the de Rham connecting map applied to $[\eta]$. If $\eta = \eta' \wedge dx/x$, then $d\varphi \wedge \eta = 0$, because $d\varphi$ is a multiple of dx. This gives the vanishing on the $d\theta_\partial$ summand.

For general X, operators supported in a collar $V \cong \partial X \times [0,1)$ embed in the algebras of the product $\partial X \times [0,1]$. The result follows from the naturality of the exact sequences and of the isomorphisms of Propositions 7.1, 7.2, 8.1 [14, p. 15]. □

Remark 8.3 (On the splitting s_0). The existence of a multiplicative splitting s_0 is asserted in [14] without details. We were unable to reconstruct it for formal Laurent series in x, since an element of $\mathcal{A}_\sigma$ has only finitely many poles and cannot carry an infinite formal series. What the proof uses is the leading-symbol computation of σ_* on E^2. For that it suffices to have a splitting that is multiplicative modulo terms of lower order and modulo $\mathcal{I}_\sigma$. The constant $\pm i$ in (3) (MN write $\imath$) is irrelevant for everything that follows, which uses only kernels and images. We therefore regard (3) as established up to this point of detail.

9 The indicial algebra $\mathcal{I}_\partial$ (MN §4)

9.1 The model: a boundary point

Let ∂X be a point; for instance $X = [0,1)$. Then $\Psi_{\mathrm{sus}}^{-\infty}(\mathrm{pt})$ is the convolution algebra $\mathcal{S}(\mathbb{R}_t)$. By (10) it is isomorphic to $\mathcal{S}(\mathbb{R}_\tau)$ with pointwise multiplication, and

$$\mathcal{I}_\partial \cong \mathcal{S}(\mathbb{R}_\tau)[[x]], \qquad A \sim \sum_{j \geq -p} x^j a_j(\tau),$$

with the product determined by the twist (Lemma 4.4). Let us make the product explicit to second order. For $a, b \in \mathcal{S}(\mathbb{R}_\tau)$ (elements of x-degree zero),

$$x^k a = a x^k + ik\, x^{k+1} a' + O(x^{k+2}),$$

since $[x^k, a] = \sum_j x^j [x,a] x^{k-1-j}$ and $[x,a] = i x^2 a' + O(x^3)$.

Lemma 9.1 (Commutators in $\mathcal{I}_\partial$)**.** *For $f, g \in \mathcal{S}(\mathbb{R}_\tau)$ and $k \in \mathbb{Z}$:*

1. $[f,g] = 0$ *when both have x-degree zero;*
2. $[x^k f,\, g] = ik\, x^{k+1} g' f + O(x^{k+2})$;
3. $[x^k f,\, x^{-k} g] = ik\, x\, (fg)' + O(x^2)$.

Proof. (1) Translation invariant operators commute. (2) $[x^k f, g] = x^k [f,g] + [x^k, g] f = ikx^{k+1} g' f + O(x^{k+2})$. (3) Write $x^k f x^{-k} = f + ik\, x f' + O(x^2)$ and $x^{-k} g x^k = g - ik\, x g' + O(x^2)$. Then $x^k f x^{-k} g - x^{-k} g x^k f = fg + ikx f' g - gf + ikx g' f + O(x^2)$. □

Proposition 9.2 ($\mathcal{I}_\partial$ for a boundary point)**.** $\mathrm{HH}_0(\mathcal{I}_\partial) \cong \mathbb{C}$, *detected by*

$$\mathrm{Tr}_\partial(A) = \overline{\mathrm{Tr}}(a_1) = \frac{1}{2\pi} \int_{\mathbb{R}} a_1(\tau)\, d\tau,$$

where a_1 is the coefficient of x^1. $\mathrm{HH}_1(\mathcal{I}_\partial) \cong \mathbb{C}$, *generated by the cycle $a \otimes b$ with a, b of x-degree zero and $\int ab'\, d\tau \neq 0$, and detected by the cocycle*

$$(i_{\log x}\, \mathrm{Tr}_\partial)(A, B) = \mathrm{Tr}_\partial(A[\log x, B]), \qquad (i_{\log x}\, \mathrm{Tr}_\partial)(a, b) = \frac{i}{2\pi} \int_{\mathbb{R}} a(\tau) b'(\tau)\, d\tau.$$

$\mathrm{HH}_k(\mathcal{I}_\partial) = 0$ *for $k \geq 2$, and $\mathcal{I}_\partial$ is H-unital.*

Proof. Spectral sequence. Filter by x-degree. The chains are formal sums in increasing powers of x, complete in that direction, and b raises x-degree. The associated graded algebra is the commutative algebra $\mathcal{S}(\mathbb{R}_\tau)[x, x^{-1}]$ (a graded piece for each power of x). By HKR (B5) and Lemma 6.4,

$$E^1 \cong \{\text{forms on } \mathbb{R}_\tau \times (\text{punctured formal disc}), \text{Schwartz in } \tau, \text{Laurent in } x\} \qquad [14, (42)].$$

By Lemma 4.4, the first differential is $-i\delta$ for the Poisson structure $\{x, \tau\} = -x^2$, whose symplectic form is $\omega = d\tau \wedge dx/x^2$ [14, p. 16]. The symplectic star operator in two dimensions is $*1 = \omega$, $*\omega = 1$ and, on 1-forms, $*\, d\tau = d\tau$, $*\, dx = dx$. Indeed $dx \wedge *d\tau = \{x, \tau\}\omega = -x^2\, d\tau \wedge dx/x^2 = dx \wedge d\tau$. So

$$E_k^2 \cong H_{\mathrm{dR}}^{2-k}\big(\text{Schwartz-in-}\tau,\ \text{Laurent-in-}x \text{ forms}\big) \cong H_c^{2-k}(\mathbb{R}_\tau \times \mathbb{S}_\partial) \cong H^{1-k}(\mathbb{S}_\partial),$$

where the Laurent de Rham cohomology in x is spanned by 1 and dx/x (Lemma 6.5) and the Schwartz de Rham cohomology in τ is $H_c^1(\mathbb{R}) = \mathbb{C}$, detected by $\int d\tau$. Concretely:

- in degree $k = 0$, a function $\sum x^j f_j$ goes to $\sum x^j f_j\, d\tau \wedge dx/x^2$, whose class is $\int f_1\, d\tau$ times $[d\tau \wedge dx/x]$;
- in degree $k = 1$, a form $\sum x^j (g_j\, d\tau + h_j\, dx)$ goes to a class determined by $\int g_0\, d\tau$.

The E^2 term sits in x-degree 1 for $k = 0$ and in x-degree 0 for $k = 1$. Higher differentials raise x-degree by $r \geq 2$ and lower k by one, so they vanish. Convergence holds by completeness in the direction of increasing powers of x, and here the corrections never increase pole orders.

The generators. Lemma 9.1(3) shows directly that the x^1-coefficient of a commutator is a derivative, so its integral vanishes. Hence Tr_∂ is a trace. Lemma 9.1(2) shows that $x^{k+1}h$ is a commutator modulo higher order for $k \neq 0$ and h a sum of products $g'f$. Such sums exhaust $\mathcal{S}(\mathbb{R})$. Indeed, if $\int h = 0$ then $h = G'$ with $G \in \mathcal{S}$; writing $G = uv$ with $u, v \in \mathcal{S}$ (every Schwartz function is such a product) gives $h = u'v + uv'$. And $\psi'\psi'$ has positive integral for any non-zero $\psi \in \mathcal{S}$. For degree one: $a \otimes b$ is a cycle by Lemma 9.1(1). By Lemma 4.4(3), $[\log x, b] = -x[t, b] + O(x^2)$ and $\mathrm{In}([t, b]) = -ib'$. Hence

$$\mathrm{Tr}_\partial(a[\log x, b]) = \overline{\mathrm{Tr}}\big(-a[t, b]\big) = -\frac{1}{2\pi}\int a(-ib')\, d\tau = \frac{i}{2\pi}\int ab'\, d\tau.$$

This is non-zero for suitable a, b, so $i_{\log x}\,\mathrm{Tr}_\partial$ is a non-zero class (it is a cocycle by Lemma 5.3).

H-unitality. The same filtration applied to the bar complex has E^1 equal to the bar homology of the commutative graded algebra. That algebra is H-unital: it is $\mathcal{S}(\mathbb{R}) \otimes \mathbb{C}[x, x^{-1}]$, and $\mathcal{S}(\mathbb{R}) = \mathcal{S}(\mathbb{R}) \cdot \mathcal{S}(\mathbb{R})$ has the contraction $(Ef)(\tau', \tau_0, \dots) = \phi(\tau' - \tau_0) f(\tau_0, \dots)$ with $\phi \in \mathcal{S}(\mathbb{R})$ equal to 1 near 0. Hence $E^1 = 0$, and the bar complex of $\mathcal{I}_\partial$ is acyclic by completeness. With $\phi \equiv 1$ near 0, the same E is in fact a contraction for the twisted product: all derivatives of $\phi(\tau' - \tau_0)$ vanish on the diagonal, so the twist corrections of Lemma 4.4 do not contribute to $d_0 E$. □

9.2 General boundary

In a product collar, the variable $x = 1/t$ commutes with operators acting only in y. Hence, componentwise,

$$\mathcal{I}_\partial \cong \bigoplus_{j=1}^{N} \Psi^{-\infty}(H_j) \widehat{\otimes} \big(\mathcal{S}(\mathbb{R}_\tau)[[x]]\big) \qquad [14, \text{p. } 16].$$

Since $\Psi^{-\infty}(H) = C^\infty(H) \widehat{\otimes} C^\infty(H)$ is of the form in Lemma 5.4, Morita invariance for smoothing kernels (Section 5.8) removes it. MN use the Künneth formula [7] for the same purpose.

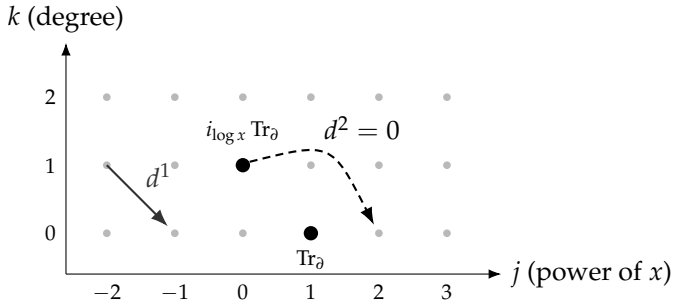


Figure 6: The x-adic spectral sequence of Proposition 9.2 (one boundary point). Gray dots: E^1, forms on $\mathbb{R}_\tau \times \mathbb{S}_\partial$ of fixed x-degree j. The first differential $d^1 = -i\delta$ raises j by one and lowers k by one. E^2 survives only at $(j,k) = (1,0)$, dual to Tr_∂, and at $(0,1)$, dual to $i_{\log x}\,\mathrm{Tr}_\partial$. A differential d^r goes from (j,k) to $(j+r, k-1)$, so for $r \geq 2$ it misses both positions.

Proposition 9.3 (MN Proposition 6, for N components)**.** *$\mathcal{I}_\partial$ is H-unital, and*

$$\mathrm{HH}_k(\mathcal{I}_\partial) \cong \bigoplus_{j=1}^{N} H^{1-k}(\mathbb{S}_\partial).$$

That is, $\mathrm{HH}_0(\mathcal{I}_\partial) \cong \mathrm{HH}_1(\mathcal{I}_\partial) \cong \mathbb{C}^N$*, and* $\mathrm{HH}_k(\mathcal{I}_\partial) = 0$ *for* $k \geq 2$*. The dual bases are* $\{\mathrm{Tr}_{\partial,j}\}$ *and* $\{i_{\log x}\,\mathrm{Tr}_{\partial,j}\}$*, where* $\mathrm{Tr}_{\partial,j}(A) = \overline{\mathrm{Tr}}(A_{(1)}|_{H_j})$ *[14, Lemma 10].*

Correction to MN 9.4 (Several boundary components). [14, (16), Prop. 6, Lemma 12] assert $\mathrm{HH}^k(\mathcal{I}_\partial) \cong \mathbb{C}$ for $k = 0, 1$. This holds only if ∂X is connected, which is never stated, although MN allow disconnected boundary in Appendix A and use two boundary components in the proof of their Proposition 5. In general the answer is $\mathbb{C}^N$ (Proposition 9.3). The consequences for $\mathcal{A}_\partial$ and $\mathcal{A}$ are worked out in Section 12. The algebras $\mathcal{A}_\partial$, $\mathcal{A}_{\partial,\sigma}$ and $\mathcal{I}_\partial$ are direct sums over the components, so for them nothing else changes. For $\mathcal{A}$, the boundary map $\mathrm{HH}^1(\mathcal{I}_\partial) \to \mathrm{HH}^2(\mathcal{A}_\sigma)$ has rank $N-1$, not 0 (Proposition 12.5).

9.3 The residual ideal and the Laurent smoothing operators

Proposition 9.5. *$\mathcal{I}$ is H-unital,* $\mathrm{HH}_0(\mathcal{I}) \cong \mathbb{C}$ *via the trace and* $\mathrm{HH}_k(\mathcal{I}) = 0$ *for* $k > 0$ *[14, end of §4]. Moreover, for* $x^{-\mathbb{Z}}\Psi_c^{-\infty}(X)$*,*

$$\mathrm{HH}_0 \cong \mathbb{C}^N, \qquad \mathrm{HH}_1 \cong \mathbb{C}^{N-1}, \qquad \mathrm{HH}_k = 0 \ (k \geq 2),$$

and the connecting map $\mathrm{HH}_1(\mathcal{I}_\partial) \to \mathrm{HH}_0(\mathcal{I})$ *is dual to* $\mathrm{Tr} \mapsto -\sum_j i_{\log x}\,\mathrm{Tr}_{\partial,j}$*.*

Proof. $\mathcal{I} = \dot{C}^\infty(X) \widehat{\otimes} \dot{C}^\infty(X)$, so the first part is Lemma 5.4. For the second part, apply Lemma 5.2 to $0 \to \mathcal{I} \to x^{-\mathbb{Z}}\Psi_c^{-\infty} \to \mathcal{I}_\partial \to 0$. Extend the trace by the *cusp trace* $\mathrm{Tr}_c(A) = \mathrm{FP}_{z=0}\,\mathrm{Tr}(Ax^z)$, the constant term at $z = 0$. For $\mathrm{Re}\, z$ large,

$$\mathrm{Tr}([A,B]x^z) = \mathrm{Tr}(A[B,x^z]) = -z\ \mathrm{Tr}\big(AL(z)x^z\big), \qquad L(z) = \frac{x^z B x^{-z} - B}{z}, \quad L(0) = [\log x, B].$$

Since $\mathrm{Tr}(AL(z)x^z)$ has at most a simple pole at $z = 0$ with residue $\mathrm{Tr}_\partial(AL(0))$, we get $\mathrm{Tr}_c([A,B]) = -\mathrm{Tr}_\partial(A[\log x, B])$. This vanishes if A or B is in $\mathcal{I}$. So the image of $[\mathrm{Tr}]$ is $-\sum_j i_{\log x}\,\mathrm{Tr}_{\partial,j}$, which is non-zero on every component by Proposition 9.2. The connecting map $\mathrm{HH}_1(\mathcal{I}_\partial) = \mathbb{C}^N \to \mathrm{HH}_0(\mathcal{I}) = \mathbb{C}$ is therefore surjective. The long exact sequence gives the result. □

This is the cusp counterpart of the b-trace defect formula and of part (iii) of the main theorem of [6]. MN do not state it; we include it because it is the simplest place where the rank $N-1$ appears.

10 The residue functionals (MN §5)

Fix x and a positive, elliptic, invertible $Q \in \Psi_c^1(X)$ (Remark 4.6). All the trace functionals of [14] come from one meromorphic function of two variables.

10.1 The double zeta function

Definition 10.1 (MN (43))**.** For $A \in x^p \Psi_c^m(X)$ put

$$Z(A;s,z) = Z_{x,Q}(A;s,z) = \tfrac{1}{2}\big(\operatorname{Tr}(Ax^zQ^{-s}) + \operatorname{Tr}(AQ^{-s}x^z)\big), \qquad Z^L(A;s,z) = \operatorname{Tr}(Ax^zQ^{-s}).$$

By (B3), Z is holomorphic for $\operatorname{Re} z > -p+1$ and $\operatorname{Re} s > m+n$. It extends meromorphically, with at most simple poles on the lines $z = -p+1-j$ and $s = m+n-k$, $j,k \in \mathbb{N}_0$ [14, Lemma 4]. The reason for the shift $-p+1$ in z is that the cusp density is $x^{-2}dx\,\nu_\partial$: the integral $\int_0 x^{z+p}x^{-2}dx$ converges for $\operatorname{Re} z > 1-p$. Near the origin $sz\,Z(A;s,z)$ is holomorphic, and we write [14, (44)]

$$sz\,Z(A;s,z) = \operatorname{Tr}_{\partial,\sigma}(A) + s\,\widehat{\operatorname{Tr}}_\partial(A) + z\,\widehat{\operatorname{Tr}}_\sigma(A) + O\big(|s|^2+|z|^2\big). \tag{17}$$

Thus:

- $\operatorname{Tr}_{\partial,\sigma}(A)$ is the coefficient of $(sz)^{-1}$, the *double residue*;
- $\widehat{\operatorname{Tr}}_\partial(A)$ is the residue in z of the regular part in s; it regularizes the boundary trace by Q;
- $\widehat{\operatorname{Tr}}_\sigma(A)$ is the residue in s of the regular part in z; it is the Wodzicki residue regularized at the boundary by x.

Question 10.2. *Why are two complex parameters needed?*

Answer. An operator in $x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}$ fails to be trace class for two independent reasons: its order may be too large, which is a divergence of the kernel at the diagonal, and its kernel may not decay at the boundary, which is a divergence of the integral over the cusp end $\int^\infty dt$. The factor Q^{-s} cures the first and x^z the second. The two filtrations have their own residual ideals $\mathcal{I}_\sigma$ and $\mathcal{I}_\partial$, and their own residues Tr_σ and $\operatorname{Tr}_\partial$. The double residue $\operatorname{Tr}_{\partial,\sigma}$ is the residue of a residue, and it lives on the doubly quotient algebra $\mathcal{A}_{\partial,\sigma}$. It is the trace dual to the top class of $Z \times \mathbb{S}_\sigma \times \mathbb{S}_\partial$: it picks the coefficient of x^1 (the residue in x, because $x \cdot dx/x^2 = dx/x$) and the homogeneity $-n$ (the residue in the fibre). □

On the diagonal $s = z$ we get $\widetilde{Z}(A;z) = Z(A;z,z)$, with

$$z^2\widetilde{Z}(A;z) = \operatorname{Tr}_{\partial,\sigma}(A) + z\big(\widehat{\operatorname{Tr}}_\partial + \widehat{\operatorname{Tr}}_\sigma\big)(A) + O(z^2).$$

Figure 7 shows the picture.

Lemma 10.3 (MN Lemma 5)**.** *$Z^L(A;s,z) - Z(A;s,z)$ is holomorphic near $(0,0)$. Hence the three functionals in* (17) *are the same for Z, Z^L and $Z^R = \operatorname{Tr}(AQ^{-s}x^z)$.*

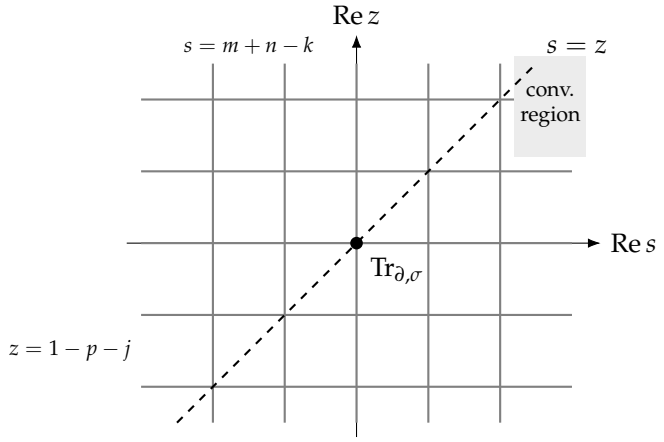


Figure 7: The poles of $Z(A; s, z)$ lie on the gray lines (drawn schematically through the origin; after shifting A by powers of x and Q, lines pass through $(0, 0)$). The crossing at the origin produces the double residue $\mathrm{Tr}_{\partial,\sigma}$. The residue along the line $z = 0$ and the residue along $s = 0$, each regularized in the other variable, give $\widehat{\mathrm{Tr}}_\partial$ and $\widehat{\mathrm{Tr}}_\sigma$. Restricting to the diagonal $s = z$ gives a double pole with leading coefficient $\mathrm{Tr}_{\partial,\sigma}$.

Proof. By the trace property for Re large, $Z^R - Z^L = \mathrm{Tr}(A[Q^{-s}, x^z])$, and

$$[Q^{-s}, x^z] = \big(Q^{-s}x^zQ^sx^{-z} - 1\big)x^zQ^{-s}.$$

The family $Q^{-s}x^zQ^sx^{-z} - 1$ is entire of order 0 and vanishes on both axes $s = 0$ and $z = 0$. So it equals $sz\,M(s, z)$ with M entire of order 0. Then $Z^R - Z^L = sz\,\mathrm{Tr}(AMx^zQ^{-s})$, which has no pole at the origin, since $\mathrm{Tr}(AMx^zQ^{-s})$ has at most simple poles in each variable. Finally $Z = \frac{1}{2}(Z^L + Z^R)$. □

Lemma 10.4 (MN Lemma 6)**.** $\mathrm{Tr}_{\partial,\sigma}$ *is a trace on* $x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X)$*. It does not depend on* x *or* Q*, and it vanishes on* $\mathcal{I}_\sigma + \mathcal{I}_\partial$*. Hence it defines a trace on* $\mathcal{A}$*,* $\mathcal{A}_\sigma$*,* $\mathcal{A}_\partial$ *and* $\mathcal{A}_{\partial,\sigma}$*.*

Proof. Trace. By Lemma 10.3 we may use Z^L. For Re large,

$$Z^L([A, B]) = \mathrm{Tr}(A[B, x^zQ^{-s}]) = \mathrm{Tr}\big(AK(s, z)x^zQ^{-s}\big), \qquad K = B - x^zQ^{-s}BQ^sx^{-z}.$$

The family K is entire, of fixed order, and vanishes at $(0, 0)$. Hence $K = sK_1 + zK_2$ with K_1, K_2 entire. This is Hadamard's lemma in two variables, $K_1 = \int_0^1 \partial_s K(\theta s, \theta z)\,d\theta$, and similarly for K_2. Then

$$sz\,Z^L([A, B]) = s \cdot sz\,\mathrm{Tr}(AK_1x^zQ^{-s}) + z \cdot sz\,\mathrm{Tr}(AK_2x^zQ^{-s}),$$

and both brackets are holomorphic near 0. So the value at $(0, 0)$ is 0.

Independence. If $x' = ax$, then $x'^z = x^z(1 + zB(z))$ with $B(z) = (a^z - 1)/z$ entire. So $sz(Z^L_{x'} - Z^L_x) = z \cdot sz\,\mathrm{Tr}(AB(z)x^zQ^{-s})$ vanishes at the origin. The case of Q is the same, with $Q'^{-s} = (1 + sC(s))Q^{-s}$.

Vanishing. If $A \in x^\infty\Psi_c^{\mathbb{Z}}$, then Ax^z is entire in z with values in a fixed space of flat operators. So Z has no pole in z, and $\mathrm{Tr}_{\partial,\sigma}(A) = 0$. If $A \in x^{-\mathbb{Z}}\Psi_c^{-\infty}$, then Z has no pole in s. □

10.2 The interior residue Tr_σ

If $A \in \mathcal{I}_\sigma$, then Z is entire in z. Its residue in s is $\mathrm{Tr}_\sigma(A) = \mathrm{Res}_{s=0} \mathrm{Tr}(AQ^{-s})$, and $\widehat{\mathrm{Tr}}_\sigma$ restricts to Tr_σ on $\mathcal{I}_\sigma$.

Lemma 10.5 (MN Lemma 7 and Proposition 7)**.** Tr_σ *is the extension by continuity of Wodzicki's residue on the double* $2X$*. If* $\sum_j a_j$ *is the full symbol of* $A \in \mathcal{I}_\sigma$*, then*

$$\mathrm{Tr}_\sigma(A) = (2\pi)^{-n} \int_{{}^cS^*X} a_{-n}\, {}^c\nu,$$

where ${}^c\nu$ *is the contraction of* ω^n *with the radial vector field. Since* ${}^c\nu = x^{-2}\nu'$ *with* ν' *smooth, the same formula defines* Tr_σ *on* $x^p\Psi_c^{\mathbb{Z}}$ *for* $p > 1$*.*

Proof. For A with kernel supported in $(X^\circ)^2$, choose $Q' \in \Psi^1(2X)$ positive and elliptic with $Q'A - QA$ smoothing. Then $\mathrm{Tr}(AQ^{-s}) - \mathrm{Tr}(AQ'^{-s})$ is regular at 0, so $\mathrm{Tr}_\sigma(A) = \mathrm{Tr}_{\mathrm{R}}(A)$. Such operators are dense in $\mathcal{I}_\sigma$, and both sides are continuous. The formula is the local formula (B8). The singular density x^{-2} comes from ω^n, which contains dx/x^2. □

10.3 The double residue

For general $A \in x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X)$ the function $\mathrm{Tr}_\sigma(x^zA)$ is defined for $\mathrm{Re}\, z$ large and is meromorphic in z [14, (52)].

Lemma 10.6 (MN Lemma 8)**.** $\mathrm{Tr}_{\partial,\sigma}(A) = \mathrm{Res}_{z=0} \mathrm{Tr}_\sigma(x^zA)$*. Explicitly,*

$$\mathrm{Tr}_{\partial,\sigma}(A) = (2\pi)^{-n} \int_{{}^cS^*_{\partial X}X} a_{-n,(1)}\, {}^c\nu_\partial,$$

where $a_{-n,(1)}$ *is the term of homogeneity* $-n$ *in the symbol of the coefficient* $A_{(1)}$ *of* x^1 *in* (12)*, and* ${}^c\nu_\partial$ *is the density on* ${}^cS^*_{\partial X}X$ *obtained by contracting* $\omega_\partial^{n-1}d\tau$ *with the radial field on* ${}^cT^*_{\partial X}X = T^*\partial X \times \mathbb{R}_\tau$*.*

Proof. By the definitions, $\mathrm{Tr}_{\partial,\sigma}(A)$ is the z-residue of the s-residue, which is $\mathrm{Res}_z \mathrm{Tr}_\sigma(x^zA)$. Insert the formula of Lemma 10.5: $\int_0 x^z\, x^j\, x^{-2}dx$ has a pole at $z = 0$ only for $j = 1$, with residue 1. □

Example 10.7 (A boundary point)**.** Let $n = 1$ and let $A = x\, c(D_t)$ near the boundary point, with c a classical symbol. The kernel of $xc(D_t)Q^{-s}$ on the diagonal is $x \cdot (2\pi)^{-1} \int c(\tau)\langle\tau\rangle^{-s}d\tau$, and the cusp density is $dt = dx/x^2$. So

$$\mathrm{Res}_z \int_0^1 x^{1+z}\frac{dx}{x^2} = 1, \qquad \mathrm{Tr}_{\partial,\sigma}(A) = \frac{1}{2\pi}\big(c_{-1}(1) + c_{-1}(-1)\big),$$

where $c \sim c_{-1}(\pm1)|\tau|^{-1}$ as $\tau \to \pm\infty$. The two points of $Z = \{\tau = \pm1\}$ contribute separately, *with positive weights*. The density ${}^c\nu_\partial$ is a density, not a form.

Example 10.8 (The four functionals on the half-line)**.** Let $n = 1$, and near the boundary point let $A = \varphi(x)\, x^k\, c(D_t)$, where φ is a cut-off equal to 1 near $x = 0$, c is a classical symbol of order m, and Q is translation invariant near the boundary with $q = \langle\tau\rangle$. The diagonal value of the kernel of $x^kc(D_t)Q^{-s}$ is $x^k(2\pi)^{-1} \int c(\tau)\langle\tau\rangle^{-s}d\tau$. Integrating against $x^z\, dx/x^2$ gives

$$Z(A; s, z) \approx \Big(\int_0^1 \varphi\, x^{k+z-2}\, dx\Big) \cdot \frac{1}{2\pi} \int_{\mathbb{R}} c(\tau)\langle\tau\rangle^{-s}\, d\tau,$$

up to terms that are holomorphic near the origin. The first factor has a pole at $z = 0$ only if $k = 1$, with residue 1. The second factor has a pole at $s = 0$ with residue $(2\pi)^{-1}(c_{-1}(1) + c_{-1}(-1))$, and regular value $\overline{\mathrm{Tr}}(c)$. So, for $k = 1$,

$$\mathrm{Tr}_{\partial,\sigma}(A) = \frac{c_{-1}(1) + c_{-1}(-1)}{2\pi}, \qquad \widehat{\mathrm{Tr}}_{\partial}(A) = \overline{\mathrm{Tr}}(c), \qquad \widehat{\mathrm{Tr}}_{\sigma}(A) = \mathrm{FP}_{z=0}\int_0^1 \varphi\, x^{z-1}dx \cdot \mathrm{Tr}_{\partial,\sigma}(A),$$

and for $k \neq 1$, $\mathrm{Tr}_{\partial,\sigma}(A) = \widehat{\mathrm{Tr}}_{\partial}(A) = 0$. This illustrates Lemma 10.6 and Corollary 10.14. The functional $\widehat{\mathrm{Tr}}_{\sigma}$ is a residue in the fibre variable with a Hadamard finite part in x, and $\widehat{\mathrm{Tr}}_{\partial}$ is a regularized trace in the fibre variable with a residue in x.

Correction to MN 10.9 (Dimension one). For $n = 1$ the set Z_j consists of two points, $\tau = \pm 1$, over each boundary point. Then $\mathrm{Tr}_{\partial,\sigma}$ splits as $\mathrm{Tr}^{+}_{\partial,\sigma} + \mathrm{Tr}^{-}_{\partial,\sigma}$, and each summand is a trace (the proof of Lemma 10.4 applies with a cut-off in τ). Hence $\mathrm{HH}^0(\mathcal{A}_\sigma) \cong \mathrm{HH}^0(\mathcal{A}_{\partial,\sigma}) \cong \mathbb{C}^{2N}$, and similarly $\mathrm{HH}^0(\mathcal{I}_\sigma) \cong H^0(Y \times \mathbb{S}_\sigma) \cong \mathbb{C}^2$ for connected X. The statements "HH^0 is one-dimensional" after [14, (16)] and [14, Lemma 12] require $n \geq 2$, exactly as Wodzicki's uniqueness of the residue does. The cohomological formulas of Theorem 1.1 for $\mathcal{I}_\sigma$, $\mathcal{A}_\sigma$, $\mathcal{A}_{\partial,\sigma}$ remain true for $n = 1$, with Z_j disconnected.

10.4 The regularized interior residue $\widehat{\mathrm{Tr}}_\sigma$

By Lemma 10.6, $\widehat{\mathrm{Tr}}_\sigma(A)$ is the regular value at $z = 0$ of $\mathrm{Tr}_\sigma(x^z A)$. Equivalently, it is the Hadamard finite part [14, (55)]:

$$\widehat{\mathrm{Tr}}_\sigma(A) = \lim_{\epsilon \downarrow 0}\Big((2\pi)^{-n} \int_{{}^{c}S^*X \cap \{x > \epsilon\}} a_{-n}\, {}^{c}\nu + (\log \epsilon)\, \mathrm{Tr}_{\partial,\sigma}(A) + \sum_{l>0} \gamma_l \epsilon^{-l} \Big). \tag{18}$$

The sign of the logarithm is right, since $\int_\epsilon^1 x^{-1}dx = -\log\epsilon$.

Lemma 10.10 (MN Lemma 9)**.** *$\widehat{\mathrm{Tr}}_\sigma$ vanishes on $\mathcal{I}_\partial$, does not depend on Q, and for $x' = ax$*

$$\widehat{\mathrm{Tr}}_\sigma(A; x') - \widehat{\mathrm{Tr}}_\sigma(A; x) = \mathrm{Tr}_{\partial,\sigma}\big(A \log a\big).$$

Proof. The residue Tr_σ vanishes on smoothing operators and does not depend on Q. For the change of x, $\mathrm{Tr}_\sigma((x'^z - x^z)A) = \mathrm{Tr}_\sigma(z x^z B(z))$ with $B(z) = z^{-1}(a^z - 1)A$ entire (MN print A^z for a^z). Its value at $z = 0$ is $\mathrm{Res}_z\, \mathrm{Tr}_\sigma(x^z B(z)) = \mathrm{Tr}_{\partial,\sigma}(B(0)) = \mathrm{Tr}_{\partial,\sigma}(A \log a)$. □

10.5 The boundary trace and the regularized suspended trace

Lemma 10.11 (MN Lemma 10)**.** *On $\mathcal{I}_\partial$, $\mathrm{Tr}_\partial(A) = \mathrm{Res}_{z=0}\mathrm{Tr}(Ax^z) = \overline{\mathrm{Tr}}(A_{(1)})$, where $\overline{\mathrm{Tr}}(B) = (2\pi)^{-1}\int \mathrm{Tr}\,\mathrm{In}(B,\tau)\, d\tau$ [14, (57)]. It is a trace and does not depend on x or Q.*

Proof. On the diagonal of X_c^2 the kernel of A is $B\nu$ with $\nu = x^{-2}\nu'$, and $B \in x^{-\mathbb{Z}}C^\infty(X)$. Write $B \sim \sum_k x^k B_{(k)}$. The residue at $z = 0$ of $\int x^z B\, x^{-2} dx\, \nu_\partial$ comes from $k = 1$ and equals $\int_{\partial X} B_{(1)}\nu_\partial$. This is the kernel of $A_{(1)}$ at $\{0\} \times \Delta_{\partial X}$ integrated over the boundary, which is $\overline{\mathrm{Tr}}(A_{(1)})$ by Fourier inversion. The trace property was proved in Proposition 9.2. Independence is as in Lemma 10.4. □

Melrose's regularization of $\overline{\mathrm{Tr}}$ to all of $\Psi^{\mathbb{Z}}_{\mathrm{sus}}(H)$ [11] works as follows [14, (58)–(59)]. For $B \in \Psi^m_{\mathrm{sus}}(H)$ and $p \geq m + \dim H + 2$, the function $h_p(\tau) = \mathrm{Tr}\big(\partial_\tau^p B(\tau)\big)$ is defined. Integrate it $p+1$ times:

$$\int_{-T}\int_0^{\tau_p} \cdots \int_0^{\tau_1} h_p\, d\tau\, d\tau_1 \cdots d\tau_p \sim \sum_j c_j T^j + P(T)\log T,$$

and put $\overline{\mathrm{Tr}}(B) = c_0/2\pi$. Changing p changes the inner integral by a polynomial, which does not affect c_0. The functional $\overline{\mathrm{Tr}}$ is a trace on $\Psi^{\mathbb{Z}}_{\mathrm{sus}}(H)$: for p large, $h_p([A,B]) = \sum_l \binom{p}{l} \mathrm{Tr}[\partial^l A, \partial^{p-l} B] = 0$, because each commutator is of total order less than $-\dim H$.

Proposition 10.12 (MN Proposition 8)**.** *Let $Q_0 = (D_t^2 + \Delta_H + 1)^{1/2}$. For $B \in \Psi^m_{\mathrm{sus}}(H)$, the function $\overline{\mathrm{Tr}}(BQ_0^{-s})$ is meromorphic, and*

$$s\,\overline{\mathrm{Tr}}(BQ_0^{-s}) = \widetilde{\mathrm{Tr}}(B) + s\,\overline{\mathrm{Tr}}(B) + O(s^2),$$

where $\widetilde{\mathrm{Tr}}(B) = (2\pi)^{-n} \int b_{-n}$ is the Wodzicki residue of B regarded as an operator on $\mathbb{R} \times H$ ($n = \dim H + 1$).

Proof. Apply the definition with BQ_0^{-s} in place of B. The function $h_p(\tau; s)$ is entire in s and has an expansion $\sum_k d_k^{\pm}(s)|\tau|^{m-p+n-1-s-k}$ as $\tau \to \pm\infty$, with holomorphic coefficients. Integrating $p+1$ times produces a simple pole at $s = 0$ from the terms that become $|T|^{-s}$. Its residue is the sum of the coefficients of $|\tau|^{-1}$ in $\mathrm{Tr}\, B(\tau)$ at $\pm\infty$, divided by 2π; this is the residue of B on $\mathbb{R} \times H$. The regular value is $\overline{\mathrm{Tr}}(B)$. □

Remark 10.13 (The functional in MN Proposition 8). MN write the residue as $\widetilde{\mathrm{Tr}}(B) = \overline{\mathrm{Tr}}([t, B])$. This cannot be right as it stands. Take $H = \mathrm{pt}$ and $B = \langle D_t \rangle^{-1}$. The residue is $(2\pi)^{-1}(1+1) = 1/\pi$, whereas $\mathrm{In}([t,B]) = -i\partial_\tau \langle\tau\rangle^{-1}$ is odd, so $\overline{\mathrm{Tr}}([t,B]) = 0$. (In general $\overline{\mathrm{Tr}}([t,B])$ picks out the difference of the $|\tau|^0$-coefficients at $\pm\infty$, not the $|\tau|^{-1}$ coefficients.) Only the residue description is used later, through Corollary 10.14, so we take it as the definition.

Corollary 10.14 (MN Corollaries 1 and 2)**.** *For $A \in x^{-\mathbb{Z}}\Psi^{\mathbb{Z}}_c(X)$:*

1. *$\mathrm{Tr}_{\partial,\sigma}(A) = \widetilde{\mathrm{Tr}}(A_{(1)})$, the residue of $\mathrm{Tr}_\partial(AQ^{-s})$ at $s = 0$ [14, (61)];*
2. *if Q is asymptotically translation invariant, $\mathrm{In}(Q) = Q_0$, then $\widehat{\mathrm{Tr}}_\partial(A) = \overline{\mathrm{Tr}}(A_{(1)})$.*

Proof. If $\mathrm{In}(Q) = Q_0$ and Q is translation invariant to all orders, then $(AQ^{-s})_{(1)} = A_{(1)}Q_0^{-s}$, and Proposition 10.12 applies. In general the residue does not depend on Q. □

Lemma 10.15 (MN Lemma 11)**.** *For two regularizers Q, Q',*

$$\widehat{\mathrm{Tr}}_\partial(A; Q') - \widehat{\mathrm{Tr}}_\partial(A; Q) = -\,\mathrm{Tr}_{\partial,\sigma}\big(A \log(Q'/Q)\big), \qquad \log(Q'/Q) = \frac{d}{ds} Q'^s Q^{-s}\Big|_{s=0} \in \Psi^0_c(X).$$

Proof. Write $Q'^{-s} = (1 + sB(s))Q^{-s}$ with $B(s) = s^{-1}(Q'^{-s}Q^s - 1)$, so that $B(0) = -\log(Q'/Q)$. Then $\mathrm{Tr}_\partial(AQ'^{-s}) - \mathrm{Tr}_\partial(AQ^{-s}) = s\,\mathrm{Tr}_\partial(AB(s)Q^{-s})$. Its value at $s = 0$ is $\mathrm{Res}_s\, \mathrm{Tr}_\partial(AB(0)Q^{-s}) = \mathrm{Tr}_{\partial,\sigma}(AB(0))$. □

Thus $\widehat{\mathrm{Tr}}_\partial$ does not depend on x but does depend on Q, while $\widehat{\mathrm{Tr}}_\sigma$ does not depend on Q but does depend on x.

11 Exterior derivations and explicit cocycles (MN §§6–7)

11.1 The two derivations

The derivations are [14, (63)–(64)]

$$D_{\log Q} A = [\log Q, A] = \frac{d}{ds} Q^s A Q^{-s}\Big|_{s=0}, \qquad D_{\log x} A = [\log x, A] = \frac{d}{dz} x^z A x^{-z}\Big|_{z=0}.$$

On symbols, $\sigma([\log Q, A]) = -i\{\log q, a\}$ lowers the order by one. Similarly $\sigma([\log x, A]) = -i\{\log x, a\} = i\,x\,\partial_\tau a$, which lowers the order by one and raises the power of x by one, by (7). Both derivations preserve $\mathcal{I}$, $\mathcal{I}_\sigma$, $\mathcal{I}_\partial$. Changing Q or x changes them by inner derivations, so the induced maps $i_{\log Q}$, $i_{\log x}$ are canonical (Lemma 5.3).

Lemma 11.1. $\mathrm{Tr}_{\partial,\sigma}([\log Q, B]) = \mathrm{Tr}_{\partial,\sigma}([\log x, B]) = 0$ *for all* B *[14, (84)].*

Proof. Put $Q^u B Q^{-u} = B + uM(u)$, with M entire and $M(0) = [\log Q, B]$. By the trace property, for Re large,

$$\mathrm{Tr}\big(Q^u B Q^{-u} x^z Q^{-s}\big) = \mathrm{Tr}\big(B x^z (x^{-z} Q^{-u} x^z Q^u) Q^{-s}\big).$$

The family $x^{-z}Q^{-u}x^zQ^u - 1$ vanishes for $u = 0$ and for $z = 0$, so it equals $uz\,N(u,z)$ with N entire. Comparing the two sides, $u\,Z^L(M(u); s, z) = uz\,\mathrm{Tr}(Bx^zNQ^{-s})$, so $Z^L(M(0); s, z) = z\,\mathrm{Tr}(Bx^zN(0,z)Q^{-s})$. Multiplying by sz and evaluating at the origin gives $\mathrm{Tr}_{\partial,\sigma}([\log Q, B]) = 0$. For $\log x$, put $x^w B x^{-w} = B + wL(w)$ and use $\mathrm{Tr}(x^wBx^{-w}x^zQ^{-s}) = \mathrm{Tr}\big(Bx^z(x^{-w}Q^{-s}x^wQ^s)Q^{-s}\big)$, where now $x^{-w}Q^{-s}x^wQ^s - 1 = ws\,P(w,s)$. The same argument gives $\mathrm{Tr}_{\partial,\sigma}([\log x, B]) = 0$. This is [14, (84)], where only the combination $[\log x - \log Q, B]$ is treated. □

11.2 Traces generate degree zero

Lemma 11.2 (MN Lemma 12, for N components, $n \geq 2$)**.** $\mathrm{HH}^0(\mathcal{I}_\sigma) = \mathbb{C}\ \mathrm{Tr}_\sigma$. *Moreover* $\mathrm{HH}^0(\mathcal{A}_\sigma)$ *and* $\mathrm{HH}^0(\mathcal{A}_{\partial,\sigma})$ *are spanned by the double residues* $\mathrm{Tr}_{\partial,\sigma,j}$ *of the components (the restriction of* $\mathrm{Tr}_{\partial,\sigma}$ *to* Z_j*), and* $\mathrm{HH}^0(\mathcal{I}_\partial)$ *is spanned by the* $\mathrm{Tr}_{\partial,j}$*.*

Proof. By Theorem 1.1 and Propositions 7.1–9.3, the dimensions are $1, N, N, N$. The functionals listed are traces (Lemmas 10.4, 10.5, 10.11, applied componentwise), and they are linearly independent: by Example 10.7 and Proposition 9.2 each one is non-zero on an element supported near a single component. □

11.3 The isomorphisms Φ_j and the action of the derivations

Using Lefschetz and Poincaré duality as in Proposition 7.1, the homology computations give isomorphisms [14, (65)–(68)]

$$\Phi_1 : H^*(Z \times \mathbb{S}_\sigma) \oplus H^*_{\mathrm{rel}}(Y \times \mathbb{S}_\sigma) \to \mathrm{HH}^*(\mathcal{A}_\sigma), \qquad \Phi_2 : H^*(Y \times \mathbb{S}_\sigma) \to \mathrm{HH}^*(\mathcal{I}_\sigma),$$
$$\Phi_3 : H^*(Z \times \mathbb{S}_\sigma \times \mathbb{S}_\partial) \to \mathrm{HH}^*(\mathcal{A}_{\partial,\sigma}), \qquad \Phi_4 : H^*(\mathbb{S}_\partial)^{\oplus N} \to \mathrm{HH}^*(\mathcal{I}_\partial).$$

(MN print S^*M for Y in (65).) The class $\Phi(\eta)$ is characterized by its values on antisymmetric chains [14, (69)]:

$$\Phi([\eta])\big(\mathsf{A}(f_0 \otimes \cdots \otimes f_p)\big) = \int *\big(\chi_0(f_0 \otimes \cdots \otimes f_p)\big) \wedge \eta, \tag{19}$$

where χ_0 is the HKR map on principal symbols and A is antisymmetrization. MN normalize the circle classes as $d\theta_\sigma = [i\,dr/r]$ and $d\theta_\partial = [i\,dx/x]$. The factor i absorbs the $-i$ in $\sigma([\cdot,\cdot]) = -i\{\cdot,\cdot\}$.

Proposition 11.3 (MN Proposition 9)**.** $i_{\log Q}\Phi_j(\xi) = \Phi_j(d\theta_\sigma \wedge \xi)$ *for* $j = 1, 2, 3$*, and* $i_{\log x}\Phi_j(\xi) = \Phi_j(d\theta_\partial \wedge \xi)$ *for* $j = 3, 4$*.*

Proof. Choose Q with principal symbol r. On leading symbols, $i_{\log Q}$ replaces $f_0 \otimes f_1 \otimes \cdots$ by $f_0\sigma([\log Q, f_1]) \otimes \cdots = -if_0H_{\log r}f_1 \otimes \cdots$. After antisymmetrization this gives

$$\chi_0 \circ i_{\log Q} \circ \mathsf{A} = -i\,\iota_{H_{\log r}} \circ \chi_0 \circ \mathsf{A}.$$

By Proposition 5.9, $\iota_{H_f} * \eta' = \pm * (df \wedge \eta')$. Inserting this into (19) moves $d\log r$ onto η, which is wedge with $d\theta_\sigma$ up to MN's normalization. The same argument with $\log x$ gives $d\theta_\partial$ on the boundary algebras, where x is a variable of the symplectic space. On $\mathcal{A}_\sigma$ and $\mathcal{I}_\sigma$ the derivation $i_{\log x}$ is not of this form, because $\log x$ is singular at the boundary. □

11.4 Explicit cocycles

For $\mathcal{I}_\partial$ the degree one generator is $i_{\log x}\operatorname{Tr}_\partial$. Its value on translation invariant elements was computed in Proposition 9.2. In general, by Lemma 4.4(3),

$$(i_{\log x}\operatorname{Tr}_\partial)(B,C) = -\sum_{k\geq 1}\frac{1}{k}\operatorname{Tr}_\partial\big(B\,x^k\operatorname{ad}_t^k(C)\big),$$

and only finitely many terms contribute to the coefficient of x^1. [14, (70)] gives a kernel formula for this cocycle. We were not able to match its index conventions with [14, Lemma 10]: with $A \sim \sum x^{-j}A_j$ the products $B_{-j}C_{j-k}$ in (70) have total power x^k, not x^1. We therefore use the formula above.

For the symbol algebras, MN describe cocycles $\tau_b(g_0,\dots,g_p) = \operatorname{Tr}_\sigma(f_0g_0[f_1,g_1]\cdots[f_p,g_p])$ built from chains $b = f_0\otimes\cdots\otimes f_p$. When $\chi_0(b)$ represents η, the class of τ_b is $\Phi_2([\eta])$. Composing with $i_{\log Q}$ and $i_{\log x}$ gives the classes with $d\theta_\sigma$ and $d\theta_\partial$ [14, p. 24]. We do not need these beyond degree two.

12 The homology of $\mathcal{A}_\partial$ and $\mathcal{A}$ (MN §8)

12.1 The boundary maps of the traces: MN (17)

Lemma 12.1 (MN Lemma 13). *For $A, B \in x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X)$,*

$$\widehat{\operatorname{Tr}}_\partial([A,B]) = \operatorname{Tr}_{\partial,\sigma}(A[\log Q, B]) = (i_{\log Q}\operatorname{Tr}_{\partial,\sigma})(A,B),$$
$$\widehat{\operatorname{Tr}}_\sigma([A,B]) = -\operatorname{Tr}_{\partial,\sigma}(A[\log x, B]) = -(i_{\log x}\operatorname{Tr}_{\partial,\sigma})(A,B).$$

(MN print $i_{\log r}$ in the second line.)

Proof. First line. $\widehat{\operatorname{Tr}}_\partial(C)$ is the regular value at $s=0$ of $\operatorname{Tr}_\partial(CQ^{-s})$, by Lemma 10.3. For $C=[A,B]$,

$$\operatorname{Tr}_\partial([A,B]Q^{-s}) = \operatorname{Tr}_\partial(A[B,Q^{-s}]) = s\,\operatorname{Tr}_\partial\big(AM(s)Q^{-s}\big), \qquad M(s) = \frac{B - Q^{-s}BQ^s}{s},$$

using that $\operatorname{Tr}_\partial = \operatorname{Res}_z\operatorname{Tr}(\cdot\,x^z)$ is a trace on operators of order below $-n$. Since $M(0) = [\log Q, B]$, the regular value at $s=0$ is $\operatorname{Res}_s\operatorname{Tr}_\partial(A[\log Q,B]Q^{-s}) = \operatorname{Tr}_{\partial,\sigma}(A[\log Q,B])$ by Corollary 10.14.

Second line. $\widehat{\operatorname{Tr}}_\sigma(C)$ is the regular value at $z=0$ of $\operatorname{Tr}_\sigma(Cx^z)$. Since Tr_σ is a trace,

$$\operatorname{Tr}_\sigma([A,B]x^z) = \operatorname{Tr}_\sigma(A[B,x^z]) = -z\,\operatorname{Tr}_\sigma\big(AL(z)x^z\big), \qquad L(z) = \frac{x^zBx^{-z} - B}{z},$$

with $L(0) = [\log x, B]$. The regular value is $-\operatorname{Res}_z\operatorname{Tr}_\sigma(A[\log x,B]x^z) = -\operatorname{Tr}_{\partial,\sigma}(A[\log x,B])$ by Lemma 10.6. □

Both right-hand sides vanish if A or B lies in $\mathcal{I}_\sigma + \mathcal{I}_\partial$. By Lemma 5.2, Lemma 12.1 computes the connecting maps of $0\to\mathcal{I}_\partial\to\mathcal{A}_\partial\to\mathcal{A}_{\partial,\sigma}\to 0$ and $0\to\mathcal{I}_\sigma\to\mathcal{A}_\sigma\to\mathcal{A}_{\partial,\sigma}\to 0$ on the traces [14, (17)–(19)]. The corresponding statements hold for each boundary component separately.

12.2 The algebra $\mathcal{A}_\partial$

Since $\mathcal{I}_\partial$ is H-unital, $0 \to \mathcal{I}_\partial \to \mathcal{A}_\partial \to \mathcal{A}_{\partial,\sigma} \to 0$ gives a long exact sequence. By Proposition 9.3 it reduces in cohomology to [14, (71)–(72)]

$$0 \to \mathrm{HH}^0(\mathcal{A}_{\partial,\sigma}) \to \mathrm{HH}^0(\mathcal{A}_\partial) \to \mathrm{HH}^0(\mathcal{I}_\partial) \xrightarrow{\partial} \mathrm{HH}^1(\mathcal{A}_{\partial,\sigma}) \to \mathrm{HH}^1(\mathcal{A}_\partial) \to \cdots$$
$$\cdots \to \mathrm{HH}^1(\mathcal{I}_\partial) \xrightarrow{\partial} \mathrm{HH}^2(\mathcal{A}_{\partial,\sigma}) \to \mathrm{HH}^2(\mathcal{A}_\partial) \to 0,$$

together with $\mathrm{HH}^k(\mathcal{A}_\partial) \cong \mathrm{HH}^k(\mathcal{A}_{\partial,\sigma})$ for $k \geq 3$.

Proposition 12.2 (MN Proposition 10)**.** *For each component H_j,*

$$\partial[\mathrm{Tr}_{\partial,j}] = [i_{\log Q}\,\mathrm{Tr}_{\partial,\sigma,j}], \qquad \partial[i_{\log x}\,\mathrm{Tr}_{\partial,j}] = -[i_{\log x} i_{\log Q}\,\mathrm{Tr}_{\partial,\sigma,j}],$$

and both maps ∂ are injective.

Proof. The first formula is Lemma 12.1 with Lemma 5.2, applied to the extension $\widehat{\mathrm{Tr}}_\partial$ of Tr_∂. For the second, the cochain $i_{\log x}\widehat{\mathrm{Tr}}_\partial$ extends $i_{\log x}\,\mathrm{Tr}_\partial$, and by Lemma 5.3

$$b(i_{\log x}\widehat{\mathrm{Tr}}_\partial) = -i_{\log x}(b\widehat{\mathrm{Tr}}_\partial) = -i_{\log x} i_{\log Q}\,\mathrm{Tr}_{\partial,\sigma}\,.$$

This vanishes when one argument is in $\mathcal{I}_\partial$, because $\mathrm{Tr}_{\partial,\sigma}$ vanishes on $\mathcal{I}_\partial$ and $\mathcal{I}_\partial$ is an ideal preserved by both derivations. By Proposition 11.3, under Φ_3 the two images are $d\theta_\sigma \cdot 1_{Z_j}$ and $-d\theta_\partial \wedge d\theta_\sigma \cdot 1_{Z_j}$. These are linearly independent over j, and non-zero because each Z_j is non-empty. □

Proposition 12.3 (MN Proposition 11)**.** $\mathrm{HH}^k(\mathcal{A}_\partial) \cong \mathrm{HH}^k(\mathcal{A}_{\partial,\sigma})/\mathbb{C}^N$ *for $k = 1, 2$, and* $\mathrm{HH}^k(\mathcal{A}_\partial) \cong \mathrm{HH}^k(\mathcal{A}_{\partial,\sigma})$ *otherwise. Dually,* $\mathrm{HH}_k(\mathcal{A}_\partial)$ *is the kernel of* $\partial : \mathrm{HH}_k(\mathcal{A}_{\partial,\sigma}) \to \mathrm{HH}_{k-1}(\mathcal{I}_\partial)$ *for $k = 1, 2$ [14, (75)].*

Proof. Insert the injectivity of Proposition 12.2 into the exact sequence. □

12.3 The algebra $\mathcal{A}$

Now consider $0 \to \mathcal{I}_\partial \to \mathcal{A} \to \mathcal{A}_\sigma \to 0$ [14, (77)]. The quotient maps give a morphism of extensions to $0 \to \mathcal{I}_\partial \to \mathcal{A}_\partial \to \mathcal{A}_{\partial,\sigma} \to 0$. By naturality of the connecting maps,

$$\partial_{\mathcal{A}} = p^* \circ \partial_{\mathcal{A}_\partial} : \mathrm{HH}^i(\mathcal{I}_\partial) \to \mathrm{HH}^{i+1}(\mathcal{A}_\sigma), \qquad p : \mathcal{A}_\sigma \to \mathcal{A}_{\partial,\sigma}. \tag{20}$$

Lemma 12.4 (The map p^*)**.** *Write* $\mathrm{HH}^k(\mathcal{A}_{\partial,\sigma}) \cong H^k(Z \times \mathbb{S}_\sigma) \oplus H^{k-1}(Z \times \mathbb{S}_\sigma)\, d\theta_\partial$ *and* $\mathrm{HH}^k(\mathcal{A}_\sigma) \cong H^k_{\mathrm{rel}}(Y \times \mathbb{S}_\sigma) \oplus H^k(Z \times \mathbb{S}_\sigma)$*. Then p^* is the identity from the first summand to the second, and $\pm\delta$ from the $d\theta_\partial$-summand to H^k_{rel}. Here $\delta : H^{k-1}(Z \times \mathbb{S}_\sigma) \to H^k_{\mathrm{rel}}(Y \times \mathbb{S}_\sigma)$ is the de Rham connecting map.*

Proof. p^* is dual to $p_* : \mathrm{HH}_k(\mathcal{A}_\sigma) \to \mathrm{HH}_k(\mathcal{A}_{\partial,\sigma})$, which is restriction plus the identity on the $d\theta_\partial$-part (Proposition 8.2). Under Poincaré duality on $Z \times \mathbb{S}_\sigma \times \mathbb{S}_\partial$ the summand without $d\theta_\partial$ in homology pairs with the summand with $d\theta_\partial$ in cohomology, and conversely. The dual of restriction $H^{2n-k}(Y \times \mathbb{S}_\sigma) \to H^{2n-k}(Z \times \mathbb{S}_\sigma)$ under Lefschetz and Poincaré duality is $\pm\delta$, by Stokes' theorem: $\int_Z a|_Z \wedge b = \pm \int_Y a \wedge d(\chi_0 \tilde{b})$ for closed a and b. This is the content of the computation in the proof of [14, Prop. 12], where it is written $i \oplus (-1)^q \partial$. □

Proposition 12.5 (MN Proposition 12, corrected)**.** *Let X be connected and $n \geq 2$. Then:*

1. $\partial_{\mathcal{A}}[\mathrm{Tr}_{\partial,j}] = [i_{\log Q}\,\mathrm{Tr}_{\partial,\sigma,j}] \leftrightarrow d\theta_\sigma \cdot 1_{Z_j} \in H^1(Z \times \mathbb{S}_\sigma)$, *and* $\partial_{\mathcal{A}}$ *is injective on* $\mathrm{HH}^0(\mathcal{I}_\partial)$;

2. $\partial_{\mathcal{A}}[i_{\log x}\,\mathrm{Tr}_{\partial,j}] = \pm\Phi_1\big(\delta(d\theta_\sigma \cdot 1_{Z_j})\big)$, *and* $\partial_{\mathcal{A}}$ *has rank* $N-1$ *on* $\mathrm{HH}^1(\mathcal{I}_\partial)$, *with kernel spanned by* $\sum_j i_{\log x}\,\mathrm{Tr}_{\partial,j} = i_{\log x}\,\mathrm{Tr}_\partial$.

Consequently $\mathrm{HH}^k(\mathcal{A})$ *is as stated in Theorem 1.1.*

Proof. (1) By (20), Proposition 12.2 and Lemma 12.4, the image is the class $d\theta_\sigma \cdot 1_{Z_j}$ in the summand $H^1(Z \times \mathbb{S}_\sigma)$ of $\mathrm{HH}^1(\mathcal{A}_\sigma)$. These are independent.

(2) Similarly the image is $\pm\delta(d\theta_\sigma \cdot 1_{Z_j})$. By exactness of $H^1(Y \times \mathbb{S}_\sigma) \to H^1(Z \times \mathbb{S}_\sigma) \xrightarrow{\delta} H^2_{\mathrm{rel}}(Y \times \mathbb{S}_\sigma)$, a combination $\sum_j c_j d\theta_\sigma 1_{Z_j}$ lies in $\ker\delta$ if and only if it is the restriction of a class on $Y \times \mathbb{S}_\sigma$. By the Künneth decomposition $H^1(Y \times \mathbb{S}_\sigma) = H^1(Y) \oplus H^0(Y)\,d\theta_\sigma$, and restriction preserves it. Since Y is connected ($n \geq 2$), $H^0(Y) = \mathbb{C}$, and the restriction of $d\theta_\sigma$ is $\sum_j d\theta_\sigma 1_{Z_j}$. So $\ker\delta \cap \mathrm{span}\{d\theta_\sigma 1_{Z_j}\}$ is the line $c_1 = \cdots = c_N$, and the rank is $N-1$.

For the groups, insert (1) and (2) into the exact sequence of $0 \to \mathcal{I}_\partial \to \mathcal{A} \to \mathcal{A}_\sigma \to 0$:

$$\mathrm{HH}^0(\mathcal{A}) \cong \mathrm{HH}^0(\mathcal{A}_\sigma) \cong \mathbb{C}^N,$$
$$\mathrm{HH}^1(\mathcal{A}) \cong \mathrm{HH}^1(\mathcal{A}_\sigma)/\mathbb{C}^N \oplus \ker\big(\partial_{\mathcal{A}}|_{\mathrm{HH}^1(\mathcal{I}_\partial)}\big) \cong H^1_{\mathrm{rel}}(Y \times \mathbb{S}_\sigma) \oplus H^1(Z \times \mathbb{S}_\sigma)/\mathbb{C}^N \oplus \mathbb{C},$$
$$\mathrm{HH}^2(\mathcal{A}) \cong \mathrm{HH}^2(\mathcal{A}_\sigma)/\mathbb{C}^{N-1}, \qquad \mathrm{HH}^k(\mathcal{A}) \cong \mathrm{HH}^k(\mathcal{A}_\sigma),\ k \geq 3.$$

The splitting in degree one depends on a choice of extension of $i_{\log x}\,\mathrm{Tr}_\partial$ to $\mathcal{A}$. One choice is $-\,\mathrm{IF}$ (Section 15). □

Correction to MN 12.6 (MN Proposition 12). MN find $\partial[i_{\log x}\,\mathrm{Tr}_\partial] = 0$, because "the form $d\theta_\sigma$ has an extension to the whole manifold" [14, p. 27]. That argument is correct for the *sum* $\sum_j d\theta_\sigma 1_{Z_j}$, which is all there is when $N = 1$. For $N \geq 2$ the individual classes $d\theta_\sigma 1_{Z_j}$ do not extend, and $\mathrm{HH}^2(\mathcal{A})$ is smaller than [14, (16)] claims by $N-1$ dimensions. The same rank appears in the b-calculus [6, Remark "Several boundary components"]. If X has c components, each with non-empty boundary, the rank is $N-c$.

Discussion. **How the index formula comes out of the boundary maps.** The index functional is a class in $\mathrm{HH}^1(\mathcal{A})$ whose restrictions to the two ideals are known: $i_{\log Q}\,\mathrm{Tr}_\sigma$ on $\mathcal{I}_\sigma$ (Atiyah–Singer) and $-i_{\log x}\,\mathrm{Tr}_\partial$ on $\mathcal{I}_\partial$ (Toeplitz). A pair of cocycles on the two ideals extends to $\mathcal{A}$ only if their images in $\mathrm{HH}^2(\mathcal{A}_{\partial,\sigma})$ agree. By Lemma 12.1 and Proposition 12.2, both images equal $[i_{\log x} i_{\log Q}\,\mathrm{Tr}_{\partial,\sigma}]$. The extension is unique up to the image of $\mathrm{HH}^1(\mathcal{A}_\sigma)$ and $\mathrm{HH}^1(\mathcal{A}_\partial)$, and analytic continuation picks one: the regularized value of $\widetilde{Z}([A,B];z)$. Evaluating it on $[A] \otimes [A^{-1}]$ splits the index into an interior symbolic term and a boundary indicial term (Section 15).

12.4 The full operator algebra

The ideal $\mathcal{I}$ is H-unital with $\mathrm{HH}_*(\mathcal{I}) = \mathbb{C}$ in degree 0 (Proposition 9.5). The connecting map $\mathrm{HH}_1(\mathcal{A}) \to \mathrm{HH}_0(\mathcal{I}) = \mathbb{C}$ is the index functional, which is non-zero: there are fully elliptic operators of non-zero index (Example 16.2). Hence $\mathrm{HH}_k(x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X)) \cong \mathrm{HH}_k(\mathcal{A})$ for $k \neq 1$, and $\mathrm{HH}_1(x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X)) = \ker \mathrm{IF} \subset \mathrm{HH}_1(\mathcal{A})$. This completes the proof of Theorem 1.1.

12.5 Examples

Example 12.7 (The cylinder)**.** Let $X = [0,1] \times S^1$, so $n = 2$ and $N = 2$. The cusp cotangent bundle is trivial, so $Y \cong [0,1] \times T^2$ and $Z = \{0,1\} \times T^2$. The cohomology of tori and $H^k_{\mathrm{rel}}([0,1] \times M) \cong H^{k-1}(M)$ give the following dimensions, for $k = 0,1,2,3,4$:

	$k=0$	1	2	3	4
$\mathrm{HH}^k(\mathcal{I}_\sigma) = H^k(T^3)$	1	3	3	1	0
$\mathrm{HH}^k(\mathcal{A}_\sigma)$	2	7	9	5	1
$\mathrm{HH}^k(\mathcal{A}_{\partial,\sigma}) = 2H^k(T^4)$	2	8	12	8	2
$\mathrm{HH}^k(\mathcal{I}_\partial)$	2	2	0	0	0
$\mathrm{HH}^k(\mathcal{A}_\partial)$	2	6	10	8	2
$\mathrm{HH}^k(\mathcal{A})$, Theorem 1.1	2	6	8	5	1
$\mathrm{HH}^k(\mathcal{A})$, [14, (16)] read literally	2	7	9	5	1

In degree one the correct quotient is by $\mathbb{C}^N$, not $\mathbb{C}$, and in degree two by the rank $N-1=1$ image of Correction 12.6. In degree zero the formula of [14, (16)] already gives 2, although the text asserts that HH^0 is one-dimensional. Note that MN's own product computation in the proof of [14, Prop. 5] (with $\partial_1 = -\partial_0$) is the cylinder case, and is consistent with these numbers.

Example 12.8 (The interval)**.** Let $X=[0,1]$, so $n=1$ and $N=2$. Then $Y=[0,1]\times\{\pm\}$, two intervals, and Z consists of four points. The same bookkeeping (Correction 10.9) gives

$$\dim \mathrm{HH}^k(\mathcal{A}_\sigma) = 4,6,2, \qquad \dim \mathrm{HH}^k(\mathcal{A}_\partial) = 4,6,2, \qquad \dim \mathrm{HH}^k(\mathcal{A}) = 4,5,1 \qquad (k=0,1,2).$$

Here $\partial_{\mathcal{A}}$ on $\mathrm{HH}^1(\mathcal{I}_\partial)=\mathbb{C}^2$ has rank one: the two classes $\delta(d\theta_\sigma(1_{j,+}+1_{j,-}))$, $j=0,1$, are opposite, since $1_{0,\pm}+1_{1,\pm}$ is the restriction of $1_{Y_\pm}$.

13 Morita invariance and bundles (MN §9)

To have operators of non-zero index one needs bundles. Embed $E\subset X\times\mathbb{C}^N$, and let $e\in M_N(C^\infty(X))$ be the orthogonal projection onto E. Then $x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X;E) = eM_N(x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X))e$, and similarly for all the quotient algebras and ideals. The idempotent e lies in no proper two-sided ideal, so each corner algebra is Morita equivalent to the trivial-bundle algebra, and the Hochschild groups are the same (B6) [14, Prop. 13]. In the symbol algebras this replaces $x^{-\mathbb{Z}}\mathcal{G}({}^cT^*X)$ by $x^{-\mathbb{Z}}\mathcal{G}({}^cT^*X;\hom E)$, and the HKR map uses the fibre trace. All trace functionals extend by composing with the fibre trace tr. From now on all algebras act on sections of E.

14 The boundary index (MN §10)

Definition 14.1 (MN (78))**.** For $A,B\in x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X;E)$ put $\mathrm{Bf}(A,B)=\lim_{z\to 0} z\,\widetilde{Z}([A,B];z)$.

The limit exists because $\mathrm{Tr}_{\partial,\sigma}([A,B])=0$ (Lemma 10.4), so the double pole of $\widetilde{Z}([A,B];z)$ is absent.

Proposition 14.2 (MN Proposition 14)**.** *For all A,B,*

$$\mathrm{Bf}(A,B) = \mathrm{Tr}_{\partial,\sigma}\big(B[\log x - \log Q, A]\big) = \big(-i_{\log x}\,\mathrm{Tr}_{\partial,\sigma} + i_{\log Q}\,\mathrm{Tr}_{\partial,\sigma}\big)(A,B) = \big(\widehat{\mathrm{Tr}}_\partial + \widehat{\mathrm{Tr}}_\sigma\big)([A,B]).$$

Bf *vanishes if A or B lies in $\mathcal{I}_\sigma$ or $\mathcal{I}_\partial$. It is a Hochschild* 1*-cocycle on $\mathcal{A}_{\partial,\sigma}$, and under Φ_3 it corresponds to $-d\theta_\partial + d\theta_\sigma$ on each component Z_j.*

Proof. By the trace property, $\operatorname{Tr}([A,B]Q^{-z}x^z) = \operatorname{Tr}\big([A,B](1+z^2C(z))x^zQ^{-z}\big)$, where

$$C(z) = z^{-2}\big(Q^{-z}x^zQ^zx^{-z} - 1\big)$$

is entire because the bracket vanishes on both axes. Hence [14, (79)]

$$\widetilde{Z}([A,B];z) = \operatorname{Tr}([A,B]x^zQ^{-z}) + \tfrac{1}{2}z^2\operatorname{Tr}\big([A,B]C(z)x^zQ^{-z}\big).$$

The second term has at most a simple pole, so it does not contribute to Bf. By the Baker–Campbell–Hausdorff formula, $C(0) = -[\log Q, \log x] = [\log x, \log Q]$. MN write $2[\log Q, \log x]$ in [14, (80)], but only the fact that C is entire matters. Next,

$$\operatorname{Tr}([A,B]x^zQ^{-z}) = \operatorname{Tr}\big(B[x^zQ^{-z}, A]\big),$$
$$[x^zQ^{-z}, A] = (x^zAx^{-z} - A)x^zQ^{-z} + x^z(Q^{-z}AQ^z - A)Q^{-z}.$$

Write $zL_1(z) = x^zAx^{-z} - A$ and $zL_2(z) = A - Q^zAQ^{-z}$, so that $L_1(0) = [\log x, A]$ and $L_2(0) = -[\log Q, A]$. Since $x^z(Q^{-z}AQ^z - A)Q^{-z} = x^zQ^{-z}\cdot zL_2(z)$, the trace property gives [14, (81)]

$$z\operatorname{Tr}([A,B]x^zQ^{-z}) = z^2\operatorname{Tr}\big(BL_1(z)x^zQ^{-z}\big) + z^2\operatorname{Tr}\big(L_2(z)Bx^zQ^{-z}\big).$$

At $z = 0$ the right side is $\operatorname{Tr}_{\partial,\sigma}(BL_1(0) + L_2(0)B) = \operatorname{Tr}_{\partial,\sigma}(B[\log x, A] - [\log Q, A]B)$. This equals $\operatorname{Tr}_{\partial,\sigma}(B[\log x - \log Q, A])$, since $\operatorname{Tr}_{\partial,\sigma}$ is a trace. The second expression follows from $\operatorname{Tr}_{\partial,\sigma}(A[D,B]) = -\operatorname{Tr}_{\partial,\sigma}([D,A]B) = -\operatorname{Tr}_{\partial,\sigma}(B[D,A])$ (Lemma 11.1). The third is Lemma 12.1. Vanishing on $\mathcal{I}_\sigma + \mathcal{I}_\partial$ holds because $\operatorname{Tr}_{\partial,\sigma}$ vanishes there and the ideals are preserved by the derivations. The cohomology class is given by Proposition 11.3. □

Since $\mathcal{I}_\sigma \cdot \mathcal{I}_\partial \subset \mathcal{I}_\sigma \cap \mathcal{I}_\partial = 0$ in $\mathcal{A}$, the ideal $\mathcal{I}_\sigma + \mathcal{I}_\partial = \mathcal{I}_\sigma \oplus \mathcal{I}_\partial$ is H-unital. The last expression in Proposition 14.2 shows, by Lemma 5.2, that Bf is the image of the trace $\operatorname{Tr}_\partial + \operatorname{Tr}_\sigma$ under the connecting map of $0 \to \mathcal{I}_\sigma + \mathcal{I}_\partial \to \mathcal{A} \to \mathcal{A}_{\partial,\sigma} \to 0$ [14, p. 28].

Proposition 14.3 (MN Proposition 15)**.** *For invertible $A \in \mathcal{A}_{\partial,\sigma}$ put* c-ind$(A) = \operatorname{Bf}(A, A^{-1})$. *This is constant along smooth paths of invertible elements, and it vanishes if A is the image of an invertible element of $\mathcal{A}$.*

Proof. Let $\widetilde{D} = [\log x - \log Q, \cdot\,]$ [14, (83)]. By Proposition 14.2, $\operatorname{Bf}(A,B) = \operatorname{Tr}_{\partial,\sigma}(B\widetilde{D}(A))$. For a smooth path A_t,

$$\frac{d}{dt}\operatorname{Bf}(A_t, A_t^{-1}) = \operatorname{Tr}_{\partial,\sigma}\big(A^{-1}\widetilde{D}(\dot{A}) - A^{-1}\dot{A}A^{-1}\widetilde{D}(A)\big) = \operatorname{Tr}_{\partial,\sigma}\big(\widetilde{D}(A^{-1}\dot{A})\big),$$

because $\widetilde{D}(A^{-1}\dot{A}) = -A^{-1}\widetilde{D}(A)A^{-1}\dot{A} + A^{-1}\widetilde{D}(\dot{A})$ and $\operatorname{Tr}_{\partial,\sigma}$ is a trace. The right side vanishes by Lemma 11.1. If A lifts to $A' \in x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}$ with an inverse B' modulo $\mathcal{I}$, then $[A', B'] \in \mathcal{I}$. So $\widetilde{Z}([A', B'];z)$ is entire and Bf $= 0$. □

Invertibility in $\mathcal{A}_{\partial,\sigma}$ means ellipticity of the full symbol near the boundary, in the Laurent sense. Liftability to an invertible element of $\mathcal{A}$ means that, in addition, a representative is elliptic in the interior and has invertible indicial family. So c-ind is an obstruction to completing boundary symbol data to a fully elliptic operator [14, Prop. 15].

Example 14.4 (A boundary index in dimension one)**.** Let $n = 1$ with one boundary point. The algebra $\mathcal{A}_{\partial,\sigma} = C^\infty(\{\pm\})[[x, q^{-1}]$ splits as $\mathcal{A}^+_{\partial,\sigma} \oplus \mathcal{A}^-_{\partial,\sigma}$ by the central idempotents $1_\pm$, the symbols that are 1 for $\pm\tau > 0$. For $k \in \mathbb{Z}$ let

$$A = x^k 1_+ + 1_-, \qquad A^{-1} = x^{-k} 1_+ + 1_-.$$

We compute $\mathrm{Bf}(A, A^{-1}) = \mathrm{Tr}_{\partial,\sigma}(A^{-1}[\log x - \log Q, A])$ with Q translation invariant near the boundary and $q = |\tau|$ there. First, $[\log x, A] = 0$ in $\mathcal{A}_{\partial,\sigma}$: $\log x$ commutes with x^k, and $[\log x, 1_\pm]$ has symbol $ix\partial_\tau 1_\pm = 0$. Second, the star product expansion gives

$$[\log Q, x^k] = -i\{\log|\tau|, x^k\} + (\text{order} \le -2) = -ik\,\frac{x^{k+1}}{\tau} + (\text{order} \le -2),$$

by (7). Hence the coefficient of x^1 in $-A^{-1}[\log Q, A]$ has homogeneous part of degree -1 equal to $ik\tau^{-1}1_+$. By Example 10.7,

$$\text{c-ind}(A) = \frac{1}{2\pi}\Big(\frac{ik}{1}\Big) = \frac{ik}{2\pi}.$$

For $A = x^k$ (that is, $x^k 1_+ + x^k 1_-$) the two points contribute $ik/2\pi$ and $ik/(2\pi\cdot(-1))$, which cancel, as they must: x^k is invertible in $\mathcal{A}$. For $k \neq 0$, $x^k 1_+ + 1_-$ cannot be lifted to a fully elliptic operator. Indeed any lift has leading indicial coefficient vanishing on a half-line of τ.

15 The index cocycle and the index formula (MN §§11–12)

15.1 The index cocycle

Definition 15.1 (MN (85))**.** For $A, B \in x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X; E)$ let $\mathrm{IF}(A, B)$ be the regular value at $z = 0$ of $\widetilde{Z}([A, B]; z)$, that is, $\mathrm{IF}(A, B) = \lim_{z\to 0}\big(\widetilde{Z}([A, B]; z) - z^{-1}\,\mathrm{Bf}(A, B)\big)$.

Lemma 15.2 (MN Lemma 14; Fedosov's formula)**.** *If A is fully elliptic and $B \in x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X; E)$ satisfies $[B] = [A]^{-1}$ in $\mathcal{A}$, then A is Fredholm on $\dot{C}^\infty(X; E)$ and* $\mathrm{Ind}(A) = \mathrm{IF}(A, B)$.

Proof. By (B2), A has a generalized inverse B_0 with $AB_0 = 1 - \Pi_1$ and $B_0A = 1 - \Pi_0$, where Π_0 projects onto $\ker A$ and Π_1 onto a complement of the range. Both are finite rank and lie in $\mathcal{I}$. Then

$$\mathrm{Ind}\, A = \mathrm{Tr}\,\Pi_0 - \mathrm{Tr}\,\Pi_1 = \mathrm{Tr}(AB_0 - B_0A) = \mathrm{Tr}([A, B_0]).$$

Since $[A, B_0] \in \mathcal{I}$, the function $\widetilde{Z}([A, B_0]; z)$ is entire with value $\mathrm{Tr}([A, B_0])$ at 0. So $\mathrm{Bf}(A, B_0) = 0$ and $\mathrm{IF}(A, B_0) = \mathrm{Ind}\, A$. Finally IF vanishes when either argument lies in $\mathcal{I}$, and $B - B_0 \in \mathcal{I}$. □

Lemma 15.2 says that IF represents the image of [Tr] under the connecting map of $0 \to \mathcal{I} \to x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}} \to \mathcal{A} \to 0$. This is Lemma 5.2 with the extension $\widehat{\mathrm{Tr}}(C) =$ the regular value of $\widetilde{Z}(C; z)$. Nistor's compatibility theorem [16] identifies this connecting map with the index map in K-theory [14, p. 29].

Proposition 15.3 (MN Proposition 16)**.** IF *descends to a* 1*-cocycle on $\mathcal{A}$, and*

$$\mathrm{IF}(A, B) = \tfrac{1}{2}\big(\widehat{\mathrm{Tr}}_\partial + \widehat{\mathrm{Tr}}_\sigma\big)\big(B\widetilde{D}(A) + \widetilde{D}(A)B\big), \qquad \widetilde{D} = [\log x - \log Q, \cdot\,].$$

Proof. By definition $\mathrm{IF}(A,B) = \frac{d}{dz}\big(z\widetilde{Z}([A,B];z)\big)|_{z=0}$. Symmetrizing the computation in the proof of Proposition 14.2 gives [14, (88)]

$$2\widetilde{Z}([A,B];z) = z\,\mathrm{Tr}\big((BL_1 + L_2B)x^zQ^{-z}\big) + z\,\mathrm{Tr}\big((BL_3 + L_4B)Q^{-z}x^z\big),$$

with $zL_3(z) = Q^{-z}AQ^z - A$ and $zL_4(z) = A - x^{-z}Ax^z$. Hence $L_1(0) = L_4(0) = [\log x, A]$ and $L_2(0) = L_3(0) = -[\log Q, A]$. Expanding $x^{\pm z}Ax^{\mp z}$ to second order shows $L_1'(0) + L_4'(0) = 0$, and similarly $L_2'(0) + L_3'(0) = 0$. By (17) and Lemma 10.3, both $z^2\,\mathrm{Tr}(Cx^zQ^{-z})$ and $z^2\,\mathrm{Tr}(CQ^{-z}x^z)$ equal $\mathrm{Tr}_{\partial,\sigma}(C) + z(\widehat{\mathrm{Tr}}_\partial + \widehat{\mathrm{Tr}}_\sigma)(C) + O(z^2)$, also for holomorphic families $C(z)$. Therefore

$$\begin{aligned}2\,\mathrm{IF}(A,B) = {}& \big(\widehat{\mathrm{Tr}}_\partial + \widehat{\mathrm{Tr}}_\sigma\big)\big(BL_1(0) + L_2(0)B + BL_3(0) + L_4(0)B\big)\\ &+ \mathrm{Tr}_{\partial,\sigma}\big(B(L_1' + L_3')(0) + (L_2' + L_4')(0)B\big).\end{aligned}$$

Since $\mathrm{Tr}_{\partial,\sigma}$ is a trace, the last term equals $\mathrm{Tr}_{\partial,\sigma}\big(B(L_1' + L_4' + L_2' + L_3')(0)\big) = 0$. The first term is $(\widehat{\mathrm{Tr}}_\partial + \widehat{\mathrm{Tr}}_\sigma)(B\widetilde{D}(A) + \widetilde{D}(A)B)$. Finally IF vanishes if an argument lies in $\mathcal{I}$. □

Correction to MN 15.4 (The sign of $\widetilde{D}$). [14, (21)] defines $\widetilde{D}(A) = [\log Q - \log x, A]$, while [14, (83)] and the proof of [14, Prop. 16] use $[\log x - \log Q, A]$. The computation above, which follows (88) of [14], gives $[\log x - \log Q, A]$. The examples in Section 16 confirm this sign independently.

15.2 Restrictions to the ideals

On $\mathcal{I}_\partial$ the Q-regularization is unnecessary. By Proposition 9.5, $\mathrm{IF}|_{\mathcal{I}_\partial} = -i_{\log x}\,\mathrm{Tr}_\partial$, the *Toeplitz cocycle*. On $\mathcal{I}_\sigma$ the x-regularization is unnecessary, and the closed-manifold computation (Proposition 3.4) gives $\mathrm{IF}|_{\mathcal{I}_\sigma} = i_{\log Q}\,\mathrm{Tr}_\sigma$, the *Atiyah–Singer cocycle*. So IF extends $-i_{\log x}\,\mathrm{Tr}_\partial$ from $\mathcal{I}_\partial$ to $\mathcal{A}$. This is why $\partial[i_{\log x}\,\mathrm{Tr}_\partial] = 0$ (Proposition 12.5). The compatibility $\partial[i_{\log Q}\,\mathrm{Tr}_\sigma] = \partial[i_{\log x}\,\mathrm{Tr}_\partial]$ in $\mathrm{HH}^2(\mathcal{A}_{\partial,\sigma})$ is the statement that both equal $[i_{\log x}i_{\log Q}\,\mathrm{Tr}_{\partial,\sigma}]$ (Proposition 12.2) [14, p. 30].

15.3 The four terms

For invertible $A \in \mathcal{A}$ MN define [14, (89)–(92)]

$$\begin{aligned}\overline{\eta}(A) &= -\widehat{\mathrm{Tr}}_\partial\big(A^{-1}[\log x, A] + [\log x, A]A^{-1}\big), & \overline{\mathrm{AS}}(A) &= \tfrac{1}{2}\widehat{\mathrm{Tr}}_\sigma\big(A^{-1}[\log Q, A] + [\log Q, A]A^{-1}\big),\\ \beta_Q(A) &= \tfrac{1}{2}\widehat{\mathrm{Tr}}_\partial\big(A^{-1}[\log Q, A] + [\log Q, A]A^{-1}\big), & \gamma(A) &= \tfrac{1}{2}\widehat{\mathrm{Tr}}_\sigma\big(A^{-1}[\log x, A] + [\log x, A]A^{-1}\big).\end{aligned}$$

(MN call β_Q simply β.) Inserting $B = A^{-1}$ in Proposition 15.3 and using $\widetilde{D} = D_{\log x} - D_{\log Q}$ gives the following.

Theorem 15.5 (Index theorem, MN (93), corrected)**.** *For every fully elliptic $A \in x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X;E)$,*

$$\mathrm{Ind}(A) = -\overline{\mathrm{AS}}(A) - \tfrac{1}{2}\overline{\eta}(A) - \beta_Q(A) + \gamma(A).$$

Equivalently, with $\mathrm{AS}(A) = -\overline{\mathrm{AS}}(A) = \frac{1}{2}\widehat{\mathrm{Tr}}_\sigma\big(A[\log Q, A^{-1}] + [\log Q, A^{-1}]A\big)$ *and* $\beta(A) = -\beta_Q(A)$,

$$\mathrm{Ind}(A) = \mathrm{AS}(A) - \tfrac{1}{2}\overline{\eta}(A) + \beta(A) + \gamma(A).$$

Proof. By Proposition 15.3 with $B = A^{-1}$,

$$\begin{aligned}2\,\mathrm{IF}(A, A^{-1}) = {}& (\widehat{\mathrm{Tr}}_\partial + \widehat{\mathrm{Tr}}_\sigma)\big(A^{-1}[\log x, A] + [\log x, A]A^{-1}\big)\\ &- (\widehat{\mathrm{Tr}}_\partial + \widehat{\mathrm{Tr}}_\sigma)\big(A^{-1}[\log Q, A] + [\log Q, A]A^{-1}\big).\end{aligned}$$

In order, the four pieces are $-\overline{\eta}$, 2γ, $-2\beta_Q$ and $-2\overline{\mathrm{AS}}$. For the second form note that

$$A[\log Q, A^{-1}] = -[\log Q, A]A^{-1}, \qquad [\log Q, A^{-1}]A = -A^{-1}[\log Q, A].$$ □

Correction to MN 15.6 (Signs in MN (93), (22)–(23), (98)–(100)). MN state $\mathrm{Ind}(A) = \overline{\mathrm{AS}}(A) - \frac{1}{2}\overline{\eta}(A) + \beta(A) + \gamma(A)$ with $\overline{\mathrm{AS}}$, β as in [14, (90), (91)]. With those definitions the signs of $\overline{\mathrm{AS}}$ and β are wrong. The following checks support this. The second and third are not independent of each other, since the circle is a closed manifold, and the sign of β rests on the derivation alone:

1. The derivation from [14, (87)] (Proposition 15.3) gives Theorem 15.5.
2. On a closed manifold $\overline{\eta} = \beta = \gamma = 0$, and MN's own [14, (3), (5)] give (Proposition 3.4)
$$\mathrm{Ind}\, A = \mathrm{Tr}_{\mathrm{R}}\big(A[\log Q, A^{-1}]\big) = -\overline{\mathrm{AS}}(A).$$
3. On the circle, Section 3.3 gives $\mathrm{Ind}\, A = -1$ and $\overline{\mathrm{AS}}(A) = +1$ for the Toeplitz-type operator there.

Consequently the symbolic term in [14, (23)] and [14, (100)] should be $\mathrm{Tr}_{\mathrm{R}}(A[\log Q, A^{-1}]) = -\mathrm{Tr}_{\mathrm{R}}(A^{-1}[\log Q, A])$. The sign of the eta term is correct; see Example 16.2. Identifying $\overline{\eta}$ with the eta invariant of [11] assumes that its parametrization of the suspended variable agrees with MN Definition 4; reversing that parametrization reverses the sign. The interval example and the Atiyah–Patodi–Singer convention support the choice made here.

15.4 The eta term

Proposition 15.7 (MN Proposition 17)**.** *Let $A \in \Psi_c^m(X;E)$ be fully elliptic, and let Q be asymptotically translation invariant. Let $A_0 = \mathrm{In}(A) \in \Psi_{\mathrm{sus}}^m(\partial X;E)$ and $t = 1/x$. Then*
$$\overline{\eta}(A) = 2\,\overline{\mathrm{Tr}}\big(A_0^{-1}[t, A_0]\big) = \frac{1}{\pi i}\overline{\int_{\mathbb{R}} \mathrm{Tr}\big(A_0(\tau)^{-1}\partial_\tau A_0(\tau)\big)d\tau},$$
which is Melrose's eta invariant $\eta(A_0)$ of [11], (B7).

Proof. A parametrix $B \in \Psi_c^{-m}$ has $\mathrm{In}(B) = A_0^{-1}$. By Lemma 4.4(3), $[\log x, A] \in x\Psi_c^{m-1}$ with $\mathrm{In}(x^{-1}[\log x, A]) = -[t, A_0]$ [14, (95)]. Since x commutes with everything to leading order, the coefficient of x^1 in $B[\log x, A]$ is $-A_0^{-1}[t, A_0]$, and in $[\log x, A]B$ it is $-[t, A_0]A_0^{-1}$. By Corollary 10.14(2),
$$\widehat{\mathrm{Tr}}_\partial(B[\log x, A]) = -\overline{\mathrm{Tr}}(A_0^{-1}[t, A_0]) = -\overline{\mathrm{Tr}}([t, A_0]A_0^{-1}) = \widehat{\mathrm{Tr}}_\partial([\log x, A]B),$$
using that $\overline{\mathrm{Tr}}$ is a trace on $\Psi_{\mathrm{sus}}^{\mathbb{Z}}$ (B7). Hence $\overline{\eta}(A) = -(-2\overline{\mathrm{Tr}}(A_0^{-1}[t, A_0])) = 2\overline{\mathrm{Tr}}(A_0^{-1}[t, A_0])$. By (10), $\mathrm{In}([t, A_0]) = -i\partial_\tau A_0$. So $2\overline{\mathrm{Tr}}(A_0^{-1}[t, A_0]) = 2(2\pi)^{-1}\overline{\int}\,\mathrm{Tr}(A_0^{-1}(-i)\partial_\tau A_0)\,d\tau$, which is the second expression. □

Correction to MN 15.8 (The proof of MN Proposition 17). The proof in [14] correctly obtains $\widehat{\mathrm{Tr}}_\partial(B[\log x, A]) = -\overline{\mathrm{Tr}}(A_0^{-1}[t, A_0]) = \widehat{\mathrm{Tr}}_\partial([\log x, A]B)$, but then concludes $\overline{\eta}(A) = -\overline{\mathrm{Tr}}(A_0^{-1}[t, A_0])$. With definition [14, (89)] the conclusion is $2\overline{\mathrm{Tr}}(A_0^{-1}[t, A_0])$, in agreement with [14, (24)], where $[\log x, A_0]$ should read $[t, A_0]$. Also, the proof cites [14, (54)]; what is needed is [14, Cor. 2], which assumes that Q is asymptotically translation invariant. For general Q, Lemma 10.15 gives $\overline{\eta}(A;Q) = \eta(A_0) + \mathrm{Tr}_{\partial,\sigma}\big((A^{-1}[\log x, A] + [\log x, A]A^{-1})\log(Q/Q_{\mathrm{ti}})\big)$, with Q_{ti} asymptotically translation invariant. The correction is compensated by β_Q, since the index does not depend on Q.

Corollary 15.9 (MN Corollary 3)**.** *Suppose that near the boundary $A = \sigma(-\partial_t + B_\partial)$, with σ a unitary bundle map and B_∂ self-adjoint on ∂X with $0 \notin \mathrm{spec}\, B_\partial$. This is the APS normal form $\sigma(\partial_u + B_\partial)$ in the inward normal variable $u = -t$. Then $\overline{\eta}(A) = \eta_{\mathrm{APS}}(B_\partial) = \sum_\lambda \mathrm{sgn}\,\lambda$ (regularized).*

Proof. $\mathrm{In}(A,\tau) = e^{i\tau t}Ae^{-i\tau t} = \sigma(i\tau + B_\partial)$. On the λ-eigenspace, $\mathrm{Tr}(A_0^{-1}\partial_\tau A_0) = i/(\lambda + i\tau) = (i\lambda + \tau)/(\lambda^2 + \tau^2)$. The odd part integrates to zero over $[-T,T]$, and $(\pi i)^{-1}\int i\lambda/(\lambda^2+\tau^2)\,d\tau = \operatorname{sgn}\lambda$. The regularized sum is the eta invariant (B7). □

15.5 Simplifications

Lemma 15.10 (MN Lemma 15)**.** *Let $A \in \Psi_c^m(X;E)$ be fully elliptic and translation invariant in a collar $\{x' < \epsilon\}$ with respect to a normal fibration. Choose $x = f(x')$ with $f(x') = x'$ near 0 and $x \equiv 1$ outside $\{x' < \epsilon\}$. Then $\gamma(A) = 0$.*

Proof. In the collar write $\log x = G(t')$ with $t' = 1/x'$. So $G(t') = -\log t'$ for large t', and G is constant ($= 0$) near $t' = 1/\epsilon$. For translation invariant A, Taylor expansion of $G(t) - G(t')$ about the diagonal gives

$$[\log x, A] \sim \sum_{j\ge 1} G^{(j)}(t')R_j,$$

with R_j translation invariant of order $m - j$ [14, p. 32]. The same holds for $B[\log x, A]$ and $[\log x, A]B$, with other translation invariant coefficients R_j', since a parametrix B can also be taken translation invariant there. The residue density of $G^{(j)}(t')R_j'$ is $G^{(j)}(t')$ times the constant residue density of R_j'. By (18), $\widehat{\mathrm{Tr}}_\sigma$ integrates it against $dt'\nu_\partial$ and takes the Hadamard finite part as $x \to 0$. For $j \ge 2$, $\int G^{(j)}dt' = [G^{(j-1)}]$ vanishes at both ends, since $G^{(j-1)} = O(t'^{1-j})$ at infinity and G is constant at the inner end. For $j = 1$, $\int_{1/\epsilon}^{T} G'\,dt' = G(T) - G(1/\epsilon) = \log x|_{x=1/T} - 0$, which is purely logarithmic and is removed by the finite part. Outside the collar $d\log x = 0$, so there is no further contribution. □

Lemma 15.11 (MN Lemma 16)**.** *Suppose that $A \in \Psi_c^m(X;E)$, $m > 0$, is fully elliptic and normal to first order at the boundary: $[A^*, A] \in x^2\Psi_c^{2m-1}(X;E)$. Let $Q = (A^*A + 1)^{1/2m}$. Then $[\log Q, A] \in x^2\Psi_c^{m-1}$, so $\beta_Q(A) = 0$, and the integral defining $\overline{\mathrm{AS}}(A)$ converges without regularization.*

Proof. $[A, A^*A] = [A, A^*]A \in x^2\Psi_c^{3m-1}$, so by the functional calculus A commutes with functions of A^*A modulo x^2. Hence $[\log Q, A] \in x^2\Psi_c^{m-1}$. The functional $\widehat{\mathrm{Tr}}_\partial$ only sees the coefficient of x^1: for $C \in x^2\Psi_c$ the function $Z(C;s,z)$ has no pole at $z = 0$, since its poles lie at $z = -1 - j$. So $\beta_Q(A) = 0$. Similarly $\mathrm{Tr}_\sigma(x^z C)$ is regular at $z = 0$ for $C \in x^2\Psi_c$, so no finite part is needed in (18). For operators of order ≤ 0, replace 1 by an operator that is positive in the interior and vanishes near the boundary [14, p. 33]. □

Theorem 15.12 (Reduced index theorem, MN (99)–(100), corrected)**.** *Let $A \in \Psi_c^m(X;E)$ be fully elliptic and translation invariant near the boundary, with normal indicial family A_0. Let Q be positive, translation invariant near the boundary and commuting with A there. Then*

$$\mathrm{Ind}(A) = \mathrm{AS}(A) - \tfrac{1}{2}\eta(A_0), \qquad \mathrm{AS}(A) = \mathrm{Tr}_{\mathrm{R}}\big(A[\log Q, A^{-1}]\big) = -\mathrm{Tr}_{\mathrm{R}}\big(A^{-1}[\log Q, A]\big),$$

where the residue integral converges and $\eta(A_0)$ is as in Proposition 15.7.

Proof. By Lemmas 15.10 and 15.11, $\gamma = \beta_Q = 0$, and by Proposition 15.7, $\overline{\eta}(A) = \eta(A_0)$. The two terms of AS agree, because $\widehat{\mathrm{Tr}}_\sigma([C_1, C_2]) = -\mathrm{Tr}_{\partial,\sigma}(C_1[\log x, C_2])$ vanishes for $C_2 = [\log Q, A] \in x^2\Psi_c$ (Lemma 12.1). □

16 Worked examples of the index formula

Example 16.1 (The closed circle)**.** This is Section 3.3. With $A = e^{i\theta}\Pi + (1-\Pi)$, $\operatorname{Ind} A = -1 = \mathrm{AS}(A)$ and $\overline{\mathrm{AS}}(A) = +1$. This fixes the sign of the symbolic term.

Example 16.2 (First order operators on an interval)**.** Let X be the compactification of $\mathbb{R}_u$ at both ends by $1/|u|$. So $X \cong [-1,1]$, $n=1$, $N=2$, and near the ends $x = 1/|u|$ and $t = |u|$. Let $a \in C^\infty(\mathbb{R})$ with $a(u) = a_\pm$ for $\pm u > R$, and consider

$$A = \partial_u + a(u) \in \Psi^1_c(X).$$

Full ellipticity. The principal symbol is $i\xi \neq 0$. At the right end $t = u$, so $A = \partial_t + a_+$ and $\operatorname{In}(A,\tau) = e^{i\tau t}(\partial_t + a_+)e^{-i\tau t} = a_+ - i\tau$. At the left end $t = -u$, so $A = -\partial_t + a_-$ and $\operatorname{In}(A,\tau) = a_- + i\tau$. Both are invertible for all real τ if and only if $a_+ a_- \neq 0$.

Index directly. A solution of $Au = 0$ is a multiple of $\exp(-\int a)$. It lies in $\dot{C}^\infty(X)$ (rapid decay as $|u| \to \infty$) if and only if $a_+ > 0 > a_-$. For the formal adjoint $-\partial_u + a$, the condition is $a_+ < 0 < a_-$. Hence

$$\operatorname{Ind} A = \tfrac{1}{2}\big(\operatorname{sgn} a_+ - \operatorname{sgn} a_-\big).$$

The formula. Take $Q = (A^*A+1)^{1/2}$. Near the ends $A^*A = -\partial_u^2 + a_\pm^2$ commutes with A, so A is normal there and Theorem 15.12 applies. The symbol of $[\log Q, A]$ is $-i\{\log\langle\xi\rangle + c_{-1} + \cdots,\ i\xi + a\}$, of order -1, because the principal symbol $i\xi$ of A does not depend on u. So $A^{-1}[\log Q, A]$ has order -2, and its residue density (homogeneity $-n = -1$) vanishes: $\mathrm{AS}(A) = 0$. By Proposition 15.7, at the right end

$$\eta = \frac{1}{\pi i}\int \frac{-i}{a_+ - i\tau}\,d\tau = -\frac{1}{\pi}\int \frac{a_+ + i\tau}{a_+^2 + \tau^2}\,d\tau = -\operatorname{sgn} a_+,$$

and at the left end $\eta = \frac{1}{\pi i}\int \frac{i}{a_- + i\tau} d\tau = \operatorname{sgn} a_-$. So

$$\mathrm{AS}(A) - \tfrac{1}{2}\eta(A_0) = 0 - \tfrac{1}{2}\big(-\operatorname{sgn} a_+ + \operatorname{sgn} a_-\big) = \tfrac{1}{2}\big(\operatorname{sgn} a_+ - \operatorname{sgn} a_-\big) = \operatorname{Ind} A.$$

This confirms the sign of the eta term in [14, (93)] and the sign of $\widetilde{D}$ in Proposition 15.3. If instead $\widetilde{D}$ had the sign of [14, (21)], the eta term would enter with $+\frac{1}{2}$ and the formula would give $-\operatorname{Ind} A$. At the right end the boundary operator in APS form is $B_\partial = -a_+$ (Corollary 15.9), and $\eta_{\mathrm{APS}}(B_\partial) = -\operatorname{sgn} a_+$, as found.

Example 16.3 (Dirac operators on manifolds with cylindrical ends)**.** Let $\eth$ be a Dirac operator for an exact cusp metric (Remark 4.6) and a product structure near the boundary, $\eth = \sigma(-\partial_t + \eth_\partial)$. If $\eth_\partial$ is invertible, $\eth$ is fully elliptic. In the $\mathbb{Z}_2$-graded setting of Section 17.2, Theorem 15.12 and Corollary 15.9 give

$$\operatorname{Ind} \eth^+ = \mathrm{AS}(\eth) + \nu - \tfrac{1}{2}\eta(\eth_\partial),$$

where ν is the interior correction of [14, (110)]. This is the APS theorem in the invertible case. The term AS is the residue form of the Atiyah–Singer integrand, by the local index theorem. MN do not evaluate it further.

17 Cyclic homology, the graded case and other algebras (MN §§13–15)

We summarize the remaining sections of [14], with comments.

17.1 Cyclic homology

For a unital algebra the Connes–Loday–Quillen sequence $\cdots \to \mathrm{HH}_n \xrightarrow{I} \mathrm{HC}_n \xrightarrow{S} \mathrm{HC}_{n-2} \xrightarrow{B} \mathrm{HH}_{n-1} \to \cdots$ relates cyclic and Hochschild homology [14, (101)–(103)]. The key point is [14, Lemma 17]: $B = 0$ for $\mathcal{A}_\sigma, \mathcal{I}_\sigma, \mathcal{A}_{\partial,\sigma}, \mathcal{I}_\partial$. The reason is an order count. By the spectral sequences, HH_q is generated by cycles of order $q - n$ (x-degree for $\mathcal{I}_\partial$), and all cycles of lower order are boundaries. Inductively $\mathrm{HC}_{m-1} \cong \bigoplus_k \mathrm{HH}_{m-1-2k}$ is generated by elements of order at most $m - 1 - n$. The operator B does not raise order, so it maps them into cycles of order less than $m - n$, which are boundaries. Hence [14, Prop. 18]

$$\mathrm{HC}_m(\mathfrak{A}) \cong \bigoplus_{k\geq 0} \mathrm{HH}_{m-2k}(\mathfrak{A}), \qquad \mathfrak{A} = \mathcal{A}_\sigma, \mathcal{I}_\sigma, \mathcal{A}_{\partial,\sigma}, \mathcal{I}_\partial, \mathcal{A}_\partial,$$

with S the projection, and periodic cyclic homology obtained by dropping the restriction $k \geq 0$. For $\mathcal{A}_\partial$ MN write $\mathrm{HC}_i(\mathcal{A}_\partial) \cong \mathrm{HC}_i(\mathcal{A}_{\partial,\sigma})/\mathbb{C}$ for $i > 0$. With Proposition 12.3, the codimension is N in general, and the statement is about dimensions: in homology $\mathrm{HH}_{1,2}(\mathcal{A}_\partial)$ are subspaces of $\mathrm{HH}_{1,2}(\mathcal{A}_{\partial,\sigma})$, not quotients.

17.2 Bundles and grading

Dirac operators map between different bundles $E^\pm$. MN therefore pass to $\mathbb{Z}_2$-graded algebras [14, §14]. A graded embedding $E \subset X \times (\mathbb{C}^N \oplus \mathbb{C}^N)$ gives a graded HKR map with the supertrace and the Grassmann connection [14, (105)]. The Hochschild groups have the same dimensions as in the ungraded case and do not depend on the embedding [14, Lemma 18, Prop. 19]. If $E^+ = E^-$ near the boundary, the index of an odd fully elliptic A is [14, Thm. 1, (111)]

$$\mathrm{Ind}(A) = \mathrm{IF}([A], [A]^{-1}) + \nu, \qquad \nu = \mathrm{Tr}\big(e^+ Q_N^{-s}\big) - \mathrm{Tr}\big(e^- Q_N^{-s}\big)\big|_{s=0},$$

a local interior correction coming from the embedding. Proposition 15.3 holds verbatim, and so do our sign corrections.

17.3 Other algebras

The method applies to $\Psi_c^{\mathbb{Z}}(X)$, $\Psi_c^0(X)$, complete symbol algebras and the b-calculus, but the groups can be infinite dimensional [14, §15]. For $\mathcal{A}_6 = x^{-\mathbb{Z}}\Psi_c^0(X)/x^{-\mathbb{Z}}\Psi_c^{-\infty}(X)$ the E^1 term agrees with that of $\mathcal{A}_{\partial,\sigma}$ for $p \leq 0$ and vanishes for $p > 0$. The E^2 term is unchanged for $p < 0$, while $E^2_{0,q}$ consists of spaces of closed forms [14, Prop. 20]. This is the precise sense in which restricting to order zero destroys finiteness. (MN state the E^1 comparison with $\mathcal{A}_{\partial,\sigma}$. Since $\mathcal{A}_6$ is the order-≤ 0 part of $\mathcal{A}_\sigma$, and the stated $E^2_{0,q}$ involves forms on Y as well as on Z, this is presumably a misprint for $\mathcal{A}_\sigma$.)

18 Comparison with the b-calculus

The companion notes [6] carry out the b-calculus analogue of [14], as MN suggest [14, p. 3]. Table 1 summarizes the differences.

	cusp calculus [14]	b-calculus [6]
vector fields	$x^2\partial_x,\ \partial_y$	$x\partial_x,\ \partial_y$
boundary variable	$t = 1/x$, translations	$\log x$, dilations
indicial family	$\mathrm{In}(A,\tau)$, $\tau \in \mathbb{R}$, symbol class	$I(A,\lambda)$, entire in $\lambda \in \mathbb{C}$
conjugation by x	$x^{-1}Ax = A + O(x)$: a *derivation* at leading order, $[x, \partial_t] = x^2$	$I(x^{-1}Ax,\lambda) = I(A,\lambda - i)$: an *automorphism*
graded algebra of $\mathcal{I}_\partial$	commutative, Poisson	skew Laurent series
method for $\mathcal{I}_\partial$	x-adic spectral sequence	weight decomposition, crossed products
invertibility in $\mathcal{A}$	$\mathrm{In}(A,\tau)$ invertible for $\tau \in \mathbb{R}$	$I(A,\lambda)$ invertible for all $\lambda \in \mathbb{C}$
$\mathrm{HH}^{0,1}(\mathcal{I}_\partial)$	$\mathbb{C}^N, \mathbb{C}^N$	$\mathbb{C}^N, \mathbb{C}^N$
rank of $\mathrm{HH}^1(\mathcal{I}_\partial) \to \mathrm{HH}^2(\mathcal{A}_\sigma)$	$N-1$ (MN: 0)	$N-1$
trace defect on $\mathcal{I}_\partial$	$b\,\mathrm{Tr}_c = -i_{\log x}\,\mathrm{Tr}_\partial$	b-trace defect formula

Table 1: Cusp versus b-calculus.

18.1 Derivation twist versus automorphism twist

In the cusp calculus, Lemma 4.4 shows that commuting x past an indicial operator produces terms of *higher* order in x, governed by the derivation ∂_τ. The associated graded algebra is therefore commutative, with Poisson bracket $\{t, \tau\} = 1$. The x-adic spectral sequence is then a Brylinski–Getzler computation on $\mathbb{R}_\tau \times \mathbb{S}_\partial$ (Proposition 9.2). In the b-calculus, $x^{-1}Ax$ has indicial family $I(A, \lambda - i)$. The complex shift by $-i$ is an automorphism that is *not* close to the identity, so the associated graded algebra is a skew Laurent series ring. The computation proceeds by a weight decomposition and the homology of crossed products by $\mathbb{Z}$ [6, §"The boundary ideal"]. The answers agree, $\mathrm{HH}_{0,1}(\mathcal{I}_\partial) \cong \mathbb{C}^N$. The generators are also closely related: Tr_∂ and $i_{\log x}\,\mathrm{Tr}_\partial$ in the cusp case, and the boundary trace Tr_∂ and the cocycle φ of [6] in the b-case.

18.2 Several boundary components

In both calculi the boundary algebras are direct sums over components, and the only global interaction is through the interior symbol algebra. The map $\mathrm{HH}^1(\mathcal{I}_\partial) \to \mathrm{HH}^2(\mathcal{A}_\sigma)$ sends $i_{\log x}\,\mathrm{Tr}_{\partial,j}$ to the de Rham coboundary of $d\theta_\sigma 1_{Z_j}$ (Proposition 12.5). Its kernel is spanned by the sum over j, because only $\sum_j 1_{Z_j} = 1_Z$ extends over the connected Y. Equivalently, by Stokes' theorem, $\sum_j \int_{Z_j} \beta = 0$ for every closed β on Y. This is the rank $N-1$ phenomenon of [6, Remark "Several boundary components"]. In the cusp case MN's argument ("$d\theta_\sigma$ extends") covers only the sum, and their statements (16) and Proposition 12 are correct exactly when $N = 1$. Proposition 9.5 shows the same rank in the simplest setting: $\mathrm{HH}_1(x^{-\mathbb{Z}}\Psi_c^{-\infty}(X)) \cong \mathbb{C}^{N-1}$, the cusp analogue of [6, Thm., part (iii)].

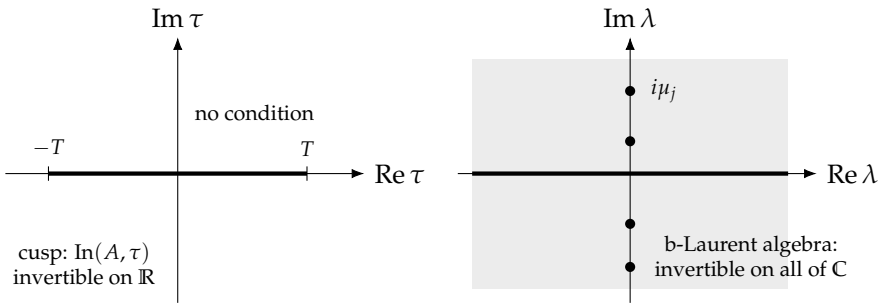


Figure 8: Left: a cusp operator is fully elliptic when $\mathrm{In}(A,\tau)$ is invertible for real τ (thick line). The eta term is the regularized integral of $\mathrm{Tr}(A_0^{-1}\partial_\tau A_0)$ over $[-T,T]$, with $T\to\infty$ by Hadamard's finite part. Right: in the b-calculus $I(A,\lambda)$ is entire, and invertibility in the b-Laurent algebra requires $I(A,\lambda)$ to be invertible on all of $\mathbb{C}$ (shaded). A Dirac operator $x\partial_x + B_\partial$ has indicial roots $i\mu_j$, $\mu_j \in \operatorname{spec} B_\partial$ (dots), so it is never invertible there. Fredholmness on L^2_b needs only the real axis, which is exactly the cusp condition after the transcendental blow-up.

18.3 Invertibility, completeness and the trace defects

In the b-calculus, invertibility in $\mathcal{A}_b$ forces the indicial family to be invertible on all of $\mathbb{C}$ (Section 4.6). Dirac operators never satisfy this, so the b-Laurent algebra cannot see APS-type indices. The cusp calculus needs invertibility only on $\mathbb{R}$, which is the APS condition $\ker B_\partial = 0$. Under the transcendental blow-up $x_b = e^{-1/x}$ the b-indicial family becomes the cusp indicial family with $\tau = \lambda$. A b-operator with indicial family invertible on $\mathbb{R}$ lifts to a fully elliptic cusp operator, and its index is given by Theorem 15.5. The trace defects also match. For translation invariant A, B, Proposition 9.2 and Proposition 9.5 give

$$\mathrm{Tr}_c([A,B]) = -\,\mathrm{Tr}_\partial(A[\log x, B]) = -\frac{i}{2\pi}\int_{\mathbb{R}} \mathrm{Tr}\big(\mathrm{In}(A,\tau)\,\partial_\tau\,\mathrm{In}(B,\tau)\big)\,d\tau.$$

The b-trace satisfies the same identity in the variable λ: ${}^b\mathrm{Tr}([A,B]) = -\,\mathrm{Res}_z\,\mathrm{Tr}(A[\log x_b, B]x_b^z)$, and $I([\log x_b, B]) = i\partial_\lambda I(B)$ [6, Lemma "Conjugation by powers of x"]. This is Melrose's b-trace defect formula. The two regularizations differ (x^z versus $x_b^z = e^{-z/x}$), but on the leading terms they agree.

Acknowledgement of sources

The mathematics in these notes is that of Melrose and Nistor [14]. The spectral sequence and HKR arguments follow [13, Ch. 10]. The Hochschild tools, the contraction lemmas and the comparison with the b-calculus follow the companion notes [6]. Corrections and the worked examples are our own and are marked as such.

References

[1] M. F. Atiyah, V. K. Patodi and I. M. Singer, *Spectral asymmetry and Riemannian geometry I*, Math. Proc. Cambridge Philos. Soc. **77** (1975), 43–69.

[2] J.-L. Brylinski, *A differential complex for Poisson manifolds*, J. Differential Geom. **28** (1988), 93–114.

[3] J.-L. Brylinski and E. Getzler, *The homology of algebras of pseudo-differential symbols and the noncommutative residue*, K-Theory **1** (1987), 385–403.

[4] A. Connes, *Non commutative differential geometry*, Publ. Math. IHÉS **62** (1985), 41–144.

[5] V. W. Guillemin, *A new proof of Weyl's formula on the asymptotic distribution of eigenvalues*, Adv. Math. **55** (1985), 131–160.

[6] *Hochschild Homology of the b-Calculus*, companion notes in this repository (file `Hochschild homology of b-pseudodifferential operators.tex`).

[7] M. Karoubi, *Formule de Künneth en homologie cyclique I et II*, C. R. Acad. Sci. Paris **303** (1986), 527–530 and 595–598.

[8] J.-L. Loday, *Cyclic Homology*, Grundlehren Math. Wiss. 301, Springer, 1992.

[9] R. Mazzeo and R. B. Melrose, *Pseudodifferential operators on manifolds with fibred boundaries*, Asian J. Math. **2** (1998), 833–866.

[10] R. B. Melrose, *The Atiyah–Patodi–Singer Index Theorem*, Research Notes in Mathematics 4, A K Peters, 1993.

[11] R. B. Melrose, *The eta invariant and families of pseudodifferential operators*, Math. Res. Lett. **2** (1995), 541–561.

[12] R. B. Melrose, *Fibrations, compactifications and algebras of pseudodifferential operators*, in *Partial Differential Equations and Mathematical Physics* (L. Hörmander and A. Melin, eds.), Birkhäuser, 1996, 246–261.

[13] R. B. Melrose, *Introduction to Microlocal Analysis*, lecture notes, chapters "Traces, determinants and the index" and "Hochschild homology".

[14] R. B. Melrose and V. Nistor, *Homology of pseudodifferential operators I. Manifolds with boundary*, preprint (1996), arXiv:funct-an/9606005.

[15] R. Nest and B. Tsygan, *Formal deformations of symplectic manifolds with boundary*, J. Reine Angew. Math. **481** (1996), 27–54.

[16] V. Nistor, *Properties of the boundary map in cyclic cohomology*, Penn State preprint, 1994 (reference [20] of [14]).

[17] *Notes on the b-Calculus and the APS Index Theorem*, based on lectures by P. Loya, companion notes in this repository.

[18] F. Trèves, *Topological Vector Spaces, Distributions and Kernels*, Academic Press, 1967.

[19] M. Wodzicki, *Local invariants of spectral asymmetry*, Invent. Math. **75** (1984), 143–178.

[20] M. Wodzicki, *Cyclic homology of differential operators*, Duke Math. J. **54** (1987), 641–647; and *Cyclic homology of pseudodifferential operators and noncommutative Euler class*, C. R. Acad. Sci. Paris **306** (1988), 321–325.

[21] M. Wodzicki, *Excision in cyclic homology and in rational algebraic K-theory*, Ann. of Math. **129** (1989), 591–639.